Polynomial Upper Bounds on Characteristic-Data Length for Multiple Context-Free Languages under Fixed Finite-Semigroup Typing

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Abstract

We study positive-data learning of multiple context-free languages that are tuple-substitutable under a fixed finite-semigroup typing and bound the total encoded length of characteristic samples relative to their MCFG presentations. Qualitative identification follows Yoshinaka's distributional framework; our contribution is quantitative. A single computable parameter, the shortest yield length t_{typ} of the full yield-type refinement, controls the encoded witness length. With the alphabet, fan-out, rule rank, and typing fixed, a reduced good grammar G of rule thickness $\bar{\theta}_G$ admits characteristic data of length $O(\|G\|^2\bar{\theta}_G)$ for prefix–suffix-window typings combined with any finite, possibly noncommutative, group, and of length $O(\|G\|^3\bar{\theta}_G)$ when the positive typing image is a local group. The proofs shorten interleaved tuple components simultaneously while preserving ordered group products. A type-deletion argument gives presentation-independent lower bounds from the shortest words of each realized type. Combined with an explicit bracketed doubling family, whose substitutability is proved here, the local-group condition characterizes polynomial characteristic-data length at fan-out one at the level of finite-semigroup pseudovarieties (with rank two allowed). At higher fan-out local groups remain sufficient and characterize polynomial thickness of the full yield-type refinement, but their necessity for characteristic-data length remains open. General input MCFGs are covered by parameterized bounds using component-pattern thickness, which may be exponential in grammar size. A separate conservative learner gives Gold identification. We exhibit a class-wide finite-feature bias for cross-serial languages and explicit characteristic samples for that family. A two-hole power-transfer lemma shows why this exponential family is excluded by every fixed finite typing at fan-out two or more.

Keywords: grammatical inference; multiple context-free grammars; positive data; finite-semigroup typing; multidimensional substitutability; Gold identification; rule rank.

1 Introduction

Positive-data grammatical inference asks when a language can be identified in the limit from positive examples alone [1, 2]. Distributional learning answers this question by treating words, or tuples of words, that occur in common contexts as candidate constituents. The approach began with substitutable CFGs [3, 12] and was extended to multiple context-free grammars (MCFGs), whose tuple-valued nonterminals express discontinuous and cross-serial dependencies [13, 14, 29, 30]. Yoshinaka's finite-signature generalization [15] provides positive-data identification and, in the corresponding bounded-rank specialization, characteristic samples consisting of polynomially many words.

The question: how long must characteristic data be? A sample of polynomial *cardinality* may still contain exponentially long words. We therefore measure a finite sample K by its total encoded length $\|K\|_+ = \sum_{w\in K}(|w|+1)$, not merely by $|K|$. This follows the polynomial *thick-data* perspective reviewed by Eyraud, Heinz and Yoshinaka [10, Section 2.5.2]: the size of a grammar and the length needed to exercise its productive rules are distinct resources. Our aim is to bound the total encoded length of characteristic data relative to an MCFG presentation of the target. Qualitative identification is inherited from Yoshinaka's framework; the contribution of this paper is quantitative.

The setting: a fixed finite-semigroup typing. We study this question for learners equipped with a finite bias. Before any data are seen, we fix a fan-out bound f and a surjective finite-semigroup homomorphism $s : \Sigma^+ \twoheadrightarrow \mathsf{S}$, called a *typing*. The learner compares observed *nonempty* tuple components only when their s-values agree; the target class $\mathcal{C}^{\mathrm{mcf}}_{f,s}$ consists of the f-MCFLs that are tuple-substitutable under this restriction (Definition 2.8). The canonical fresh-identity completion $\widehat{s} : \Sigma^* \to \mathsf{S}^\sharp$ is used to type grammar yields and evaluate rule templates, including empty terminal factors. The substitutability condition itself compares nonempty tuple components using s. Fixing the typing in advance separates the choice of a finite bias from learning under that bias [5]. Prefix–suffix windows, values in a finite group, their products, and DFA transition morphisms are typical instances. Other biases restrict derivations by regular control sets [6, 7] or restrict the evidence used to compare factors [9, 11].

The typing is a *class-level* bias: it is supplied once, before the target is known, and is not chosen to recognize each target separately. Finite features, such as morphological compatibility or case licensing, can constrain substitution between discontinuous constituents while the MCFG accounts for unbounded dependencies. For example, fix one window–group typing $s = (s_{1,1}, g)|_{\Sigma^+_{\mathrm{feat}}}$ for *every* $R \subseteq \{0,1\}^2$, and consider

$$L_R = \{\mathtt{a}^m\mathtt{p}_i\mathtt{b}^n\mathtt{c}^m\mathtt{q}_j\mathtt{d}^n : m,n \geq 1,\ (i,j) \in R\}.$$

Every L_R belongs to $\mathcal{C}^{\mathrm{mcf}}_{2,s}$, and for nonempty R these are non-context-free cross-serial languages (Corollary 6.4). The group component g records the two feature *values*, not the compatibility relation R; neither R nor a target grammar is an input to the learner, which must recover both the licensing information and the unbounded crossing dependencies from positive examples. Explicit characteristic samples of encoded length at most $36|R| + 22$ exist for $R \neq \emptyset$, while every sample characteristic for a single set-driven operator on the whole family must exhibit every licensed pair (Proposition 6.5). The nonrectangular instance $R = \{(0,0),(0,1),(1,0)\}$ shows that untyped and boundary-only substitutability both fail here (Proposition 6.3).

Why noncommutative and local-group typings? A finite permutation automaton naturally supplies a group-valued feature; its transition group need not be commutative (two transpositions of three states already generate S_3). For discontinuous constituents, the order in which features compose can matter. The agreement family L^A_R of Proposition 6.6 compares group values of separated blocks under one fixed window–$(A \times A)$ typing. For $A = S_3$, agreement belongs to this class, but no window–abelian typing admits it at fan-out two, while Theorem 4.14 provides class-uniform $O(\|G\|^2\bar{\theta}_G)$ data for such typings. A local group is a finite semigroup T such that eTe is a group for every idempotent $e \in T$. Local groups further allow group contributions from sliding factors, such as the parity of occurrences of a bigram, beyond letterwise group morphisms together with fixed boundary windows (Proposition B.5).

Why bound the original presentation? A natural route is to refine the grammar by types and apply untyped estimates to the refinement. This route can be expensive. The full typed construction may contain as many as $|V||\mathsf{S}^\sharp|^f$ nonterminals and $|P||\mathsf{S}^\sharp|^{fr}$ rules, and its thickness can be exponential: for a fixed two-element typing s_{c} there are rank-two CFG presentations G_n of $\{\mathsf{a},\mathsf{c}\}^+$ with $\|G_n\| = O(n)$ and rule thickness $\theta_{G_n} \le 2$, whose trimmed full yield-type refinement has thickness $\theta_{\widetilde{G}_{n,0}} \ge 2^n$ (Proposition 4.8). We therefore bound characteristic data directly in terms of the original presentation, and ask when the algebra of the fixed typing permits short witnesses *before* this cost is paid.

The technical obstacle. To obtain short representatives of prescribed root types, we replace selected long subderivations by shorter derivations that need not preserve the componentwise group values of the replaced subtrees. In a multi-component MCFG derivation, one subtree may contribute several blocks to different root components, interleaved with blocks from other subtrees. For noncommutative groups, the effect of simultaneous replacements depends on the order of these blocks. The key quantitative step is an *owner-wise simultaneous selection* of replacements that preserves the ordered group product of every root component, even though individual replacement blocks may change their group values (Proposition 4.10).

Main results. Two input-sensitive thickness scales appear. For a reduced good MCFG G, the primary scale is its *rule thickness* $\bar\theta_G = \max\{1, \theta_G\}$: the maximum, over rules, of the shortest accepted sentence using that rule (Section 4.1). For an arbitrary standard MCFG H, the normalization bounds use instead the *component-pattern thickness* $\bar t_{\mathrm{pat}}(H_{\mathrm{red}})$, where H_{red} is obtained by trimming unreachable or unproductive nonterminals and rules from H. This parameter measures the shortest way to realize each feasible pattern of nonempty tuple components (Definition 4.24). The two scales are not interchangeable for inputs with empty components; Table 2 compares them with the auxiliary quantities used in the proofs. The central reduction (Theorem 4.4) measures the smallest typed tuple yields by $t_{\mathrm{typ}}(G,s)$ and bounds total characteristic-data length by $N_{\mathrm{typ}}(1 + 2\bar\theta_G + r_{\max} t_{\mathrm{typ}})$, where $N_{\mathrm{typ}} = O(\|G\|)$. The window–group and local-group theorems then bound this one parameter by $O(\|G\|\bar\theta_G)$ and $O(\|G\|^2\bar\theta_G)$, respectively. With the alphabet, fan-out f, rule-rank bound $r_{\max}$ and typing fixed, write $\bar\theta = \bar\theta_G$ for a reduced good grammar G, and $n = \max\{1, \|H\|\}$, $t = \bar t_{\mathrm{pat}}(H_{\mathrm{red}})$ for a standard input grammar H. The principal bounds on total encoded characteristic-data length for the set-driven Const/Link/Comp reconstruction operator are:

Fixed typing / rank	Reduced good $(\|G\|, \bar\theta)$	Any standard input (n,t)	Source
Rule rank ≤ 1; any typing	—	$O(n^3)$	Thm. 5.1
Prefix–suffix window	$O(\|G\|^2\bar\theta)$	$O(n^3 t)$	Thm. 4.13
Window with finite group	$O(\|G\|^2\bar\theta)$	$O(n^3 t)$	Thm. 4.14
Positive image in **LG**	$O(\|G\|^3\bar\theta)$	$O(n^4 t)$	Thm. 4.19

The bounds are *class-uniform*: they hold for every $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$ with a presentation of the stated form, with constants depending only on f, $r_{\max}$, $|\Sigma|$ and the typing. For noncommutative groups these contain $M_{A,f} = U_f \max\{3, |A|^f\}^{2^{f^2}}$, where $U_f = f^{f+1} f!$; this grows doubly exponentially in f. The abelian case is sharper (Theorem 4.15). The arbitrary-input column follows by normalization controlled by component-pattern thickness (Corollary 4.26). A separate conservative learner gives Gold identification.

The shortest word in each realized s-type of the target is a presentation-independent lower bound: all those types must occur in every characteristic sample (Proposition 4.6). This yields a lower–upper sandwich (Corollary 4.7). A pseudovariety of finite semigroups is a class closed under finite direct

products, subsemigroups and homomorphic images. At fan-out one, the local-group bound here and an explicit bracketed doubling family (Proposition 4.22) together characterize the pseudovarieties with uniform polynomial characteristic-data length: precisely the subpseudovarieties of **LG** (Remark 4.23). The family was introduced in the companion CFG manuscript [25, Proposition 6.4]; to keep the present paper independent of that manuscript, its substitutability is proved in Appendix G. At fan-out two or more, **LG** remains sufficient and still characterizes polynomial full-refinement thickness (Corollary 4.20). This doubling family cannot be transferred to higher arity: the family is not (f, s)-tuple-substitutable for any fixed finite typing at any $f \geq 2$ (Theorem 6.13).

Further results include a local-group sufficient condition strictly beyond $\mathbf{LI} \vee \mathbf{Grp}$ (Proposition B.5), a rule-only computable witness budget (Proposition 5.2), and an intrinsically rank-two example (Theorem 6.16). Theorem 5.4 complements the class-uniform bounds: it gives a polynomial bound for targets that *have* a representation $L = \widehat{s}^{-1}(\mathsf{Acc}) \cap \varphi^{-1}(\gamma + \mathsf{H})$ for an abelian-group morphism φ and a subgroup H, even when the auxiliary abelian group is infinite.

Relation to prior work. The table compares the guarantees used here.

Source	Established scope	Present additional question
Yoshinaka (2008) [12]	(k, ℓ)-substitutable CFGs; characteristic-data size with a grammar-thickness parameter	Higher-fan-out, fixed-group, and LG-typed shortening
Yoshinaka (2009, 2011) [13, 14]	Multidimensionally substitutable MCFGs and positive-data learning; polynomial identification in the corresponding grammar regime	Original-thickness bounds under supplied finite-semigroup typing
Yoshinaka (2015) [15]	Finite-signature and disjoint-sort learning principles, including finite witness cardinality under fixed bounded-rank signatures	Encoded-length upper bounds that survive the typing refinement

The qualitative learning principle, the Const/Link/Comp construction and finite witness cardinality are due to Yoshinaka [14, 15]; the finite-semigroup structure theory and local-group/delay representations are classical [36–38]; the rank-one obstruction uses [21, 22]. Our contribution is the noncommutative owner-wise shortening, the resulting window–group and local-group bounds (Theorems 4.14 and 4.19), and the normalization controlled by component-pattern thickness. The bracketed doubling family of Proposition 4.22 first appears in the related CFG manuscript [25]; its substitutability is proved in Appendix G, so no result of that manuscript is a premise here. Other mildly context-sensitive formalisms are learned from positive data in [19], and MCFGs with membership queries or dual context sets in [16, 18, 20]; the MCFL substitution lemma of [23] is a membership condition, not a learning criterion.

Relation to the companion CFG manuscript. The author's companion manuscript [25] treats the fan-out-one case, context-free grammars under a fixed finite-semigroup typing. Table 1 separates its results from those of the present paper; numbers in the second column refer to the companion. No result of the companion is used as a premise: every statement needed here is proved in this paper. At fan-out one, the qualitative reconstruction, the typed-thickness witness bound, the window–group bound and the bracketed doubling family are already in the companion and are not claimed as new. New at fan-out one are the local-group typings and their bounds, the general rare-type lower bound, and hence the pseudovariety boundary of Remark 4.23; everything else in the table is specific to fan-out $f \geq 2$ or to arbitrary MCFG presentations.
The expressive results of the companion (the counter and Dyck obstructions, the nonlinear example and the comparison with Clark's congruential grammars) are not used here.

Table 1: Results of the companion CFG manuscript [25] (fan-out one) and of the present paper.

Topic	Companion [25]	Present paper	Status here
Typed setting, set-driven operator, typed refinement, Gold learner	Lemma 2.2; Sections 4–5 (CFG)	Section 3; Proposition 3.11, Theorem 3.13, Corollary 3.16	Tuple version; qualitative results credited to Yoshinaka in both; not claimed new
Typed-thickness witness bound	Lemma 6.2, Corollary 6.3 (binary CFG normal form)	Theorem 4.4 (reduced good MCFG)	Tuple version; same idea at $f = 1$
Locally trivial typings as fixed windows	Proposition 3.5	Used in Lemma 4.16	Classical; not new
Window and window–group bounds	Lemma 7.1, Theorem 7.2, Lemma 7.4, Corollary 7.5 (one word, one ordered product)	Theorems 4.13, 4.14; Corollary 4.17	Not new at $f = 1$. New for $f \geq 2$: owner-wise simultaneous replacement across interleaved components (Proposition 4.10, Appendix A)
Local-group typings	Not treated (only $\mathbf{LI} \vee \mathbf{Grp}$, and $U_1 \notin \mathbf{LI} \vee \mathbf{Grp}$ in Lemma 7.6)	Definition 4.18, Theorems B.4, 4.19; Proposition B.5; Appendix B	New at every f, including $f = 1$; the underlying semigroup algebra is classical
Bracketed doubling family under s_U	Proposition 6.4, Corollary 6.6, Appendix C	Proposition 4.22, Appendix G	Not new; reproved (adapted parser argument) for self-containment
Rare-type lower bound	Proposition 6.4(iii)–(iv) for the single word w_n^*; typed fiber minimum in Section 6	Lemma 4.5, Proposition 4.6, Corollary 4.7	Generalization to every realized type and every f
Fan-out-one pseudovariety boundary	Sufficiency for $\mathbf{LI} \vee \mathbf{Grp}$ only; necessity not claimed (Remark 7.7)	Remark 4.23	New: the local-group upper bound combined with the doubling family
Exponential refinement thickness	Corollary 6.6 (every binary-normal-form presentation of L_n)	Proposition 4.8, Corollary 4.20	New: grammars for Σ^+ and the $\mathbf{LG}$ dichotomy at every f
Rank one	Theorem 8.3 (linear CFG targets)	Theorem 5.1	At $f = 1$ a variant of the companion bound; new for $f \geq 2$
Only in the present paper	—	Proposition 4.25, Proposition 5.2, Theorem 5.4, Section 6 (incl. Theorems 6.13 and 6.16)	New

Scope of the claims. All bounds are upper bounds on the total encoded length of characteristic samples for the specified set-driven operator, relative to the stated presentation; they bound neither an intrinsic grammar size nor a stage of Gold convergence, and their exponents are not claimed optimal (see the end of Section 7). Lower bounds on data come only from distinguishing targets (Propositions 4.6 and 6.5): thick refinements (Proposition 4.8, Corollary 4.20) and component-pattern blow-up (Proposition F.2) limit the witness construction, not the data. The typing is a class-level bias fixed before the target is seen; refining it enlarges the class ($s \preceq t$ implies $\mathcal{C}^{\mathrm{mcf}}_{f,s} \subseteq \mathcal{C}^{\mathrm{mcf}}_{f,t}$) but changes the reconstruction operator, so we neither infer nor optimize it. Accordingly, the typing-class separations of Section 6 separate fixed biases rather than giving lower bounds for individual targets, and Theorem 5.4 covers only targets with the stated abelian-filter representation, not all of $\mathcal{C}^{\mathrm{mcf}}_{f,s}$. The fan-out-one characterization (Remark 4.23) holds at the level of pseudovarieties, not for each individual non-local-group typing, and whether **LG** is necessary at fan-out two or more is open (Open Problem 7.1).

Organization. Section 2 fixes the setting, Section 3 defines the reconstruction operator and its qualitative guarantees, Sections 4 and 5 prove the quantitative bounds, and Section 6 compares the typed classes with multidimensional substitutability and develops the examples. The appendices prove the technical shortening, normalization and reconstruction statements.

2 Preliminaries

Throughout, Σ is a finite alphabet, ε the empty word and $\Sigma^+ := \Sigma^* \setminus \{\varepsilon\}$. If $\Sigma = \emptyset$, the only languages are $\emptyset$ and $\{\varepsilon\}$ and are handled directly. For a finite $K \subseteq \Sigma^*$ put $\|K\|_+ := \sum_{w \in K}(|w|+1)$, and for a tuple $\vec{u} = (u_1, \ldots, u_d)$ put $\|\vec{u}\|_1 := \sum_i |u_i|$. Every polynomial bound below states whether it concerns cardinality or $\|\cdot\|_+$. The table fixes symbols used across sections; all others (sliding-factor labellings, auxiliary morphisms, padding lengths) are defined where they occur.

Symbol	Meaning
$G,\ H$	a (reduced good) MCFG; an arbitrary standard input MCFG
B, C, X, Y	nonterminals; X, Y also denote typed nonterminals where specified
$\|G\|$, θ_G, t_G	encoding size (Definition 2.5), rule thickness, nonterminal thickness
s, $\widehat{s}$, S	positive typing, its fresh-identity completion, its positive image
$\preceq$	kernel refinement (Definition 2.3), also between monoid morphisms on Σ^*
A	a finite group in group-theoretic lemmas (otherwise locally defined); A an abelian group in Section 5.3
g	a monoid morphism $\Sigma^* \to A$ into a finite group
T	a derivation tree

2.1 Positive-Data Learning

Definition 2.1 (Text, TxtEx identification, conservativeness)**.** Fix a pause symbol $\perp \notin \Sigma$. A *text* for $L \subseteq \Sigma^*$ is an infinite sequence over $L \cup \{\perp\}$ in which every element of L occurs; the *content* of a finite prefix is the set of elements of Σ^* occurring in it. A *learner* is a computable map from finite prefixes to grammars. It *TxtEx-identifies* L [1] if on every text for L its outputs are, from some stage on, one fixed grammar H with $L(H) = L$; it TxtEx-identifies a class if it does so for each member. It is *conservative* if it changes its hypothesis only on a datum $w \in \Sigma^*$ not generated by the current hypothesis. We fix a hypothesis H_0 with $L(H_0) = \emptyset$.

Definition 2.2 (Set-driven operator and characteristic sample)**.** A *set-driven reconstruction operator* $\mathcal{B}$ maps each finite $K \subseteq \Sigma^*$ to a grammar $\mathcal{B}(K)$ depending only on the set K; it is *sample-consistent*

if $K \subseteq L(\mathcal{B}(K))$ for all K. A finite $\mathcal{F} \subseteq L$ is a *characteristic sample for $\mathcal{B}$ and L* if $L(\mathcal{B}(K)) = L$ for every finite K with $\mathcal{F} \subseteq K \subseteq L$ (cf. [4]). This constrains generated languages only: $\mathcal{B}(K)$ need not stabilize syntactically as K grows.

2.2 Fixed Finite-Semigroup Typing

Definition 2.3 (Typing, fresh completion, refinement)**.** A *typing* is a surjective semigroup morphism $s : \Sigma^+ \twoheadrightarrow \mathsf{S}$ onto a finite semigroup, given by the multiplication table of S and the images of the letters, so that $s(w)$ is computable in time $O(|w|)$; S is its *positive image*. (A morphism into a larger finite semigroup is replaced by its corestriction to the image.) Let $\mathsf{S}^\sharp := \mathsf{S} \uplus \{\mathbf{1}_\sharp\}$ be obtained by adjoining a *fresh* identity, even if S already has one, and let $\widehat{s} : \Sigma^* \twoheadrightarrow \mathsf{S}^\sharp$ be the monoid morphism with $\widehat{s}(\varepsilon) = \mathbf{1}_\sharp$ and $\widehat{s}|_{\Sigma^+} = s$. For $\vec{w} \in (\Sigma^*)^d$ put $\widehat{s}^{(d)}(\vec{w}) := (\widehat{s}(w_1), \dots, \widehat{s}(w_d))$; on $(\Sigma^+)^d$ this is $s^{(d)}(\vec{w})$. For typings s, t write $s \preceq t$ if $\ker t \subseteq \ker s$ on Σ^+, and let (s, t) be the typing $w \mapsto (s(w), t(w))$ onto its image in $\mathsf{S} \times \mathsf{T}$; thus $s, t \preceq (s, t)$. Write s_{triv} for the one-element typing.

The fresh completion evaluates grammar templates (including empty terminal gaps) and types tuple yields in the refinement of Section 3.2; its new identity is not a positive-word type. Algebraic conditions such as **LG** concern S, not $\mathsf{S}^\sharp$. A finite-monoid morphism $q : \Sigma^* \to N$ determines the positive typing $q|_{\Sigma^+}$: by Lemma E.1(i), it gives the same target class and set-driven reconstruction rules as the fresh completion of that typing. Lemma E.1(ii) additionally identifies their trimmed yield-type refinements for *good* grammars. This last identification is not asserted for arbitrary standard grammars with empty-component yields.

Definition 2.4 (Prefix–suffix typing)**.** For $k, \ell \geq 0$ put $m_0 := \max\{1, k + \ell\}$, and for $|w| \geq m_0$ let $\mathrm{pref}_k(w)$, $\mathrm{suf}_\ell(w)$ be the prefix and suffix of the indicated lengths (length-zero windows are ε). Define

$$\widehat{s}_{k,\ell}(w) = \begin{cases} w, & |w| < m_0, \\ \langle \mathrm{pref}_k(w), \mathrm{suf}_\ell(w) \rangle, & |w| \geq m_0, \end{cases}$$

with disjoint tags for short words and pairs, and put $M_{k,\ell} := \widehat{s}_{k,\ell}(\Sigma^*)$. For short x, y and pairs $\langle u, v \rangle, \langle u', v' \rangle$ define

$$\begin{aligned} x \cdot y &:= \widehat{s}_{k,\ell}(xy), & x \cdot \langle u, v \rangle &:= \langle \mathrm{pref}_k(xu), v \rangle, \\ \langle u, v \rangle \cdot y &:= \langle u, \mathrm{suf}_\ell(vy) \rangle, & \langle u, v \rangle \cdot \langle u', v' \rangle &:= \langle u, v' \rangle. \end{aligned}$$

The *boundary typing* is $s_{k,\ell} := \widehat{s}_{k,\ell}|_{\Sigma^+}$; a typing s with $s \preceq s_{k,\ell}$ is (k, ℓ)*-boundary-determined*, or briefly a *boundary typing*.

A case check on the lengths of $x, y \in \Sigma^*$ gives $\widehat{s}_{k,\ell}(xy) = \widehat{s}_{k,\ell}(x) \cdot \widehat{s}_{k,\ell}(y)$. Since $\widehat{s}_{k,\ell}$ is onto $M_{k,\ell}$, the product is associative and $\widehat{s}_{k,\ell}(\varepsilon)$ is an identity, so $\widehat{s}_{k,\ell}$ is a finite-monoid morphism. The short-word tag separates ε from $\widehat{s}_{k,\ell}(\Sigma^+)$, so $\widehat{s}_{k,\ell}$ is isomorphic to the completion $\widehat{s_{k,\ell}}$.

2.3 Multiple Context-Free Grammars

Definition 2.5 (Standard MCFG presentation and size)**.** An MCFG [29, 30] is $G = (V, \Sigma, P, S, \mathrm{fan})$ with finite V, $S \in V$, $\mathrm{fan}(S) = 1$ and $\mathrm{fan}(A) \geq 1$. A rule of rank $r \geq 0$ is $A \to \rho(B_1, \dots, B_r)$ with template $\rho = (\alpha_1, \dots, \alpha_{d_0})$, $d_0 = \mathrm{fan}(A)$, $d_j = \mathrm{fan}(B_j)$, where each α_i is a word over $\Gamma_\rho := \Sigma \uplus \Xi_1 \uplus \cdots \uplus \Xi_r$, $\Xi_j := \{\xi_j^1, \dots, \xi_j^{d_j}\}$, and every variable occurs at most once in $\alpha_1 \cdots \alpha_{d_0}$ (*linearity*). For $\vec{u}_j \in (\Sigma^*)^{d_j}$, $\rho(\vec{u}_1, \dots, \vec{u}_r)$ substitutes $u_{j,\kappa}$ for ξ_j^κ; for $r = 0$ the rule is a *constant rule*. The tuple languages $L_A(G) \subseteq (\Sigma^*)^{\mathrm{fan}(A)}$ form the least family closed under the rules, and

$L(G) := \{w : (w) \in L_S(G)\}$. We write $A \Rightarrow_G^* \vec{u}$ for a *terminal derivation* (a finite derivation tree rooted at A, evaluated bottom-up); one from S is *successful*. G is an f-MCFG if $\mathrm{fan}(A) \le f$ for all A, and has rule-rank cap $r_{\max}$ if all ranks are at most $r_{\max}$. Such a presentation, with no further restriction, is called *standard*. Its *size* is

$$\|G\| := |V| + \sum_{(A \to \rho(B_1,\dots,B_r)) \in P} \Big(1 + r + \textstyle\sum_i |\alpha_i|\Big),$$

where $|\alpha_i|$ counts terminal and variable occurrences. All bounds use this symbol-level size; a bit-level encoding with binary nonterminal names changes it by at most a factor $O(\log \|G\|)$.

$\mathsf{MCFL}(f, r)$ is the class of languages of MCFGs of fan-out $\le f$ and rule rank $\le r$; f-MCFL $= \bigcup_{r<\infty} \mathsf{MCFL}(f, r)$, and the minimum rule rank of a language is taken over all its fan-out-$\le f$ presentations. For nonterminals of equal fan-out d, $A \to B$ abbreviates $A \to (\xi_1^1, \dots, \xi_1^d)(B)$, and $A \to \vec{c}$ a constant rule. Rule variables are always ξ_j^κ; terminals in examples are in `typewriter` font. Fan-out and rank are distinct resources. At fixed fan-out at least two the rule-rank hierarchy does not collapse in general, although adjacent levels can coincide; in particular $\mathsf{MCFL}(2,2) = \mathsf{MCFL}(2,3)$ [24, 32]. By contrast, fan-out-one CFGs can be binarized. The reconstruction below therefore assumes neither a binary normal form nor a known rank cap.

Definition 2.6 (Grammar properties)**.** A rule is *nondeleting* if every variable occurs exactly once in its template; a nondeleting rule is *nonpermuting* if for each j the variables $\xi_j^1, \dots, \xi_j^{d_j}$ occur in this order in $\alpha_1 \cdots \alpha_{d_0}$, and *nonmerging* if moreover no α_i contains a factor $\xi_j^\kappa \xi_j^{\kappa+1}$. Following [14, Section 2.3], a rule is *λ-free* if no α_i is empty. A rule, template or grammar is *good* if it is λ-free, nondeleting, nonpermuting and nonmerging. A nonterminal A is *reachable* if there is a chain $S = A_0, \dots, A_m = A$ in which each A_{i+1} is a child in a rule for A_i, and *productive* if $L_A(G) \neq \emptyset$; G is *reduced* if every nonterminal is both.

In a λ-free grammar every component of every derived tuple is nonempty (induction on derivations). λ-freeness concerns template components and is distinct from $\varepsilon \notin L(G)$.

2.4 Tuple Contexts and Substitutability

A *sentence context of arity d* is a word $E = v_0 \Box_1 v_1 \cdots v_{d-1} \Box_d v_d$ with each hole occurring once, $v_0, v_d \in \Sigma^*$ and $v_1, \dots, v_{d-1} \in \Sigma^+$; $\mathcal{E}_d$ is the set of these *admissible* contexts [14, Section 3.1]. $E[\vec{w}]$ fills $\Box_i$ by w_i, $|E|_{\mathrm{fix}} := \sum_{i=0}^d |v_i|$, and E *accepts* $\vec{w}$ in L if $E[\vec{w}] \in L$. The *d-tuple distribution* of $\vec{x} \in (\Sigma^*)^d$ is $\mathcal{D}_L^{(d)}(\vec{x}) := \{E \in \mathcal{E}_d : E[\vec{x}] \in L\}$; at fan-out one it is the two-sided distribution $\{(u, v) : uxv \in L\}$.

Definition 2.7 (Tuple occurrence)**.** For $w = a_1 \cdots a_n$ number the cut positions $⓪ a_1 ① \cdots a_n ⓝ$ and write $w[i : i'] := a_{i+1} \cdots a_{i'}$. An *arity-$d$ tuple occurrence* in w is a sequence $0 \le i_1 < i_1' < i_2 < i_2' < \cdots < i_d < i_d' \le n$; it determines $\vec{x} = (w[i_1 : i_1'], \dots, w[i_d : i_d']) \in (\Sigma^+)^d$ and the context $E \in \mathcal{E}_d$ obtained by replacing these intervals by $\Box_1, \dots, \Box_d$, so that $E[\vec{x}] = w$. For finite K, $\mathrm{Obs}_d(K) := \{\vec{x} \in (\Sigma^+)^d : E[\vec{x}] \in K \text{ for some } E \in \mathcal{E}_d\}$ is the set of tuples arising from such occurrences, and for a fixed fan-out bound $f \ge 1$, put $\mathrm{Obs}(K) := \bigcup_{1 \le d \le f} \mathrm{Obs}_d(K)$.

Definition 2.8 ((f, s)-tuple substitutability and the target class)**.** Let $f \ge 1$ and let s be a typing. A language $L \subseteq \Sigma^*$ is *(f, s)-tuple-substitutable* if for every $1 \le d \le f$ and all $\vec{x}, \vec{y} \in (\Sigma^+)^d$,

$$s^{(d)}(\vec{x}) = s^{(d)}(\vec{y}) \ \text{ and } \ \mathcal{D}_L^{(d)}(\vec{x}) \cap \mathcal{D}_L^{(d)}(\vec{y}) \neq \emptyset \implies \mathcal{D}_L^{(d)}(\vec{x}) = \mathcal{D}_L^{(d)}(\vec{y}).$$

$\mathcal{C}^{\mathrm{mcf}}_{f,s}$ is the class of (f,s)-tuple-substitutable f-MCFLs; a member of $\mathcal{C}^{\mathrm{mcf}}_{f,s}$ for a boundary typing s is *boundary-typed*. For $\vec{x},\vec{y} \in (\Sigma^+)^d$ write $\vec{x} \equiv^d_{L,s} \vec{y}$ (or $\vec{x} \equiv_{L,s} \vec{y}$) if $s^{(d)}(\vec{x}) = s^{(d)}(\vec{y})$ and $\mathcal{D}^{(d)}_L(\vec{x}) = \mathcal{D}^{(d)}_L(\vec{y})$; this is an equivalence relation.

Empty components are never compared, following [14, Section 3.1 and footnote 3] (see [8] for CFGs), and $\varepsilon \in L$ is handled at the start symbol; otherwise already $\mathtt{a}^+$ would fail, since $\mathtt{a}$ and ε share a context but not their distributions. The bias consists of both s and f:

(a) If $s \preceq t$, equal t-types are equal s-types, so $\mathcal{C}^{\mathrm{mcf}}_{f,s} \subseteq \mathcal{C}^{\mathrm{mcf}}_{f,t}$; in particular $\mathcal{C}^{\mathrm{mcf}}_{f,s_{\mathrm{triv}}} \subseteq \mathcal{C}^{\mathrm{mcf}}_{f,s}$. This compares classes only: the operators $\mathcal{B}_{f,s}$ and $\mathcal{B}_{f,t}$ of Section 3 differ in general (they coincide when $\ker s = \ker t$), and no monotonicity of characteristic-data length in the typing is claimed.

(b) $\mathcal{C}^{\mathrm{mcf}}_{f+1,s} \cap f\text{-MCFL} \subseteq \mathcal{C}^{\mathrm{mcf}}_{f,s}$, but the class is not monotone in f: for trivial typing $\{w\#w^R\}$ is 1- but not 2-dimensionally substitutable [14, Section 3.1, Example 5].

(c) For every $\mathsf{Acc} \subseteq \mathsf{S}^\sharp$, $\widehat{s}^{-1}(\mathsf{Acc}) \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$: it is regular, and componentwise equal types give equal total types in every context.

Proposition 2.9 (Intersection and abelian filtering)**.** *Fix f and s.*

(i) If L_1, L_2 are (f,s)-tuple-substitutable, so is $L_1 \cap L_2$.

(ii) If L is (f,s)-tuple-substitutable, A is an abelian group, $\varphi : \Sigma^* \to \mathsf{A}$ a monoid morphism, $\mathsf{H} \leq \mathsf{A}$ and $\gamma \in \mathsf{A}$, then $L \cap \varphi^{-1}(\gamma + \mathsf{H})$ is (f,s)-tuple-substitutable.

Whenever such an intersection is an f-MCFL, it belongs to $\mathcal{C}^{\mathrm{mcf}}_{f,s}$; in particular so does every f-MCFL $\widehat{s}^{-1}(\mathsf{Acc}) \cap \varphi^{-1}(\gamma + \mathsf{H})$. Such factorizations are not supplied to the learner.

Proof. (i) A context accepting equal-type $\vec{x},\vec{y}$ in $L_1 \cap L_2$ accepts both in each L_i, so their distributions agree in each L_i, hence in the intersection. (ii) If some context E accepts $\vec{x},\vec{y}$ in $\varphi^{-1}(\gamma + \mathsf{H})$, then $\delta := \sum_i (\varphi(y_i) - \varphi(x_i)) \in \mathsf{H}$, and by commutativity $\varphi(E'[\vec{y}]) - \varphi(E'[\vec{x}]) = \delta$ for every $E' \in \mathcal{E}_d$, so coset membership transfers in both directions. Thus $\varphi^{-1}(\gamma + \mathsf{H})$ is (f, s_{triv})-, hence (f,s)-tuple-substitutable; apply (i). The last two claims follow from (i), (ii) and item (c) above. □

3 Reconstruction and Qualitative Learning

This section fixes the reconstruction operator analysed later and records its qualitative guarantees, none of which is claimed as new. For fixed f, rank cap and typing, our operator follows the distributional reconstruction pattern for substitutable languages over finite signatures in [15]. We use the MCFG signature of [15, Example 3] with tuple sorts refined by s-types (cf. [15, Section 8]). The related qualitative results are

Here	Related prior result	Treatment here
Prop. 3.10 (soundness)	[15, Lemma 2]	—
Prop. 3.11 (completeness)	[15, Lemma 3]	arbitrary same-type representatives
Thm. 3.13 (exact reconstruction)	[15, Lemmas 2–3]	as for Prop. 3.11
Cor. 3.16 (identification)	[15, Theorem 1]	unbounded rank over the class

The table records qualitative analogues rather than literal identities of reconstruction operators. The concrete grammar statements and their adaptations are verified directly in Appendices D and E; no embedding into the signature framework is used as a premise. For a fixed sample the uncapped operator is the finite cap $\|K\|_+$, so soundness needs no addition; but identifying all of

$\mathcal{C}^{\mathrm{mcf}}_{f,s}$, whose members have unbounded minimum rank, uses a sample-dependent cap, outside the bounded-arity assumption of [15]. Representatives that are not derivable from their nonterminal are used in Proposition 5.3 and hence Theorem 5.4. Finally, [15] bounds the *number* of characteristic words by the number of target rules and nonterminals; their *length*, which this paper bounds, is not addressed there.

3.1 Set-Driven Reconstruction

All interaction between templates and contexts rests on one observation.

Lemma 3.1 (Induced child contexts)**.** *Let $\rho = (\alpha_1, \dots, \alpha_e)$ be a good template of rank r with child arities $d_1, \dots, d_r$, let $E \in \mathcal{E}_e$, let $1 \le j \le r$, and fix $\vec{u}_i \in (\Sigma^+)^{d_i}$ for $i \neq j$. Let E_j be the word obtained from $E[(\alpha_1, \dots, \alpha_e)]$ by replacing ξ^κ_j by $\square_\kappa$ and every other variable ξ^ι_i by $u_{i,\iota}$. Then $E_j \in \mathcal{E}_{d_j}$ and $E_j[\vec{x}] = E[\rho(\vec{u}_1, \dots, \vec{u}_{j-1}, \vec{x}, \vec{u}_{j+1}, \dots, \vec{u}_r)]$ for every $\vec{x} \in (\Sigma^*)^{d_j}$.*

Proof. The identity holds because substitutions commute. By linearity and nondeletion each $\square_\kappa$ occurs exactly once in E_j, and by nonpermutation in the order $\kappa = 1, \dots, d_j$, since E places $\alpha_1, \dots, \alpha_e$ from left to right. Consider ξ^κ_j and $\xi^{\kappa+1}_j$. If both lie in one α_i, nonmerging puts a terminal or a variable ξ^ι_i with $i \neq j$ between them, which becomes a nonempty word. Otherwise they lie in $\alpha_i, \alpha_{i'}$ with $i < i'$, and the material between them contains the separator $v_i \in \Sigma^+$ of E. So every internal separator of E_j is nonempty. □

Lemma 3.2 (Composition preserves equivalence)**.** *Let $L \subseteq \Sigma^*$, let ρ be a good template of rank r, and let $\vec{x}_j, \vec{u}_j \in (\Sigma^+)^{d_j}$ satisfy $\vec{u}_j \equiv_{L,s} \vec{x}_j$ for every j. Then $\rho(\vec{u}_1, \dots, \vec{u}_r) \equiv_{L,s} \rho(\vec{x}_1, \dots, \vec{x}_r)$.*

Definition 3.3 (Composition witness)**.** Let K be finite and $r \ge 1$. A *composition witness of rank r* in K is a triple $(\vec{z}, \rho, (\vec{u}_1, \dots, \vec{u}_r))$ in which ρ is a good template of rank r with output arity $e \le f$ and child arities $d_1, \dots, d_r \le f$, $\vec{z} \in \mathrm{Obs}_e(K)$, $\vec{u}_j \in (\Sigma^+)^{d_j}$, and $\vec{z} = \rho(\vec{u}_1, \dots, \vec{u}_r)$.

Equivalently, a composition witness is a segmentation of the components of an observed $\vec{z}$ into terminal gaps and nonempty intervals labelled by variables, subject to the good-template conditions. If $E[\vec{z}] \in K$, Lemma 3.1 gives $E_j[\vec{u}_j] = E[\vec{z}] \in K$ with E_j admissible, so every $\vec{u}_j \in \mathrm{Obs}(K)$. The children occupy disjoint, nonempty parts of $\vec{z}$, so

$$r \le \textstyle\sum_j \|\vec{u}_j\|_1 \le \|\vec{z}\|_1 \le \max_{w \in K} |w| \le \|K\|_+. \tag{3.1}$$

Definition 3.4 (Capped and uncapped reconstruction operators)**.** Fix f and a typing s. For $r_{\max} \ge 1$ let $\mathcal{B}^{\le r_{\max}}_{f,s}(K) = \widehat{G}_{\le r_{\max}}(K)$ be the grammar with start symbol $\widehat{S}$, a nonterminal $[\vec{x}]$ of arity d for each $\vec{x} \in \mathrm{Obs}_d(K)$, $1 \le d \le f$, and the rules

(Start) $\widehat{S} \to [(w)]$ for $w \in K \cap \Sigma^+$, and $\widehat{S} \to \varepsilon$ if $\varepsilon \in K$;

(Const) $[\vec{x}] \to \vec{x}$ for every $\vec{x} \in \mathrm{Obs}(K)$;

(Comp) $[\vec{z}] \to \rho([\vec{u}_1], \dots, [\vec{u}_r])$ for every composition witness of rank $1 \le r \le r_{\max}$;

(Link) $[\vec{x}] \to [\vec{y}]$ for $\vec{x}, \vec{y} \in \mathrm{Obs}_d(K)$ with $s^{(d)}(\vec{x}) = s^{(d)}(\vec{y})$ and $E[\vec{x}], E[\vec{y}] \in K$ for some $E \in \mathcal{E}_d$.

For $K \neq \emptyset$ put $\mathcal{B}_{f,s}(K) := \widehat{G}(K) := \widehat{G}_{\le \|K\|_+}(K)$; $\mathcal{B}_{f,s}(\emptyset)$ has no rules. Fixed encoding orders make both operators deterministic and set-driven.

By (3.1), $\mathcal{B}_{f,s}(K)$ contains a Comp rule for every composition witness in K. Both operators are sample-consistent, via $\widehat{S} \to [(w)] \to (w)$ and $\widehat{S} \to \varepsilon$. All non-start templates are λ-free, so $\widehat{S} \Rightarrow^* \varepsilon$ only by the rule $\widehat{S} \to \varepsilon$. For trivial typing and $\varepsilon \notin K$, Comp/Const, Link and Start correspond to the rule Types I–III of [14, Section 4.1], and $\mathcal{B}_{f,s}$ corresponds to the operator of the unbounded-rank learner $\mathcal{A}(p, *)$ of [14, Corollary 2].

3.2 Yield-Typed Refinement and Witnesses

Definition 3.5 (Yield-type refinement)**.** Let $G = (V, \Sigma, P, S, \mathrm{fan})$ be a finite nondeleting MCFG and s a typing. For a template ρ and $\mathbf{t}_j \in (\mathsf{S}^\sharp)^{d_j}$, let $\mathrm{out}_\rho(\mathbf{t}_1, \ldots, \mathbf{t}_r) \in (\mathsf{S}^\sharp)^{d_0}$ have ith component the image of α_i under the monoid morphism $\Gamma_\rho^* \to \mathsf{S}^\sharp$ extending $a \mapsto \widehat{s}(a)$ and $\xi_j^\kappa \mapsto (\mathbf{t}_j)_\kappa$. The *full yield-type refinement* $G^{\widehat{s}}$ has a fresh start symbol $\widetilde{S}$, nonterminals $A_{\mathbf{t}}$ for $A \in V$ and $\mathbf{t} \in (\mathsf{S}^\sharp)^{\mathrm{fan}(A)}$, start rules $\widetilde{S} \to S_{(\eta)}$ for $\eta \in \mathsf{S}^\sharp$, and for every rule $A \to \rho(B_1, \ldots, B_r)$ (including $r = 0$) and all $\mathbf{t}_j \in (\mathsf{S}^\sharp)^{\mathrm{fan}(B_j)}$ the rule $A_{\mathrm{out}_\rho(\mathbf{t}_1,\ldots,\mathbf{t}_r)} \to \rho(B_{1,\mathbf{t}_1}, \ldots, B_{r,\mathbf{t}_r})$. The *trimmed refinement* $\widetilde{G}_0$ keeps exactly the nonterminals and rules occurring in some successful derivation from $\widetilde{S}$. Unless $G^{\widehat{s}}$ is named, *typed nonterminals* and *typed rules* are the non-start nonterminals and rules of $\widetilde{G}_0$, and $L_X := \{\vec{u} : X \Rightarrow^*_{\widetilde{G}_0} \vec{u}\}$.

Typed rules have the templates of their sources, so $\widetilde{G}_0$ is good whenever G is; each L_X is nonempty, and its tuples have nonempty components if G is λ-free.

Proposition 3.6 (Yield-type invariants)**.** *(i) If $A_{\mathbf{t}} \Rightarrow^* \vec{u}$ in $G^{\widehat{s}}$, then $\widehat{s}^{(\mathrm{fan}(A))}(\vec{u}) = \mathbf{t}$, and erasing indices gives a G-derivation of $\vec{u}$.*

(ii) Every terminal derivation $A \Rightarrow^*_G \vec{u}$ lifts uniquely to $A_{\widehat{s}^{(\mathrm{fan}(A))}(\vec{u})} \Rightarrow^*_{G^{\widehat{s}}} \vec{u}$; a successful derivation of w lifts uniquely through $\widetilde{S} \to S_{(\widehat{s}(w))}$.

(iii) $L(\widetilde{G}_0) = L(G^{\widehat{s}}) = L(G)$.

Lemma 3.7 (Canonical contexts for typed nonterminals)**.** *Let G be good. Every typed nonterminal X of arity d has a context $\chi(X) \in \mathcal{E}_d$ with $\chi(X)[\vec{u}] \in L(G)$ for all $\vec{u} \in L_X$; if $\widetilde{S} \to X$ is in $\widetilde{G}_0$, one may take $\chi(X) = \Box_1$.*

Lemma 3.8 (Typed nonterminals are distributionally homogeneous)**.** *Let G be good of fan-out $\le f$ with $L := L(G) \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$. For every typed nonterminal X of arity d, L_X lies in a single $\equiv^d_{L,s}$-class, denoted $\mathrm{Cl}_{L,s}(X)$. This semantic class notation is distinct from the syntactic reconstruction nonterminal $[\vec{x}]$ of Definition 3.4.*

The proofs of these three statements are in Appendix E. The homogeneity of Lemma 3.8 is weaker than that of congruential MCFGs [17], whose contexts may have empty separators.

Definition 3.9 (Canonical tuples, contexts and the witness set)**.** Let G be good. For every typed nonterminal X fix a *canonical tuple* $\omega(X) \in L_X$ and a *canonical context* $\chi(X)$ as in Lemma 3.7, with $\chi(X) = \Box_1$ whenever $\widetilde{S} \to X$ is in $\widetilde{G}_0$. Let $\mathsf{W}(\widetilde{G}_0)$ consist of the words

$$\chi(X)[\omega(X)] \ \ (X \text{ typed}), \qquad \chi(X)[\rho(\omega(Y_1), \ldots, \omega(Y_q))] \ \ (X \to \rho(Y_1, \ldots, Y_q) \text{ typed}, \ q \ge 0).$$

Since $\omega(X)$ and $\rho(\omega(Y_1), \ldots, \omega(Y_q))$ belong to L_X, $\mathsf{W}(\widetilde{G}_0)$ is a finite subset of $L(G)$, nonempty if $L(G) \neq \emptyset$, and $|\mathsf{W}(\widetilde{G}_0)|$ is at most the number of typed nonterminals and rules. Canonical objects are properties of the target, not inputs of the operator: qualitatively any choice works, while Section 4 chooses short ones.

3.3 Qualitative Guarantees

Proposition 3.10 (Soundness of every hypothesis)**.** *Let L be (f, s)-tuple-substitutable and $K \subseteq L$ finite. For every $r_{\max} \ge 1$, if $[\vec{x}] \Rightarrow^* \vec{u}$ in $\widehat{G}_{\le r_{\max}}(K)$, then $\vec{u} \equiv_{L,s} \vec{x}$. Consequently $L(\mathcal{B}^{\le r_{\max}}_{f,s}(K)) \subseteq L$ and $L(\mathcal{B}_{f,s}(K)) \subseteq L$.*

Proposition 3.11 (Completeness from a witness assignment)**.** *Let G be a finite good MCFG of fan-out $\le f$, let r_G be the maximum rank of a typed rule of $\widetilde{G}_0$ ($r_G := 0$ if none), and assign to each typed nonterminal $X = A_{\mathbf{t}}$ a tuple $\iota(X) \in (\Sigma^+)^{\mathrm{fan}(A)}$ with $\widehat{s}^{(\mathrm{fan}(A))}(\iota(X)) = \mathbf{t}$ and a context $\chi(X) \in \mathcal{E}_{\mathrm{fan}(A)}$, with $\chi(X) = \square_1$ if $\widetilde{S} \to X$ is in $\widetilde{G}_0$. Suppose K contains $\chi(X)[\iota(X)]$ for each X and $\chi(X)[\rho(\iota(Y_1), \dots, \iota(Y_q))]$ for each typed rule $X \to \rho(Y_1, \dots, Y_q)$, $q \ge 0$. Then*

$$L(G) \subseteq L(\mathcal{B}_{f,s}^{\le r_{\max}}(K)) \quad (r_{\max} \ge \max\{1, r_G\}), \qquad L(G) \subseteq L(\mathcal{B}_{f,s}(K)).$$

Neither substitutability nor $\iota(X) \in L_X$ is assumed; $\iota = \omega$ is the case $\mathsf{W}(\widetilde{G}_0) \subseteq K$.

Exact reconstruction needs a good presentation of the target; the following normalization controls size, fan-out and rank, but not thickness (for reduced λ-free inputs, see Proposition C.1).

Proposition 3.12 (Good normalization with rank preservation)**.** *Let G_{in} be a finite standard f-MCFG of rule rank at most $r_{\max}$. There is a finite good f-MCFG G_{good} of rule rank at most $r_{\max}$ with $L(G_{\mathrm{good}}) = L(G_{\mathrm{in}}) \setminus \{\varepsilon\}$ and, for fixed $f, r_{\max}$, $\|G_{\mathrm{good}}\| = O_{f,r_{\max}}(\|G_{\mathrm{in}}\|)$. If $L(G_{\mathrm{in}}) \cap \Sigma^+ \neq \emptyset$, G_{good} is reduced.*

Theorem 3.13 (Exact reconstruction for a fixed typing)**.** *Fix $f \ge 1$ and a typing s, and let $L \in \mathcal{C}_{f,s}^{\mathrm{mcf}}$. If $L \cap \Sigma^+ = \emptyset$ put $\mathcal{F}_L := L$ and $r_G := 0$. Otherwise let G be a reduced good fan-out-$\le f$ presentation of $L \cap \Sigma^+$ (Proposition 3.12), let r_G be as in Proposition 3.11, and put $\mathcal{F}_L := \mathsf{W}(\widetilde{G}_0) \cup (\{\varepsilon\} \cap L)$. Then $\mathcal{F}_L \subseteq L$ is finite, and every finite K with $\mathcal{F}_L \subseteq K \subseteq L$ satisfies*

$$L(\mathcal{B}_{f,s}(K)) = L, \qquad L(\mathcal{B}_{f,s}^{\le r_{\max}}(K)) = L \quad \text{for every } r_{\max} \ge \max\{1, r_G\}.$$

If $L \in \mathsf{MCFL}(f, r)$, G can be chosen of rank $\le r$, so $r_{\max} = \max\{1, r\}$ suffices (also when $r = 0$).

Theorem 3.14 (Polynomial construction under a rule-rank cap)**.** *Fix $f, r_{\max} \ge 1$ and s. The encoding of $\widehat{G}_{\le r_{\max}}(K)$ is computable in time $(1 + \|K\|_+)^{O(f r_{\max})}$; its fan-out is at most f and its rank at most $r_{\max}$.*

Definition 3.15 (Conservative sequential learner)**.** The learner $\mathcal{A}_{f,s}$ (respectively $\mathcal{A}_{f,s}^{\le r_{\max}}$) starts with H_0, keeps its hypothesis on $\perp$ and on every datum the hypothesis generates, and otherwise replaces it by $\mathcal{B}_{f,s}(K)$ (respectively $\mathcal{B}_{f,s}^{\le r_{\max}}(K)$), where K is the content read so far. This is the policy of [14, Algorithm 1].

Corollary 3.16 (Gold identification)**.** *Fix $f \ge 1$ and s.*

(i) *$\mathcal{A}_{f,s}$ TxtEx-identifies $\mathcal{C}_{f,s}^{\mathrm{mcf}}$.*

(ii) *For fixed $r_{\max}$, $\mathcal{A}_{f,s}^{\le r_{\max}}$ TxtEx-identifies $\mathcal{C}_{f,s}^{\mathrm{mcf}} \cap \mathsf{MCFL}(f, r_{\max})$, and processes each datum in time polynomial in $\|K\|_+$ for the content K read so far.*

Proofs are in Appendix D (Proposition 3.12: Appendix F). None of these statements bounds the length of characteristic data or the stage of convergence.

4 Characteristic-Data Bounds

Throughout this section f, $r_{\max} \ge 1$, Σ and the typing $s : \Sigma^+ \twoheadrightarrow \mathsf{S}$ are fixed; $G = (V, \Sigma, P, S, \mathrm{fan})$ is a reduced good MCFG of fan-out $\le f$ and rule rank $\le r_{\max}$, and $\widetilde{G}_0$ is its trimmed $\widehat{s}$-refinement. Under a rank cap, Theorem 3.13 already supplies $O(\|G\|)$ characteristic words; the task is to bound their total length *in terms of* G. All bounds follow one scheme.

(1) *Witness length* (Lemma 4.2). If, for an auxiliary finite-monoid morphism λ refining s, every tuple of every nonterminal can be exchanged for a tuple of the same nonterminal and the same λ-vector of total length at most β, then $\mathsf{W}(\widetilde{G}_0)$ is characteristic and has total length $O(\|G\|(\bar{\theta}_G + \beta))$.

(2) *Minimal counterexamples* (Section 4.3). In a shortest derivation with a prescribed root λ-vector, a large antichain of heavy subtrees would admit a simultaneous replacement preserving that vector; so antichains are small, and Lemma 4.9 turns this into $\beta = O(\|G\|\bar{\theta}_G)$ or $O(\|G\|^2\bar{\theta}_G)$.

(3) *Regimes.* Only λ and the replacement step change: window typings keep marked boundary letters; window–group typings in addition preserve ordered group products (Proposition 4.10); local-group typings reduce to these through sliding-factor coordinates (Section 4.5).

The full yield-type refinement is not used as the primary thickness measure, since it may be exponentially thicker than G (Proposition 4.8). Corollary 4.20 characterizes the semigroup pseudovarieties admitting a *uniform polynomial bound* for this refinement; it does not decide whether a particular grammar exhibits exponential blow-up. Section 4.6 transfers all bounds to arbitrary standard input grammars.

4.1 Thickness Scales and Witness Length

For a rule R of G let

$$\theta_G(R) := \min\{\, |w| : w \in L(G) \text{ has a successful derivation using } R \,\},$$

the length of the derived *sentence*, not of the derivation. Put $\theta_G := \max_{R\in P} \theta_G(R)$, $\bar{\theta}_G := \max\{1, \theta_G\}$ and $t_G := \max_{A\in V} \min\{\|\vec{w}\|_1 : \vec{w} \in L_A(G)\}$ (rule and nonterminal thickness; Yoshinaka writes τ_G for θ_G [14, Section 4.3]). Table 2 lists these scales together with the pattern scale of Section 4.6 and three additional quantities; t_{typ} is the central intermediate parameter for characteristic-data bounds, while θ_{typ} and κ_G compare the full refinement and distributional representatives.

Table 2: Thickness scales: two input parameters, a central typed parameter, and two auxiliary quantities.

Scale	What it measures / role	Comparison or obstruction
θ_G	Shortest accepted sentence using each rule, maximized over rules; *primary for reduced good G.*	$t_G \le \theta_G \le 2\|G\| \max\{1, t_G\}$ (Lemma 4.1).
t_G	Largest minimum total yield length of a nonterminal; proof-scale for good forms.	Polynomially equivalent to θ_G on reduced good presentations.
$t_{\mathrm{pat}}(H)$	Largest minimum yield length over *realized nonempty-component support patterns*; *primary for arbitrary H.*	$t_H \le t_{\mathrm{pat}}(H)$, with equality for reduced λ-free H; the family E_n has $\theta_{E_n} = O(n)$ but $t_{\mathrm{pat}}(E_n) \ge 2^n$ (Prop. F.2).
t_{typ}	Largest shortest tuple yield over non-start typed nonterminals; *central parameter* in Theorem 4.4.	$t_{\mathrm{typ}} \le \theta_{\mathrm{typ}} \le 2\bar{\theta}_G + r_{\max} t_{\mathrm{typ}}$ (Lemma 4.3).
θ_{typ}	Rule thickness of the trimmed full yield-type refinement $\widetilde{G}_0$.	$\theta_{\mathrm{typ}} \ge \theta_G$; it can be $\ge 2^n$ with $\theta_G \le 2$ (Prop. 4.8).
κ_G	Shortest representative of each typed distribution class, maximized over classes; auxiliary for abelian-filtered targets.	$\kappa_G \le \theta_{\mathrm{typ}}$; representatives need not be derivable from the typed nonterminal (Prop. 5.3).

If a target L contains ε, every statement below is applied to $L \cap \Sigma^+$ and the datum ε is added, as in Theorem 3.13; this adds one to the encoded length.

Lemma 4.1 (Thickness lemma)**.** *Let G be reduced and good.*

(i) *Every rule writes at most θ_G terminal symbols explicitly.*

(ii) *$t_G \le \theta_G$: every $A \in V$ has some $\vec{u} \in L_A(G)$ with $\|\vec{u}\|_1 \le \theta_G$.*

(iii) *$\theta_G \le 2\|G\| \max\{1, t_G\}$.*

(iv) *Every $A \in V$ of fan-out d has a context $E_A \in \mathcal{E}_d$ with $|E_A|_{\mathrm{fix}} \le \theta_G$ that results from a successful derivation by deleting one occurrence of an A-subtree; $E_S = \square_1$, and $E_A[\vec{u}] \in L(G)$ for all $\vec{u} \in L_A(G)$. For every typed nonterminal $X = A_{\mathbf{t}}$, the choice $\chi(X) := E_A$ meets Definition 3.9 and is* derivational *for X: it results from a successful derivation of $\widetilde{G}_0$ by deleting an occurrence of an X-subtree.*

Proof. For a rule R with left-hand side A fix a successful derivation D_R of length $\theta_G(R)$ that uses R; it exists because G is reduced. By nondeletion every terminal written by R, and every component of the tuple produced at that occurrence of A, survives exactly once in the sentence. This gives (i), and (ii) because every productive A has a rule.

(iv) For $A = S$ take $\square_1$. Otherwise delete the A-subtree of D_R and put ordered holes in place of its components; iterating Lemma 3.1 along the path to the root (siblings have nonempty yields) shows $E_A \in \mathcal{E}_d$, and its fixed letters belong to the sentence of D_R. Inserting any A-derivation gives a successful derivation, so E_A accepts $L_A(G) \supseteq L_X$ (Proposition 3.6(i)). A typed nonterminal with a start link has the form $S_{\mathbf{t}}$, and $E_S = \square_1$. For derivationality, insert the erasure of any $\widetilde{G}_0$-derivation D_X from X into the hole; the unique lift of the resulting successful derivation (Proposition 3.6(ii)) labels every node by its nonterminal and the $\widehat{s}$-vector of its yield, as D_X does, so it contains D_X at the hole, and deleting that X-subtree leaves E_A.

(iii) Follow a shortest dependency path from S to A; it repeats no nonterminal, so its rules are distinct and differ from R. Complete every sibling on the path and every child of R by a shortest terminal tuple (length $\le t_G$) and apply R. The distinct rules used write at most $\|G\|$ terminals and have at most $\|G\|$ child occurrences, so the sentence has length at most $\|G\| + \|G\| t_G \le 2\|G\| \max\{1, t_G\}$. □

Thus θ_G and t_G are polynomially equivalent on reduced good presentations; Yoshinaka's efficiency criterion measures data in target size and rule thickness [14, Definition 1]; see also [10].

For a finite-monoid morphism $\lambda : \Sigma^* \to N$ we write $s \preceq \lambda$ if $\ker(\lambda|_{\Sigma^+}) \subseteq \ker s$; since $\widehat{s}$ separates ε, this is the relation of Definition 2.3 whenever λ also separates ε from Σ^+. For $\beta \ge 0$ consider

($\mathrm{R}^{\lambda}_{\beta}$) for every $B \in V$ and $\vec{u} \in L_B(G)$ there is $\vec{u}' \in L_B(G)$ with $\lambda^{(\mathrm{fan}(B))}(\vec{u}') = \lambda^{(\mathrm{fan}(B))}(\vec{u})$ and $\|\vec{u}'\|_1 \le \beta$.

Put

$$C_s(G) := |V||\mathsf{S}^\sharp|^f + |P||\mathsf{S}^\sharp|^{f r_{\max}} \;\le\; |\mathsf{S}^\sharp|^{f r_{\max}} \|G\|,$$

using $|V| + |P| \le \|G\|$; it bounds the numbers of typed nonterminals and typed rules together, since a typed rule is determined by its source rule and the type vectors of its at most $r_{\max}$ children.

Lemma 4.2 (Witness length)**.** *Let $\lambda : \Sigma^* \to N$ be a finite-monoid morphism with $s \preceq \lambda$.*

(i) (Type-preserving exchange.) *If $X = B_{\mathbf{t}}$ is a typed nonterminal, $\vec{u} \in L_X$, $\vec{u}' \in L_B(G)$ and $\lambda^{(\mathrm{fan}(B))}(\vec{u}') = \lambda^{(\mathrm{fan}(B))}(\vec{u})$, then $\vec{u}' \in L_X$. Hence under ($\mathrm{R}^{\lambda}_{\beta}$) there are canonical tuples with $\|\omega(X)\|_1 \le \beta$ for every X.*

(ii) (Length.) *If* $\|\omega(X)\|_1 \le b$ *for every* X *and* $\chi(X) := E_B$ *as in Lemma 4.1(iv), every word of* $\mathsf{W}(\widetilde{G}_0)$ *has length at most* $2\bar{\theta}_G + r_{\max} b$*, and*

$$|\mathsf{W}(\widetilde{G}_0)| \le C_s(G), \qquad \|\mathsf{W}(\widetilde{G}_0)\|_+ \le C_s(G)\,(1 + 2\bar{\theta}_G + r_{\max} b).$$

(iii) (Refined thickness.) *In (ii), each rule word* $\chi(X)[\rho(\omega(Y_1), \dots, \omega(Y_q))]$ *is the yield of a successful derivation of* $\widetilde{G}_0$ *using the typed rule* $X \to \rho(Y_1, \dots, Y_q)$*; hence* $\theta_{\widetilde{G}_0} \le 2\bar{\theta}_G + r_{\max} b$*.*

(iv) (Characteristic sample.) *If* $L(G) = L \cap \Sigma^+$ *for some* $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$*, then* $\mathsf{W}(\widetilde{G}_0) \cup (\{\varepsilon\} \cap L)$ *is a characteristic sample for* $\mathcal{B}^{\le r_{\max}}_{f,s}$ *and for* $\mathcal{B}_{f,s}$*.*

Items (i)–(iii) do not use substitutability.

Proof. (i) All components are nonempty because G is λ-free, so $s \preceq \lambda$ and Proposition 3.6(i) give $\widehat{s}^{(\mathrm{fan}(B))}(\vec{u}') = \widehat{s}^{(\mathrm{fan}(B))}(\vec{u}) = \mathbf{t}$. By Proposition 3.6(ii) a G-derivation of $\vec{u}'$ lifts to a $G^{\widehat{s}}$-derivation from X. Inserted at an occurrence of X in a successful derivation of $\widetilde{G}_0$ it gives a successful derivation, because typed rules depend only on typed symbols; so all its rules are retained and $\vec{u}' \in L_X$. For the second claim apply $(\mathrm{R}^{\lambda}_{\beta})$ to some $\vec{u} \in L_X \ne \emptyset$.

(ii) A witness word is $\chi(X)[\omega(X)]$ or a rule word. The context has at most $\bar{\theta}_G$ fixed letters, a typed rule has the template of its source and so writes at most $\bar{\theta}_G$ letters (Lemma 4.1(i)), and there are at most $r_{\max}$ child tuples. There is one word per typed nonterminal and per typed rule.

(iii) Apply the typed rule to $\widetilde{G}_0$-derivations of the $\omega(Y_j) \in L_{Y_j}$, erase the labels, and insert the result into the successful G-derivation from which E_B was obtained. As in the proof of Lemma 4.1(iv), the unique lift contains the typed derivation at the hole, and every rule of a successful derivation of $G^{\widehat{s}}$ is retained. A start link $\widetilde{S} \to S_{\mathbf{t}}$ is used by the successful derivation of $\omega(S_{\mathbf{t}})$, of length at most b.

(iv) Proposition 3.11 with $\iota = \omega$ ($r_G \le r_{\max}$, and $\chi(S_{\mathbf{t}}) = E_S = \Box_1$) and Proposition 3.10 give $L(\mathcal{B}(K)) \cap \Sigma^+ = L \cap \Sigma^+$ for every finite K with $\mathsf{W}(\widetilde{G}_0) \subseteq K \subseteq L$; the Start rule handles ε. □

4.2 Typed Thickness and Characteristic Data

For a reduced good grammar G, let $\widetilde{G}_0$ be its trimmed full yield-type refinement (Definition 3.5). Define

$$t_{\mathrm{typ}}(G, s) := \max_{X} \min_{\vec{u} \in L_X} \|\vec{u}\|_1, \qquad \theta_{\mathrm{typ}}(G, s) := \max\{1, \theta_{\widetilde{G}_0}\},$$

where X ranges over the non-start typed nonterminals, and let $N_{\mathrm{typ}}(G, s)$ denote their number plus the number of non-start typed rules. In particular $N_{\mathrm{typ}} \le C_s(G) \le |\mathsf{S}^{\sharp}|^{f r_{\max}} \|G\|$. These parameters are computable in time polynomial in $\|G\|$ for fixed $f, r_{\max}, s$, with numerical outputs represented in binary (Proposition 5.2). Every rule of G has a retained typed copy, and erasing typed labels from a successful derivation recovers a G-derivation; therefore $\theta_G \le \theta_{\widetilde{G}_0} \le \theta_{\mathrm{typ}}$.

Lemma 4.3 (Comparison of the two typed thickness measures)**.** *For every reduced good* G*,*

$$t_{\mathrm{typ}} \le \theta_{\mathrm{typ}} \le 2\bar{\theta}_G + r_{\max} t_{\mathrm{typ}}.$$

Proof. Apply Lemma 4.1(ii) to the reduced good $\widetilde{G}_0$ for the first inequality. For the second, choose each canonical tuple $\omega(X)$ to have the shortest total length, at most t_{typ}, and use Lemma 4.2(iii). The start links are covered by that lemma as well. □

Theorem 4.4 (Characteristic data from typed thickness)**.** *Let $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$ and G be a reduced good fan-out-$\le f$, rank-$\le r_{\max}$ grammar with $L(G) = L \cap \Sigma^+$. Take shortest canonical tuples and the derivational contexts E_B of Lemma 4.1(iv). Then $K = \mathsf{W}(\widetilde{G}_0) \cup (\{\varepsilon\} \cap L)$ is a characteristic sample for both $\mathcal{B}^{\le r_{\max}}_{f,s}$ and $\mathcal{B}_{f,s}$, with*

$$|\mathsf{W}(\widetilde{G}_0)| \le N_{\mathrm{typ}}, \qquad \|\mathsf{W}(\widetilde{G}_0)\|_+ \le N_{\mathrm{typ}}\bigl(1 + 2\bar{\theta}_G + r_{\max} t_{\mathrm{typ}}\bigr).$$

For the full sample K, the length bound has the additional term $\mathbf{1}_{\{\varepsilon \in L\}}$. The numerical length estimate holds for any typing; substitutability is used only for the characteristic-sample claim.

Proof. The chosen shortest tuples satisfy $\|\omega(X)\|_1 \le t_{\mathrm{typ}}$. Apply Lemma 4.2(ii),(iv), counting non-start typed nonterminals and rules instead of the upper bound $C_s(G)$. □

For trivial typing, every nonterminal of the λ-free G has one realized type vector, so $t_{\mathrm{typ}} = t_G$ and $N_{\mathrm{typ}} = |V| + |P| \le \|G\|$. The theorem recovers the $O(\|G\|\bar{\theta}_G)$ untyped scale, although Yoshinaka's original sample has the sharper bounds $|K_G| \le |P|$ and $\|K_G\| \le |P|\theta_G$ [14, Lemma 7 and Theorem 1]. A nontrivial typing may instead make t_{typ} exponential.

Presentation-independent lower bounds from rare types. For $L \subseteq \Sigma^*$ and $m \in \mathsf{S}$ put

$$\mu_m(L) := \min\{|w| : w \in L \cap \Sigma^+,\ s(w) = m\}, \qquad (\min \emptyset := \infty).$$

For a nonempty set of realized types $M_L := s(L \cap \Sigma^+)$, put $\mu^\star(L, s) := \max_{m \in M_L} \mu_m(L)$; set $\mu^\star(L, s) = 0$ when $M_L = \emptyset$. A typed start nonterminal $S_{(m)}$ exists for each $m \in M_L$, and $L_{S_{(m)}} = \{(w) : w \in L \cap \Sigma^+,\ s(w) = m\}$ (Proposition 3.6). Thus, whenever $L \cap \Sigma^+ \ne \emptyset$,

$$\mu^\star(L, s) \le t_{\mathrm{typ}}(G, s) \qquad \text{for every reduced good presentation } G. \tag{4.1}$$

Lemma 4.5 (Deleting an entire yield type)**.** *If $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$ and $m \in \mathsf{S}$, then $L^{\setminus m} := L \setminus s^{-1}(m)$ also belongs to $\mathcal{C}^{\mathrm{mcf}}_{f,s}$. Moreover, $L \setminus \{\varepsilon\} \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$.*

Proof. The regular language $\widehat{s}^{-1}(\mathsf{S}^\sharp \setminus \{m\})$ is (f, s)-tuple-substitutable, since equal componentwise s-types have equal total types in every admissible sentence context (item (c) after Definition 2.8). Intersections preserve tuple substitutability (Proposition 2.9(i)), and f-MCFLs are closed under regular intersection [29]. For the last assertion, intersect L with Σ^+. □

Proposition 4.6 (Rare types force positive data)**.** *Let $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$, and let $m \in M_L$. Every characteristic sample of L for $\mathcal{B}^{\le r}_{f,s}$ ($r \ge 1$) or for $\mathcal{B}_{f,s}$ contains a word of type m. The same is true for any set-driven operator having characteristic samples for both L and $L^{\setminus m}$, in particular for an operator having characteristic samples for the entire class $\mathcal{C}^{\mathrm{mcf}}_{f,s}$. If $\varepsilon \in L$, its sample for either canonical operator must also contain ε; the same holds for a set-driven operator admitting a characteristic sample for $L \setminus \{\varepsilon\}$ as well. Consequently, for the canonical operators and for class-wide set-driven operators, every characteristic sample F satisfies*

$$\|F\|_+ \ge \sum_{m \in M_L} \bigl(\mu_m(L) + 1\bigr) + \mathbf{1}_{\{\varepsilon \in L\}}.$$

Proof. If a sample $F \subseteq L$ avoids type m, then $F \subseteq L^{\setminus m} \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$ by Lemma 4.5. Soundness (Proposition 3.10) for the latter target gives $L(\mathcal{B}(F)) \subseteq L^{\setminus m} \subsetneq L$; the same applies to the bounded-rank reconstruction. For a general set-driven operator, let F' be characteristic for $L^{\setminus m}$. Then

$F \cup F'$ lies inside both L and $L^{\setminus m}$, and contains the respective characteristic samples. The required outputs would be both distinct languages, a contradiction. The same argument, using $L \setminus \{\varepsilon\}$ from Lemma 4.5, forces the datum ε when it belongs to L, provided the operator has a characteristic sample for that language; the canonical operators satisfy this by Theorem 3.13. Distinct positive types demand distinct sample words, and the minimum cost of each such word is $\mu_m(L) + 1$. □

Corollary 4.7 (Characteristic-data sandwich)**.** *Suppose $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$, $L \cap \Sigma^+ \neq \emptyset$, and G is a reduced good fan-out-$\leq f$, rank-$\leq r_{\max}$ presentation of $L \cap \Sigma^+$. Write $\mathrm{cs}^{\leq r_{\max}}_{f,s}(L)$ for the minimum total encoded length of a characteristic sample for $\mathcal{B}^{\leq r_{\max}}_{f,s}$. Then*

$$\sum_{m \in M_L} (\mu_m(L) + 1) + \mathbf{1}_{\{\varepsilon \in L\}} \;\leq\; \mathrm{cs}^{\leq r_{\max}}_{f,s}(L) \;\leq\; N_{\mathrm{typ}}(1 + 2\bar{\theta}_G + r_{\max} t_{\mathrm{typ}}(G, s)) + \mathbf{1}_{\{\varepsilon \in L\}}.$$

The lower bound is independent of the presentation and applies to every set-driven operator with characteristic samples throughout $\mathcal{C}^{\mathrm{mcf}}_{f,s}$.

Proof. Proposition 4.6 supplies the lower bound; Theorem 4.4 supplies the upper bound. □

Proposition 4.8 (Exponential blow-up of full yield-type refinement)**.** (a) *Let $e \neq z$ be idempotents of S with $ez = ze = z$, and let $u, v \in \Sigma^+$ with $s(u) = e$ and $s(v) = z$. For $n \geq 1$ let G_n be the fan-out-one, rank-two grammar*

$$S \to SS \mid a \ (a \in \Sigma) \mid A_0 \mid \cdots \mid A_n, \qquad A_0 \to u, \qquad A_i \to A_{i-1}A_{i-1} \mid v \quad (1 \leq i \leq n),$$

where u, v are constant templates. Then G_n is reduced and good, $L(G_n) = \Sigma^+$, $\|G_n\| = O_{\Sigma,u,v}(n)$, $\theta_{G_n} \leq \max\{2, 2|u|, 2|v|\}$, and both $t_{\mathrm{typ}}(G_n, s) \geq |u|2^n$ and $\theta_{\widetilde{G}_{n,0}} \geq |u|2^n$.

(b) *In particular, for $\Sigma = \{\mathtt{a}, \mathtt{c}\}$ and the two-element typing $s_{\mathtt{c}}$ onto $\{1, \zeta\}$ ($\zeta^2 = \zeta$) with $s_{\mathtt{c}}(\mathtt{a}) = 1$, $s_{\mathtt{c}}(\mathtt{c}) = \zeta$, taking $u = \mathtt{a}$, $v = \mathtt{c}$ gives $\|G_n\| = O(n)$, $\theta_{G_n} \leq 2$ and $\theta_{\widetilde{G}_{n,0}} \geq 2^n$.*

Proof. All templates are nonempty and use each variable once in order, so G_n is good; it is reduced and generates Σ^+ via $S \to SS \mid a$. Each rule has a successful use of length at most $\max\{2, 2|u|, 2|v|\}$: complete every A_i by v ($i \geq 1$) or u ($i = 0$). A yield of A_n is a product of leaf values $s(u) = e$ and $s(v) = z$; since $e^2 = e$ and z is absorbing in $\{e, z\}$, its type is e iff no leaf uses v. Hence the unique yield of $(A_n)_{(e)}$ is u^{2^n}, proving $t_{\mathrm{typ}}(G_n, s) \geq |u|2^n$; the typed rule $S_{(e)} \to (A_n)_{(e)}$ is retained (via $S \to A_n$), and every successful derivation using it has length at least $|u|2^n$. In (b), $e = 1$ and $z = \zeta$. □

The proofs below therefore work on G itself.

4.3 Short Representatives by Minimal Counterexamples

Let $\lambda : \Sigma^* \to N$ be a finite-monoid morphism. A vector $\mathbf{t} \in N^d$ is *realizable at B* (of fan-out d) if $\mathbf{t} = \lambda^{(d)}(\vec{u})$ for some $\vec{u} \in L_B(G)$. A *λ-minimal* derivation for $(B, \mathbf{t})$ is a terminal derivation from B with root λ-vector $\mathbf{t}$ that minimizes first the total yield length and then the number of nodes. For a threshold b, a node is *heavy* if its yield has total length greater than b.

Lemma 4.9 (Heavy-node count)**.** *Let $b \geq \bar{\theta}_G$ and $c \geq 0$, and let T be λ-minimal for $(B, \mathbf{t})$. If every antichain of heavy nodes of T has at most c elements, then*

$$\|\textit{root tuple of } T\|_1 \leq (r_{\max} + 1)\, b \max\{1, c\,|V|\,|N|^f\}.$$

Proof. Label each node by its nonterminal and the λ-vector of its yield; there are at most $|V||N|^f$ labels. If an ancestor and a proper descendant had the same label, replacing the ancestor's subtree by the descendant's would keep the root λ-vector (templates evaluate λ homomorphically), would not lengthen the root yield (by linearity the descendant's yield is part of the ancestor's), and would remove nodes, contradicting minimality. So every root-to-leaf path has at most $|V||N|^f$ nodes. If the root is light its yield has length at most b. Otherwise the heavy nodes form an ancestor-closed subtree whose leaves form an antichain, so there are at most $c|V||N|^f$ heavy nodes. Every terminal of the root yield is written by a heavy node (at most $\bar{\theta}_G \le b$ each, Lemma 4.1(i)) or lies in a light child of a heavy node (at most $r_{\max}$ children of length at most b). □

Group constants. For a finite abelian group H, $\mathsf{d}(H)$ is the maximal length of a zero-sum-free sequence over H (one with no nonempty subsequence summing to the identity; thus $\mathsf{d}(H)+1$ is the Davenport constant; $\mathsf{d}(\{1\}) = 0$). For a finite group A and $f \ge 1$ put

$$N_{A,f} = \max\{3, |A|^f\}, \quad U_f = f^{f+1} f!, \quad M_{A,f} = U_f\, N_{A,f}^{2^{f^2}}, \qquad c_{A,f} := \begin{cases} \mathsf{d}(A^f), & A \text{ abelian}, \\ M_{A,f} - 1, & \text{otherwise}. \end{cases}$$

U_f bounds the number of block patterns of one owner, and $N_{A,f} \ge 3$ is needed by the order classification in Appendix A.1.

Simultaneous block replacement. A *block-replacement family* of width f over A (Definition A.1) consists of $d \le f$ rows, each an ordered product in A of fixed constants and *blocks*; each block has an *owner* in $[m] := \{1, \dots, m\}$, an original value and a replacement value, and each owner owns at most f blocks, which may interleave with other owners' blocks arbitrarily. $\mathrm{val}_c(J)$ is the value of row c when exactly the blocks of owners in $J \subseteq [m]$ take their replacement values.

Proposition 4.10 (Simultaneous block replacement)**.** *In a block-replacement family of width f over a finite group A with $m > c_{A,f}$ owners there is a nonempty $J \subseteq [m]$ with $\mathrm{val}_c(J) = \mathrm{val}_c(\varnothing)$ for every row c. Individual block or owner values need not be preserved.*

Proof. This is Lemma A.2. For abelian A one selects a nonempty zero-sum subsequence of the owners' difference vectors in $A^d \le A^f$. For arbitrary A no factors are commuted: among owners with a common block pattern (at most U_f patterns) one homogenizes all pairwise order comparisons (Lemma A.7), which decomposes every row into consecutive rank chunks (Lemma A.4); a pigeonhole on prefix-product boundary states then yields an interval of owners preserving all ordered row products (Lemma A.5). □

Owner contributions cannot be commuted: if the two rows are $\beta_{1,1}\beta_{2,1}$ and $\beta_{2,2}\beta_{1,2}$, with value $\sigma = (12)$ on the blocks of owner 1 and $\tau = (23)$ on those of owner 2 in S_3, the row values are $\sigma\tau \ne \tau\sigma$. Example A.6 shows a replacement preserving rows of opposite owner orientation.

Lemma 4.11 (Short representatives for window–group invariants)**.** *Fix $k, \ell \ge 0$, a finite group A (possibly trivial) and a morphism $g : \Sigma^* \to A$. Put $\lambda := (\widehat{s}_{k,\ell}, g) : \Sigma^* \to N := M_{k,\ell} \times A$ and $h := f(k+\ell+1)$. Then $(\mathrm{R}_\beta^\lambda)$ holds with*

$$\beta = b_{k,\ell,A} := (r_{\max}+1)\,\bar{\theta}_G\,(c_{A,f}+h)\,|V||N|^f \;\le\; (r_{\max}+1)(c_{A,f}+h)|N|^f\,\|G\|\,\bar{\theta}_G.$$

Proof. Let $\vec{u} \in L_B(G)$, $\mathbf{t} := \lambda^{(\mathrm{fan}(B))}(\vec{u})$, and let T be λ-minimal for $(B, \mathbf{t})$, with heavy threshold $b = \bar{\theta}_G$. In each root component of length at least $m_0 = \max\{1, k+\ell\}$ mark the first k and the last

ℓ positions (one arbitrary position if $k = \ell = 0$), and in each shorter component mark all positions: at most h marks. Let Q be an antichain of heavy nodes. Incomparable subtrees have disjoint yields, so at most h nodes of Q contain a mark; let Q_0 be the others.

Suppose $|Q_0| > c_{A,f}$. For $v \in Q_0$ with nonterminal C_v fix a tuple of $L_{C_v}(G)$ of total length at most $\bar{\theta}_G$ (Lemma 4.1(ii)). By linearity and nondeletion each $v \in Q_0$ contributes at most f disjoint contiguous blocks to the root components, and the g-value of each root component is the ordered product of the g-values of fixed material and of these blocks. This is a block-replacement family over A with owners Q_0, so Proposition 4.10 gives a nonempty $J \subseteq Q_0$ whose simultaneous replacement preserves every root g-value. The replaced blocks contain no mark: short root components are untouched, and in a long component they lie strictly between the marked prefix and suffix, which keep their letters and positions relative to the ends, while the component keeps its $k + \ell$ marked letters (at least one if $k = \ell = 0$) and stays long. So the root $\widehat{s}_{k,\ell}$-vector is preserved as well, and the result is a derivation from B with root λ-vector $\mathbf{t}$ that is strictly shorter, since each replaced yield exceeded $\bar{\theta}_G$. This contradicts minimality.

Hence every heavy antichain has at most $c_{A,f} + h$ elements, and Lemma 4.9 with $c = c_{A,f} + h \geq 1$ gives the bound; the second inequality uses $|V| \leq \|G\|$. □

Corollary 4.12 (Window representatives)**.** *For $k, \ell \geq 0$, $(\mathrm{R}_\beta^{\widehat{s}_{k,\ell}})$ holds with $\beta = b_{k,\ell} := (r_{\max} + 1)\bar{\theta}_G\, f(k + \ell + 1)\, |V||M_{k,\ell}|^f$.*

Proof. Take A trivial in Lemma 4.11: then $c_{A,f} = 0$ and $N \cong M_{k,\ell}$. □

4.4 Window and Window–Group Typings

Theorem 4.13 (Polynomial data for prefix–suffix-determined typings)**.** *Let $s \preceq s_{k,\ell}$, let $L \in \mathcal{C}_{f,s}^{\mathrm{mcf}}$ and $L(G) = L \cap \Sigma^+$. Then $\mathsf{W}(\widetilde{G}_0) \cup (\{\varepsilon\} \cap L)$ is a characteristic sample for $\mathcal{B}_{f,s}^{\leq r_{\max}}$ with $|\mathsf{W}(\widetilde{G}_0)| \leq C_s(G) = O(\|G\|)$ and*

$$\|\mathsf{W}(\widetilde{G}_0)\|_+ \leq K_{k,\ell}\, \|G\|^2 \bar{\theta}_G, \qquad K_{k,\ell} := |\mathsf{S}^\sharp|^{f r_{\max}} \big(3 + r_{\max}(r_{\max} + 1) f(k + \ell + 1) |M_{k,\ell}|^f\big).$$

Proof. Corollary 4.12 and Lemma 4.2(i),(ii),(iv) with $\lambda = \widehat{s}_{k,\ell}$ and $b = b_{k,\ell}$ give the sample and $\|\mathsf{W}\|_+ \leq C_s(G)(1 + 2\bar{\theta}_G + r_{\max} b_{k,\ell})$; insert $C_s(G) \leq |\mathsf{S}^\sharp|^{f r_{\max}} \|G\|$, $1 + 2\bar{\theta}_G \leq 3\|G\|\bar{\theta}_G$ and $|V| \leq \|G\|$. □

Theorem 4.14 (Polynomial characteristic data for window–finite-group typing)**.** *Let A be a finite group, $g : \Sigma^* \to A$ a morphism, and $\widehat{s} \preceq (\widehat{s}_{k,\ell}, g)$; let $L \in \mathcal{C}_{f,s}^{\mathrm{mcf}}$ and $L(G) = L \cap \Sigma^+$. Then $\mathsf{W}(\widetilde{G}_0) \cup (\{\varepsilon\} \cap L)$ is a characteristic sample for $\mathcal{B}_{f,s}^{\leq r_{\max}}$ with $|\mathsf{W}(\widetilde{G}_0)| \leq C_s(G) = O(\|G\|)$ and*

$$\|\mathsf{W}(\widetilde{G}_0)\|_+ \leq K_{k,\ell,A}\, \|G\|^2 \bar{\theta}_G,$$
$$K_{k,\ell,A} := |\mathsf{S}^\sharp|^{f r_{\max}} \big(3 + r_{\max}(r_{\max} + 1)(c_{A,f} + f(k + \ell + 1))(|M_{k,\ell}||A|)^f\big).$$

The group may be noncommutative and its blocks may interleave across components.

Proof. As for Theorem 4.13, with Lemma 4.11 and $b = b_{k,\ell,A}$. □

Theorem 4.15 (Polynomial data for window–abelian typing)**.** *Here $\exp(A)$ denotes the exponent of the finite group A, i.e., the least common multiple of its element orders. If in Theorem 4.14 the group A is abelian, then $c_{A,f} = \mathsf{d}(A^f) = O(\exp(A)\, f \log |A|)$ (Remark A.3), so $K_{k,\ell,A}$ is polynomial in f apart from the factors $|\mathsf{S}^\sharp|^{f r_{\max}}$ and $(|M_{k,\ell}||A|)^f$; for nonabelian A, $c_{A,f} = M_{A,f} - 1$ grows doubly exponentially in f.*

Proof. Lemma A.2(a). □

At fan-out one, polynomial time and thick-data learning for fixed windows is due to Yoshinaka [12, Theorem 1]; the bounds above are relative to a reduced good presentation and are not claimed to improve on that result after normalization. What is new is that window and ordered group values are preserved *simultaneously* across up to f interleaved components. This higher-rank regime is needed at the language level: Theorem 6.16 gives a non-context-free $(1,1)$-typed target of minimum fan-out two and minimum rule rank two, whereas rank one is covered by Theorem 5.1.

4.5 An Algebraic Criterion for Polynomial Characteristic Data

LI is the pseudovariety of finite semigroups T satisfying $eTe = \{e\}$ for every idempotent $e \in T$, **Grp** that of finite groups, and $\mathbf{LI} \vee \mathbf{Grp}$ consists of the divisors of products $T \times A$ with $T \in \mathbf{LI}$ and A a finite group [27, 28]. The following lemma makes the window–group hypothesis intrinsic to the positive image S; the semigroup facts are classical [26].

Lemma 4.16 (Locally trivial–group typings admit window–group refinements)**.** *Let $\Sigma \neq \emptyset$. If $\mathsf{S} \in \mathbf{LI} \vee \mathbf{Grp}$, there are $k \geq 1$, a finite group A and a morphism $g : \Sigma^* \to A$ with $\widehat{s} \preceq (\widehat{s}_{k,k}, g)$; if $\mathsf{S} \in \mathbf{LI}$, A can be trivial.*

Proof. Choose a finite $S' \in \mathbf{LI}$, a finite group A, a subsemigroup $T \leq S' \times A$ and a surjective morphism $\pi : T \to \mathsf{S}$ (for $\mathsf{S} \in \mathbf{LI}$: $S' = \mathsf{S}$, $A = \{1\}$). Lifting each $s(a)$ to $(\sigma_a, g_a) \in T$ defines a morphism $\rho = (\sigma, g) : \Sigma^+ \to T$ with $\pi\rho = s$; put $g(\varepsilon) = 1_A$. The subsemigroup $S_0 := \sigma(\Sigma^+) \leq S'$ is locally trivial; let $n := |S_0|$. Two facts:

(a) $S_0^n \subseteq S_0 E(S_0) S_0$. Let t_i be the prefix products of $s_1 \cdots s_n$. If $t_i = t_j$ with $i < j$, then $t_i = t_i u = t_i u^\omega$ for $u = s_{i+1} \cdots s_j$, so the product is $t_i u^\omega y$ with $y \in S_0^1$; otherwise the t_i exhaust S_0, one of them is an idempotent e, and the product is ey. In both cases it is xey with $x, y \in S_0^1$, and $xey = (xe)e(ey)$.

(b) $euf = ef$ for $e, f \in E(S_0)$ and $u \in S_0^1$: $euf = eu(fef) = (e(uf)e)f = ef$.

For $|w| \geq 2n$ write $w = pxz$ with $|p| = |z| = n$. By (a), $\sigma(p) = aeb$ and $\sigma(z) = cfd$ with $a, b, c, d \in S_0$ and e, f idempotent, and by (b) $\sigma(w) = ae\,(b\,\sigma(x)c)\,fd = aefd$ (with $\sigma(\varepsilon) := 1$), which depends only on $p = \mathrm{pref}_n(w)$ and $z = \mathrm{suf}_n(w)$. Shorter words are recorded exactly by $\widehat{s}_{n,n}$, which also separates ε. Hence equal $(\widehat{s}_{n,n}, g)$-values imply equal ρ-values and equal s-values; take $k = n$. □

Corollary 4.17 (Algebraic sufficient condition for polynomial data)**.** *If $\mathsf{S} \in \mathbf{LI} \vee \mathbf{Grp}$, then for every $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$ and every G as above with $L(G) = L \cap \Sigma^+$, $\mathsf{W}(\widetilde{G}_0) \cup (\{\varepsilon\} \cap L)$ is a characteristic sample for $\mathcal{B}^{\leq r_{\max}}_{f,s}$ of cardinality $O(\|G\|)$ and total length $O_{f,r_{\max},s,\Sigma}(\|G\|^2 \bar{\theta}_G)$.*

Proof. Lemma 4.16 and Theorem 4.14; k, A and g depend only on s. □

A finite semigroup is a *local group*, $S \in \mathbf{LG}$, if eSe is a group for every idempotent $e \in S$. This is always tested on the positive image S, never on $\mathsf{S}^\sharp$. By Proposition B.5, $\mathbf{LI} \vee \mathbf{Grp} \subsetneq \mathbf{LG}$.

Definition 4.18 (Local-group typing)**.** Fix $k \geq 1$, a finite group A and a map $\nu : \Sigma^{k+1} \to A$ (the *sliding-factor labelling*). For $|w| \geq 2k$ let $\sigma_k(w) = (F_1, \ldots, F_{|w|-k})$ be the factors of length $k+1$ of w in order of their starting positions, and put $\gamma_{k,\nu}(w) := \nu(F_1)\nu(F_2)\cdots\nu(F_{|w|-k})$ (ordered product in A); put $\gamma_{k,\nu}(w) := 1_A$ for $|w| < 2k$. The *local-group typing* is $\lambda_{k,\nu}(w) := (\widehat{s}_{k,k}(w), \gamma_{k,\nu}(w))$; by Lemma B.1 it is a morphism onto a finite monoid $M_{k,\nu}$. The typing s is *local-group-determined* if $\widehat{s} \preceq \lambda_{k,\nu}$ for some k, A and ν.

A local-group typing can encode a group contribution determined by a sliding factor, rather than by each letter alone. For example, $\lambda_{1,\nu}$ with $\nu(\mathtt{ab}) = 1 \in \mathbb{Z}_2$ and zero on other pairs records the parity of occurrences of $\mathtt{ab}$. Even on $\{\mathtt{a}, \mathtt{b}\}$ this is not determined by any fixed boundary window combined with a finite-group morphism (Proposition B.5(ii)). The local-group characterization below expresses this additional power without requiring a new learner or a target-specific typing.

Theorem 4.19 (Polynomial characteristic data for local-group typing)**.** *Let $\mathsf{S} \in \mathbf{LG}$, $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$ and $L(G) = L \cap \Sigma^+$. Then $\mathsf{W}(\widetilde{G}_0) \cup (\{\varepsilon\} \cap L)$ is a characteristic sample for $\mathcal{B}^{\leq r_{\max}}_{f,s}$ with*

$$|\mathsf{W}(\widetilde{G}_0)| = O(\|G\|), \qquad \|\mathsf{W}(\widetilde{G}_0)\|_+ = O_{f,r_{\max},s,\Sigma}(\|G\|^3 \bar{\theta}_G).$$

Proof. By Theorem B.4, $\widehat{s} \preceq \lambda := \lambda_{k,\nu}$ for $k = |\mathsf{S}|$ and some A, ν depending only on s. Lemma B.6 gives $(\mathrm{R}^{\lambda}_{\beta})$ with $\beta = \beta_\lambda = O(\|G\|^2 \bar{\theta}_G)$: heavy subtrees are first exchanged for short tuples with the same window vector (Corollary 4.12), and then a nonempty family is selected whose ordered sliding-factor products remain unchanged (Lemma B.2 and Proposition 4.10). Lemma 4.2(i) gives $t_{\mathrm{typ}} \leq \beta_\lambda = O(\|G\|^2 \bar{\theta}_G)$, and Theorem 4.4 gives $\|\mathsf{W}\|_+ \leq N_{\mathrm{typ}}(1 + 2\bar{\theta}_G + r_{\max}\beta_\lambda) = O(\|G\|^3 \bar{\theta}_G)$. □

The Rees and sliding-factor coordinates are classical; their combination with owner-wise block shortening is the quantitative step. For window and window–group typings the quadratic bounds of Theorems 4.13 and 4.14 are sharper.

The algebraic boundary for full refinement thickness. For $f \geq 1$, a pseudovariety $\mathbf{V}$ has *polynomial refinement thickness at fan-out f* if for every nonempty finite Σ, every $r_{\max} \geq 1$ and every surjective $s : \Sigma^+ \twoheadrightarrow \mathsf{S}$ with $\mathsf{S} \in \mathbf{V}$ there is a polynomial $q = q_{f,r_{\max},s,\Sigma}$ such that every reduced good fan-out-$\leq f$, rank-$\leq r_{\max}$ grammar G over Σ satisfies $\theta_{\mathrm{typ}}(G, s) \leq q(\|G\|, \bar{\theta}_G)$.

Corollary 4.20 (Refinement-thickness dichotomy)**.** *For every $f \geq 1$ and every pseudovariety $\mathbf{V}$ of finite semigroups, $\mathbf{V}$ has polynomial refinement thickness at fan-out f if and only if $\mathbf{V} \subseteq \mathbf{LG}$; for $\mathbf{V} \subseteq \mathbf{LG}$ one may take $\theta_{\mathrm{typ}}(G, s) = O_{f,r_{\max},s,\Sigma}(\|G\|^2 \bar{\theta}_G)$. The equivalence persists if G ranges only over presentations of $L \cap \Sigma^+$ with $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$, and also holds when θ_{typ} is replaced throughout by t_{typ}. For each single typing with $\mathsf{S} \notin \mathbf{LG}$ the polynomial bound already fails at fan-out one and rule rank two.*

Proof. If $\mathsf{S} \in \mathbf{LG}$, the proof of Theorem 4.19 gives $(\mathrm{R}^{\lambda}_{\beta})$ with $\beta_\lambda = O(\|G\|^2 \bar{\theta}_G)$, and Lemma 4.2(i),(iii), which do not use substitutability, give $\theta_{\widetilde{G}_0} \leq 2\bar{\theta}_G + r_{\max}\beta_\lambda$.

If $\mathsf{S} \notin \mathbf{LG}$, Theorem B.4(ii) gives a subsemigroup $\{e, z\} \cong U_1$: idempotents $e \neq z$ with $ez = ze = z$. By surjectivity choose $u, v \in \Sigma^+$ with $s(u) = e$, $s(v) = z$. Proposition 4.8(a) gives fan-out-one, rank-two grammars G_n with $\|G_n\| = O(n)$, $\bar{\theta}_{G_n} = O(1)$ and $\theta_{\mathrm{typ}}(G_n, s) \geq 2^n$; they present $\Sigma^+ = \widehat{s}^{-1}(\mathsf{S}) \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$. Finally, if $\mathbf{V} \not\subseteq \mathbf{LG}$, pick $\mathsf{S} \in \mathbf{V} \setminus \mathbf{LG}$; it is the positive image of the evaluation typing $\mathsf{S}^+ \to \mathsf{S}$ over the alphabet $\Sigma := \mathsf{S}$. For t_{typ}, the positive direction follows from Lemma 4.2(i) and the same $(\mathrm{R}^{\lambda}_{\beta})$ estimate; the negative direction follows from the strengthened Proposition 4.8(a). □

Remark 4.21 (Where the refinement obstruction lies). The grammars of Proposition 4.8 all present Σ^+. Every element of the fixed positive image S is represented by a positive word of length at most $|\mathsf{S}|$: a shortest word cannot repeat a nonempty prefix product without allowing a shortening. Hence $\mu^\star(\Sigma^+, s) \leq |\mathsf{S}|$, whereas $t_{\mathrm{typ}}(G_n, s) \geq 2^n$. The exponential thickness is internal to the refinement, not at its start symbols. This explains why it alone is not a lower bound on characteristic-data length: the latter requires distinguishing target languages. Short distributional representatives can also coexist with thick refinements (Proposition 5.3).

A bracketed doubling family. Let $\Gamma_{\mathrm{br}} = \{\mathtt{a}, \lfloor, \rfloor, \lceil, \rceil\}$, let $U_1 = \{1, \zeta\}$ with identity 1 and $\zeta^2 = \zeta$, and let $s_U : \Gamma_{\mathrm{br}}^+ \twoheadrightarrow U_1$ send $\lceil, \rceil$ to ζ and all other letters to 1; thus $s_U(w) = 1$ exactly for the nonempty ceiling-free words. For $n \geq 1$ let G_n be the fan-out-one grammar

$$S_0 \to E_n, \qquad E_0 \to \mathtt{a}, \qquad E_i \to \lfloor E_{i-1} E_{i-1} \rfloor \mid \lceil E_{i-1} \rceil \quad (1 \leq i \leq n),$$

put $L_n := L(G_n)$, $W_0 := \mathtt{a}$ and $W_i := \lfloor W_{i-1} W_{i-1} \rfloor$. This family was introduced in the companion CFG manuscript [25, Proposition 6.4]; every property used below is proved here.

Proposition 4.22 (Bracketed doubling family)**.** *For every $n \geq 1$:*

(i) *G_n is reduced and good, of fan-out one and rule rank two, with $\|G_n\| = O(n)$ and $\theta_{G_n} \leq 4n$.*

(ii) *L_n is Clark–Eyraud substitutable: for all $x, y \in \Gamma_{\mathrm{br}}^+$, $\mathcal{D}^{(1)}_{L_n}(x) \cap \mathcal{D}^{(1)}_{L_n}(y) \neq \emptyset$ implies $\mathcal{D}^{(1)}_{L_n}(x) = \mathcal{D}^{(1)}_{L_n}(y)$. Hence $L_n \in \mathcal{C}^{\mathrm{mcf}}_{1,s}$ for every typing s on Γ_{br}^+; in particular $L_n \in \mathcal{C}^{\mathrm{mcf}}_{1,s_U}$.*

(iii) *W_n is the only word of L_n of s_U-type 1, and $|W_n| = 3 \cdot 2^n - 2$.*

(iv) *Every characteristic sample F of L_n for $\mathcal{B}^{\leq r}_{1,s_U}$ ($r \geq 1$) or $\mathcal{B}_{1,s_U}$, or for any set-driven operator having characteristic samples throughout $\mathcal{C}^{\mathrm{mcf}}_{1,s_U}$, contains W_n; hence $\|F\|_+ \geq 3 \cdot 2^n - 1$.*

Proof. (i) Every template is nonempty and uses each variable once; at fan-out one, nonpermutation and nonmerging are automatic. Every nonterminal is reachable from S_0 and productive, and the $2n + 2$ rules have bounded length, so $\|G_n\| = O(n)$. Put $u_0 := \mathtt{a}$ and $u_j := \lceil u_{j-1} \rceil$, so $|u_j| = 2j + 1$. A rule with left-hand side E_i is used in the successful derivation of $\lceil^{n-i} t \rceil^{n-i}$, where t is $\mathtt{a}$, u_i, or $\lfloor u_{i-1} u_{i-1} \rfloor$; its length is at most $2n + 2i \leq 4n$. The rule $S_0 \to E_n$ is used by u_n, of length $2n + 1$.

(ii) The first claim is the second assertion of Proposition G.4 in Appendix G. It is the condition of Definition 2.8 at $f = 1$ for the trivial typing, and the finite language L_n is a 1-MCFL; item (a) after Definition 2.8 gives $L_n \in \mathcal{C}^{\mathrm{mcf}}_{1,s}$.

(iii) A derivation yields a ceiling-free word if and only if it uses no rule $E_i \to \lceil E_{i-1} \rceil$. Such a derivation from E_i is the complete binary tree of height i and yields W_i, and $|W_i| = 2|W_{i-1}| + 2$ with $|W_0| = 1$.

(iv) By (ii), $L_n \in \mathcal{C}^{\mathrm{mcf}}_{1,s_U}$, and by (iii), $1 \in M_{L_n}$ with $\mu_1(L_n) = |W_n|$. Proposition 4.6 with $m = 1$ forces a word of type 1, which can only be W_n. □

Remark 4.23 (Fan-out one: a pseudovariety boundary for data length). Fix rule-rank bound $r_{\max} \geq 2$. At fan-out one, a finite-semigroup pseudovariety $\mathbf{V}$ has uniform polynomial encoded characteristic-data bounds for every fixed typing onto a member of $\mathbf{V}$ and every reduced good rank-$\leq r_{\max}$ presentation if and only if $\mathbf{V} \subseteq \mathbf{LG}$. Sufficiency follows from Theorem 4.19 with $f = 1$. Conversely, if $\mathbf{V} \not\subseteq \mathbf{LG}$, then $U_1 = \{1, \zeta\} \in \mathbf{V}$ by Theorem B.4(ii) and pseudovariety closure. The typing s_U of Proposition 4.22 maps onto U_1, and the reduced good rank-two grammars G_n have size and rule thickness $O(n)$. By Proposition 4.22(iv), every characteristic sample for the fixed $\mathcal{B}^{\leq r_{\max}}_{1,s_U}$, and for every set-driven operator having characteristic samples for the entire class, has total encoded length at least $3 \cdot 2^n - 1$. This is a statement at the level of pseudovarieties; no corresponding claim is made for every individual non-local-group typing on its own alphabet.

4.6 Unrestricted Standard Input MCFGs

The input grammar need not be good: deleting, permuting and merging rules and empty components are all allowed. What then matters is the cost of realizing each nonempty-component pattern.

Definition 4.24 (Component-pattern thickness)**.** For a productive nonterminal A of a standard MCFG H let $\mathrm{supp}(\vec{u}) := \{i : u_i \neq \varepsilon\}$ and $\mathsf{Pat}_H(A) := \{\mathrm{supp}(\vec{u}) : \vec{u} \in L_A(H)\}$. Put

$$t_{\mathrm{pat}}(H) := \max_{\substack{A \text{ productive} \\ I \in \mathsf{Pat}_H(A)}} \min_{\substack{\vec{u} \in L_A(H) \\ \mathrm{supp}(\vec{u}) = I}} \|\vec{u}\|_1, \qquad \bar{t}_{\mathrm{pat}}(H) := \max\{1, t_{\mathrm{pat}}(H)\},$$

with $\max \emptyset := 0$. If H is reduced and λ-free, only the full pattern occurs and $t_{\mathrm{pat}}(H) = t_H$.

Proposition 4.25 (Good normalization controlled by pattern thickness)**.** *Let H be a standard fan-out-$\leq f$, rank-$\leq r$ MCFG with $L(H) \cap \Sigma^+ \neq \emptyset$, and H_{red} its trimmed form. There is a reduced good MCFG G for $L(H) \cap \Sigma^+$ with the same fan-out and rank bounds and*

$$\|G\| = O_{f,r}(\|H\|), \qquad t_G \leq t_{\mathrm{pat}}(H_{\mathrm{red}}), \qquad \bar{\theta}_G = O_{f,r}(\|H\|\, \bar{t}_{\mathrm{pat}}(H_{\mathrm{red}})).$$

Proof. Take G from Proposition 3.12. By Lemma F.1, each nonterminal language of G is an image of a union of support-pattern fibres of H_{red} under component deletion, permutation and concatenation, which do not increase total length; so $t_G \leq t_{\mathrm{pat}}(H_{\mathrm{red}})$ (Appendix C). Lemma 4.1(iii) gives the bound on $\bar{\theta}_G$. □

Corollary 4.26 (Total characteristic-data length from any standard MCFG)**.** *Let H be any finite standard fan-out-$\leq f$, rank-$\leq r$ MCFG for $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$; put $n := \max\{1, \|H\|\}$ and $t := \bar{t}_{\mathrm{pat}}(H_{\mathrm{red}})$. The operator $\mathcal{B}^{\leq r}_{f,s}$ has a characteristic sample $K \subseteq L$ with $|K| = O(n)$ and*

$$\|K\|_+ = \begin{cases} O(n^3 t), & \text{if } \widehat{s} \preceq (\widehat{s}_{k,\ell}, g) \text{ for some } k, \ell \text{ and finite-group morphism } g \text{ (possibly trivial)}, \\ O(n^4 t), & \text{if } \mathsf{S} \in \mathbf{LG}, \end{cases}$$

with constants depending on f, r, s, Σ (and k, ℓ, g), not on H.

Proof. If $L \subseteq \{\varepsilon\}$ take $K = L$. Otherwise Proposition 4.25 gives G with $\|G\| = O(n)$ and $\bar{\theta}_G = O(nt)$; apply Theorem 4.14 or Theorem 4.19. □

5 Further Quantitative Results

This section gives three results that complement Section 4 and are not used there: a bound in grammar size alone for rank one under every typing (Section 5.1); polynomial-time computation of the short witness sets together with a computable data budget (Section 5.2); and characteristic samples from short *distributional* representatives, which yield polynomial data for abelian-filtered targets under every typing (Section 5.3). Notation is that of Section 4; in particular $C_s(G) = |V||\mathsf{S}^\sharp|^f + |P||\mathsf{S}^\sharp|^{f r_{\max}} \leq |\mathsf{S}^\sharp|^{f r_{\max}} \|G\|$.

5.1 Rule Rank One

Theorem 5.1 (Rank-one characteristic-data bound)**.** *Fix f and a typing s.*

(i) *If G is a reduced good fan-out-$\leq f$, rank-≤ 1 presentation of $L \cap \Sigma^+$ with $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$, then $\mathsf{W}(\widetilde{G}_0) \cup (\{\varepsilon\} \cap L)$ is a characteristic sample for $\mathcal{B}^{\leq 1}_{f,s}$ and*

$$\|\mathsf{W}(\widetilde{G}_0)\|_+ \leq |\mathsf{S}^\sharp|^f (3 + |\mathsf{S}^\sharp|^f) \|G\|^2 \bar{\theta}_G \leq 2|\mathsf{S}^\sharp|^f (3 + |\mathsf{S}^\sharp|^f) \|G\|^3.$$

(ii) *For any standard fan-out-$\le f$, rank-≤ 1 grammar G_{in} for $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$, the operator $\mathcal{B}^{\le 1}_{f,s}$ has a characteristic sample K with $\|K\|_+ = O_{f,s}(\|G_{\mathrm{in}}\|^3)$.*

Proof. (i) A terminal derivation in a rank-≤ 1 grammar is a chain of unary rules ending in a constant rule. In a shortest chain no nonterminal repeats (splicing out the segment between two occurrences does not lengthen the yield, by linearity), so its rules are distinct. Hence $t_G \le \|G\|$, and the proof of Lemma 4.1(iii), whose path rules together with R are distinct and whose child chain is distinct, gives $\theta_G \le 2\|G\|$, so $\bar{\theta}_G \le 2\|G\|$. In $\widetilde{G}_0$, which also has rank ≤ 1, a shortest chain from a typed nonterminal X repeats no typed nonterminal, so it has at most $|V||\mathsf{S}^\sharp|^f$ rules, each writing at most $\bar{\theta}_G$ letters (Lemma 4.1(i)). So $\|\omega(X)\|_1 \le b := |V||\mathsf{S}^\sharp|^f\bar{\theta}_G$ for suitable canonical tuples. Lemma 4.2(ii),(iv) with $r_{\max} = 1$ gives the sample and $\|\mathsf{W}\|_+ \le C_s(G)(1 + 2\bar{\theta}_G + b)$; use $C_s(G) \le |\mathsf{S}^\sharp|^f\|G\|$, $1 + 2\bar{\theta}_G \le 3\bar{\theta}_G$ and $|V| \le \|G\|$.

(ii) If $L \subseteq \{\varepsilon\}$ take $K = L$. Otherwise Proposition 3.12 gives a reduced good rank-≤ 1 presentation G of $L \cap \Sigma^+$ with $\|G\| = O_f(\|G_{\mathrm{in}}\|)$; apply (i). □

Here the special rank-one argument $\bar{\theta}_G \le 2\|G\|$ is essential for the cubic bound; applying only the general thickness lemma would lose a power of $\|G\|$. Equivalently, the shortest typed chains above give $t_{\mathrm{typ}} \le |V||\mathsf{S}^\sharp|^f\bar{\theta}_G$. No thickness parameter is needed at rank one, and the typing affects only the constant; this extends the rank-one untyped scale [14, Lemma 7]. Rank two needs thickness: the size-$O(n)$ grammar $S \to A_n$, $A_i \to A_{i-1}A_{i-1}$, $A_0 \to \mathsf{c}$ generates $\{\mathsf{c}^{2^n}\}$, which lies in $\mathcal{C}^{\mathrm{mcf}}_{f,s}$ for every f and s (over one letter, a shared sentence context forces equal total component length, hence equal distributions), and every characteristic sample for $\mathcal{B}^{\le r_{\max}}_{f,s}$ contains c^{2^n}, because it is a nonempty subset of the target ($L(\mathcal{B}(\emptyset)) = \emptyset$).

5.2 Computing Short Characteristic Data

Let G be reduced and good of fan-out $\le f$ and rank $\le r_{\max}$. For a typed rule $R : X \to \rho(Y_1, \dots, Y_q)$ let c_R be the number of terminals written by ρ and $\operatorname{lhs} R := X$; let $\mathcal{R}$ be the set of typed rules of $\widetilde{G}_0$ (rank zero included). Put

$$\ell_{\mathrm{in}}(X) := \min_{\vec{u} \in L_X} \|\vec{u}\|_1, \qquad \ell_{\mathrm{out}}(X) := \min\{|\chi|_{\mathrm{fix}} : \chi \text{ derivational for } X\},$$

(derivational as in Lemma 4.1(iv); so $\ell_{\mathrm{out}}(X) = 0$ if $\widetilde{S} \to X$ is a rule), and

$$b(R) := c_R + \sum_{j=1}^{q} \ell_{\mathrm{in}}(Y_j), \qquad \mathsf{B}_{\mathrm{rule}}(G, s) := \sum_{R \in \mathcal{R}} \big(1 + \ell_{\mathrm{out}}(\operatorname{lhs} R) + b(R)\big).$$

Proposition 5.2 (Effective short characteristic data and a rule-only budget)**.** *Fix f, $r_{\max}$ and s. Given G, the following are computable in time polynomial in $\|G\|$, numbers being written in binary.*

(i) *$\widetilde{G}_0$, all values $\ell_{\mathrm{in}}(X)$, $\ell_{\mathrm{out}}(X)$ and $\mathsf{B}_{\mathrm{rule}}(G, s)$, and $\theta_G(R)$ for every rule R of G (hence θ_G and t_G).*

(ii) *For each typed nonterminal X, a rule $R_X : X \to \rho_X(Y_1, \dots, Y_q)$ such that the recursion $\omega(X) := \rho_X(\omega(Y_1), \dots, \omega(Y_q))$ is well-founded and gives $\omega(X) \in L_X$ with $\|\omega(X)\|_1 = \ell_{\mathrm{in}}(X)$; and a derivational context $\chi(X)$ with $|\chi(X)|_{\mathrm{fix}} = \ell_{\mathrm{out}}(X)$, equal to $\Box_1$ whenever $\widetilde{S} \to X$ is a rule. Both are given by derivation DAGs of size polynomial in $\|G\|$.*

For these choices,

$$\mathsf{W}(\widetilde{G}_0) = \{\chi(\operatorname{lhs} R)[\rho_R(\omega(Y_1), \dots, \omega(Y_q))] : R \in \mathcal{R}\},$$
$$|\mathsf{W}(\widetilde{G}_0)| \le |\mathcal{R}|, \qquad \|\mathsf{W}(\widetilde{G}_0)\|_+ \le \mathsf{B}_{\mathrm{rule}}(G, s),$$

the words can be written out in time polynomial in $\|G\| + \|\mathsf{W}(\widetilde{G}_0)\|_+$*, and every bound on* $\mathsf{W}(\widetilde{G}_0)$ *proved in Sections 4.1–4.5, Section 5.1 and Appendix B holds for this set; in particular it is characteristic whenever* $L(G) = L \cap \Sigma^+$ *with* $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$*.*

Proof. For fixed $f, r_{\max}, s$ the full refinement has $O(C_s(G))$ symbols and rules, and trimming is a linear-time productivity and reachability computation.

Inside values. Give a derivation the weight (terminal length, number of nodes), ordered lexicographically. A rule R maps child weights (x_j, n_j) to $(c_R, 1) + \sum_j (x_j, n_j)$, which is strictly larger than every argument. Knuth's generalization of Dijkstra's algorithm, in Nederhof's weighted-deduction form [35] (see also [34]), therefore computes the minimum weight of every typed nonterminal together with a minimizing rule R_X whose children are finalized before X. This order makes the recursion for ω well-founded, the chosen rules form a DAG, and the first coordinate of the weight is $\ell_{\mathrm{in}}(X) = \|\omega(X)\|_1$.

Outside values. Run Dijkstra's algorithm on the graph with an edge $X \to Y_j$ of weight $c_R + \sum_{i \ne j} \ell_{\mathrm{in}}(Y_i)$ for every typed rule $R : X \to \rho(Y_1, \dots, Y_q)$, from the sources X with $\widetilde{S} \to X$ at distance 0. A path from a source to X yields, by composing its templates and filling the siblings with $\omega(Y_i)$, a context of X of that fixed length; it is admissible by Lemma 3.1 and derivational because it arises from a successful derivation by deleting the X-subtree. Conversely every derivational context of X comes from a root-to-X path whose siblings have yields at least ℓ_{in}. Hence the distance is $\ell_{\mathrm{out}}(X)$, and the shortest-path tree with the inside DAG represents $\chi(X)$. The same two passes on G give $t_G = \max_A \ell_{\mathrm{in}}(A)$ and $\theta_G(R) = \ell_{\mathrm{out}}(A) + c_R + \sum_j \ell_{\mathrm{in}}(B_j)$ for $R : A \to \rho(B_1, \dots, B_q)$, since outside and inside parts of a derivation can be chosen independently. All values are at most exponential in $\|G\|$, so have polynomial binary length.

Witness set and budget. By definition of ω, $\chi(X)[\omega(X)]$ is the rule word of R_X, and every other word of Definition 3.9 is a rule word. The rule word of R has length exactly $\ell_{\mathrm{out}}(\operatorname{lhs} R) + b(R)$; summing $|w| + 1$ over $\mathcal{R}$ gives $\mathsf{B}_{\mathrm{rule}}$, and passing to distinct words can only lower it. Expanding a DAG takes time linear in the length of the word it represents.

Inherited bounds. Each cited bound is proved by exhibiting *some* $\omega(X) \in L_X$ and the derivational context E_B (Lemma 4.1(iv)); the computed minima are no longer and the number of words is unchanged. The characteristic property is Lemma 4.2(iv). □

The budget $\mathsf{B}_{\mathrm{rule}}$ is a presentation-sensitive *upper certificate*, often far below the worst-case bounds; it is not the size of a minimum characteristic sample. For $\{\mathsf{c}^{2^n}\}$ of Section 5.1 the words are exponentially longer than $\|G\|$, but their DAGs have polynomial size.

5.3 Distributional Representatives and Abelian-Filtered Targets

Proposition 3.11 does not require a representative to be derivable from its nonterminal. One may therefore use the shortest member of the whole typed distribution class. Let G be a reduced good fan-out-$\le f$, rank-$\le r$ presentation of $L \cap \Sigma^+$ with $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$. For a typed nonterminal X let $\mathrm{Cl}_{L,s}(X)$ be the $\equiv_{L,s}$-class containing L_X (Lemma 3.8) and put

$$\kappa_G(X) := \min_{\vec{v} \in \mathrm{Cl}_{L,s}(X)} \|\vec{v}\|_1 \le \ell_{\mathrm{in}}(X), \qquad \kappa_G := \max_X \kappa_G(X).$$

Proposition 5.3 (Relaxed distributional witnesses)**.** *For each typed nonterminal $X = B_{\mathbf{t}}$ choose $\iota(X) \in \mathrm{Cl}_{L,s}(X)$ with $\|\iota(X)\|_1 = \kappa_G(X)$ and put $\chi(X) := E_B$ (Lemma 4.1(iv)). The words $\chi(X)[\iota(X)]$ and $\chi(X)[\rho(\iota(Y_1), \dots, \iota(Y_q))]$, over all typed nonterminals and typed rules $X \to \rho(Y_1, \dots, Y_q)$, form a set $K^\dagger \subseteq L$ such that $K^\dagger \cup (\{\varepsilon\} \cap L)$ is characteristic for $\mathcal{B}_{f,s}^{\leq r}$ and $\mathcal{B}_{f,s}$, and*

$$|K^\dagger| \leq C_s(G), \qquad \|K^\dagger\|_+ \leq C_s(G)(1 + 2\bar{\theta}_G + r\kappa_G).$$

Proof. Fix $\vec{u}_X \in L_X$ for every X. Since $\iota(X) \equiv_{L,s} \vec{u}_X$ and E_B accepts $\vec{u}_X$, it accepts $\iota(X)$; so $\chi(X)[\iota(X)] \in L$. By Lemma 3.2, $\rho(\iota(Y_1), \dots, \iota(Y_q)) \equiv_{L,s} \rho(\vec{u}_{Y_1}, \dots, \vec{u}_{Y_q}) \in L_X$, so the rule words also lie in L. Each $\iota(X)$ has $\widehat{s}$-vector $\mathbf{t}$, nonempty components, and $\chi(S_{\mathbf{t}}) = E_S = \square_1$, so Proposition 3.11 applies with this ι and gives $L \cap \Sigma^+ \subseteq L(\mathcal{B}(K))$ for $K \supseteq K^\dagger$; Proposition 3.10 gives the converse for $K \subseteq L$, and the Start rule handles ε. The length bound is Lemma 4.2(ii) with $b = \kappa_G$, whose proof uses only $\|\iota(X)\|_1 \leq b$. □

Unlike Proposition 5.2, the words $\chi(X)[\iota(X)]$ cannot in general be omitted. For fixed f, r, s, a family G_n with $\|G_n\|\bar{\theta}_{G_n} = n^{O(1)}$ that requires superpolynomial characteristic data must therefore have $\kappa_{G_n} = n^{\omega(1)}$; this is a necessary condition, not a proof that such a family exists.

Theorem 5.4 (Polynomial data for abelian-filtered targets)**.** *Fix a nonempty alphabet Σ, a typing s and $f, r \geq 1$. There is a constant C_{ab} depending only on Σ, s, f, r such that the following holds for every abelian group A, morphism $\varphi : \Sigma^* \to \mathsf{A}$, subgroup $\mathsf{H} \leq \mathsf{A}$, $\gamma \in \mathsf{A}$ and $\mathsf{Acc} \subseteq \mathsf{S}^\sharp$. If $L := \widehat{s}^{-1}(\mathsf{Acc}) \cap \varphi^{-1}(\gamma + \mathsf{H})$ and $L \cap \Sigma^+$ has a reduced good fan-out-$\leq f$, rank-$\leq r$ presentation G, then $L \in \mathcal{C}_{f,s}^{\mathrm{mcf}}$ and $\mathcal{B}_{f,s}^{\leq r}$ has a characteristic sample K with*

$$\|K\|_+ \leq C_{\mathrm{ab}} \left(\|G\|\bar{\theta}_G\right)^{3|\Sigma|+2}.$$

Proof. $L \in \mathcal{C}_{f,s}^{\mathrm{mcf}}$ by Proposition 2.9. By Proposition 5.3 and $C_s(G) \leq |\mathsf{S}^\sharp|^{fr}\|G\|$ it suffices to show $\kappa_G \leq C'(T+1)^{3|\Sigma|+1}$ with $T := (r+1)t_G + \|G\| \leq (r+2)\|G\|\bar{\theta}_G$ and C' depending only on Σ, s, f.

A lattice of admissible Parikh shifts. Let $\psi : \Sigma^* \to \mathbb{N}^\Sigma$ be the Parikh map, $\psi(w) = (|w|_a)_{a\in\Sigma}$, and put $\Psi(\vec{u}) := \sum_i \psi(u_i)$. Let $\overline{\varphi} : \mathbb{Z}^\Sigma \to \mathsf{A}$ be the homomorphism satisfying $\varphi = \overline{\varphi} \circ \psi$, and set $\Lambda := \overline{\varphi}^{-1}(\mathsf{H})$. For each $C \in V$ fix $\omega_C \in L_C(G)$ with $\|\omega_C\|_1 \leq t_G$. For $\vec{u} \in L_C(G)$ the context E_C accepts $\vec{u}$ and ω_C, and $L \subseteq \varphi^{-1}(\gamma + \mathsf{H})$, so

$$\Psi(\vec{u}) - \Psi(\omega_C) \in \Lambda. \tag{$*$}$$

For a rule $R : C \to \rho(B_1, \dots, B_q)$ let $\mathbf{c}_R \in \mathbb{N}^\Sigma$ be its terminal-letter Parikh vector (distinct from the scalar terminal count c_R in Section 5.2) and put $\delta_R := \Psi(\omega_C) - \mathbf{c}_R - \sum_j \Psi(\omega_{B_j})$. Applying $(*)$ to $\rho(\omega_{B_1}, \dots, \omega_{B_q})$ gives $\delta_R \in \Lambda$, and $\|\delta_R\|_1 \leq T$. Let $\Gamma_T := \{z \in \Lambda : \|z\|_1 \leq T\}$ and $\Lambda_T := \langle \Gamma_T \rangle$. For $\vec{u} = \rho(\vec{u}_1, \dots, \vec{u}_q)$ with $\vec{u}_j \in L_{B_j}(G)$, $\Psi(\vec{u}) - \Psi(\omega_C) = \sum_j (\Psi(\vec{u}_j) - \Psi(\omega_{B_j})) - \delta_R$, so induction on derivations gives $\Psi(\vec{u}) - \Psi(\omega_C) \in \Lambda_T$ for all $\vec{u} \in L_C(G)$.

A short representative. Let $X = C_{\mathbf{t}}$ have arity $d \leq f$ and $\vec{u} \in L_X$. For each i fix, once and for all, a semilinear representation (a finite union of linear sets) of the Parikh image of the regular language $\widehat{s}^{-1}(t_i) \cap \Sigma^+$, and choose a linear set $c_i + \sum_{p\in P_i} \mathbb{N}p$ of it containing $\psi(u_i)$; the number and size of these generators depend only on Σ and s. Write $\Psi(\vec{u}) - \Psi(\omega_C) = \sum_{z\in\Gamma_T} m_z z$ with $m_z \in \mathbb{Z}$. Then the system

$$\sum_{i=1}^{d} \sum_{p\in P_i} \eta_{i,p}\, p - \sum_{z\in\Gamma_T} (\mu_z^+ - \mu_z^-)\, z = \Psi(\omega_C) - \sum_{i=1}^{d} c_i$$

has a solution in nonnegative integers (the coefficients of $\psi(u_i)$ and $\mu_z^\pm := \max\{\pm m_z, 0\}$). It has $m = |\Sigma|$ equations, $n \leq D + 2(2T+1)^{|\Sigma|}$ unknowns and entries of absolute value at most

$a \le D(T+1)$, with D depending only on Σ, s, f. By Papadimitriou's bound [39] it has a nonnegative integer solution with all entries at most $n(ma)^{2m+1} = O((T+1)^{3|\Sigma|+1})$. For each i the vector $c_i + \sum_p \eta_{i,p} p$ is the Parikh vector of some nonempty $v_i \in \widehat{s}^{-1}(t_i)$; put $\vec{v} := (v_1, \ldots, v_d)$. Then $\|\vec{v}\|_1 = O((T+1)^{3|\Sigma|+1})$, $\widehat{s}^{(d)}(\vec{v}) = \mathbf{t}$, and $\Psi(\vec{v}) - \Psi(\vec{u}) \in \Lambda_T \subseteq \Lambda$. For every admissible E, $E[\vec{v}]$ and $E[\vec{u}]$ have the same $\widehat{s}$-value, and $\varphi(E[\vec{v}]) - \varphi(E[\vec{u}]) = \overline{\varphi}(\Psi(\vec{v}) - \Psi(\vec{u})) \in \mathsf{H}$; so $E[\vec{v}] \in L$ iff $E[\vec{u}] \in L$, i.e. $\vec{v} \equiv_{L,s} \vec{u}$ and $\kappa_G(X) \le \|\vec{v}\|_1$. □

The theorem holds even when $\mathsf{S} \notin \mathbf{LG}$: its hypothesis concerns the target, not the typing. No polynomial-time construction is claimed for ineffectively presented groups A.

6 Comparison with Multidimensional Substitutability

Yoshinaka's multidimensional substitutability [13, 14] is the direct positive-data baseline. This section shows that (i) trivial typing recovers it and window typing recovers the CFG (k, ℓ) hierarchy of [12], while a fixed finite typing admits non-context-free targets outside the baseline (Section 6.1); (ii) the algebraic regimes of Section 4 are separated at the level of targets (Section 6.2); and (iii) a boundary-typed target can require rule rank two at every finite fan-out (Section 6.4). Since every (f, s) condition includes arity one, $\mathcal{C}^{\mathrm{mcf}}_{f,s} \cap \mathrm{CFL} \subseteq \mathcal{C}^{\mathrm{mcf}}_{1,s}$.

6.1 Yoshinaka's Baseline and Finite Feature Licensing

Definition 6.1 (Multidimensional substitutability)**.** A language L is *fD-substitutable* if for every $1 \le d \le f$, all $E, F \in \mathcal{E}_d$ and all $\vec{x}, \vec{y} \in (\Sigma^+)^d$, $E[\vec{x}], E[\vec{y}], F[\vec{x}] \in L \Rightarrow F[\vec{y}] \in L$ (the *substitution implication*); $\mathsf{YSub}(f)$ is the class of these languages. This is Yoshinaka's definition, with nonempty components and nonempty internal separators [14, Section 3.1 and footnote 3].

The substitution implication is (f, s_{triv})-tuple substitutability, so $\mathcal{C}^{\mathrm{mcf}}_{f,s_{\mathrm{triv}}} = \mathsf{YSub}(f) \cap f\text{-MCFL} \subseteq \mathcal{C}^{\mathrm{mcf}}_{f,s}$ for every s. Yoshinaka's word-level (k, ℓ) condition [12] requires, for $u \in \Sigma^k$, $v \in \Sigma^\ell$ and nonempty uy_1v, uy_2v, that $w_1uy_1vw_1', w_1uy_2vw_1' \in L$ imply $w_2uy_1vw_2' \in L \iff w_2uy_2vw_2' \in L$. Two nonempty words have equal $s_{k,\ell}$-type iff they are identical or both have length at least $\max\{1, k+\ell\}$ with the same first k and last ℓ letters, i.e. iff they are of the forms uy_1v, uy_2v above (or equal). Hence $(1, s_{k,\ell})$-tuple substitutability is exactly Yoshinaka's (k, ℓ) condition; the componentwise version at higher fan-out is a definition of this paper.

A language L is *strictly 2-local* if there is a set Π of pairs over $\Sigma \cup \{\vdash, \dashv\}$ such that a word w lies in L iff every pair of adjacent symbols of $\vdash w \dashv$ lies in Π.

Lemma 6.2 (Strictly 2-local languages are boundary typed)**.** *Every strictly 2-local L is $(f, s_{1,1})$-tuple-substitutable for every f; in particular $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s_{1,1}}$, and $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$ whenever $s_{1,1} \preceq s$.*

Proof. Let $\vec{x}, \vec{y} \in (\Sigma^+)^d$ have equal $s_{1,1}$-types, so corresponding components are identical letters or share first and last letters, and let $E_0[\vec{x}], E_0[\vec{y}] \in L$. Suppose $E[\vec{x}] \in L$. Since the internal separators of E are nonempty, every adjacent pair of $\vdash E[\vec{y}] \dashv$ either lies inside some y_i, and then occurs in $E_0[\vec{y}] \in L$, or joins a first or last letter of some y_i to a symbol of $\vdash E \dashv$, or lies in $\vdash E \dashv$, and then occurs in $\vdash E[\vec{x}] \dashv$. So $E[\vec{y}] \in L$; symmetry and $s_{1,1} \preceq s$ finish the proof (regular languages are MCFLs). □

Proposition 6.3 (Cross-serial dependencies with finite feature licensing)**.** *Let* $\Sigma_{\mathrm{feat}} = \{\mathtt{p}_0, \mathtt{p}_1, \mathtt{q}_0, \mathtt{q}_1, \mathtt{a}, \mathtt{b}, \mathtt{c}, \mathtt{d}\}$, $\mathcal{R}_{\mathrm{feat}} = \{(0,0), (0,1), (1,0)\}$, *and*

$$L_{\mathrm{feat}} = \{\mathtt{a}^m \mathtt{p}_i \mathtt{b}^n \mathtt{c}^m \mathtt{q}_j \mathtt{d}^n : m, n \ge 1,\ (i,j) \in \mathcal{R}_{\mathrm{feat}}\}.$$

Let $g : \Sigma^*_{\text{feat}} \to \mathbb{Z}_2^2$ map $\mathtt{p}_1 \mapsto (1,0)$, $\mathtt{q}_1 \mapsto (0,1)$ and all other letters to 0, and let $s := (s_{1,1}, g)|_{\Sigma^+_{\text{feat}}}$ (corestricted to its image). Then $L_{\text{feat}} \in \mathsf{MCFL}(2,2) \setminus \mathrm{CFL}$ and $L_{\text{feat}} \in \mathcal{C}^{\text{mcf}}_{f,s} \setminus \mathsf{YSub}(f)$ for every $f \geq 2$; moreover L_{feat} is not $(1, s_{1,1})$-tuple-substitutable.

Proof. Grammar. With fan-out-two A, B and rules $A \to (\mathtt{a}, \mathtt{c}) \mid (\mathtt{a}\xi^1_1, \mathtt{c}\xi^2_1)(A)$, $B \to (\mathtt{b}, \mathtt{d}) \mid (\mathtt{b}\xi^1_1, \mathtt{d}\xi^2_1)(B)$ and $S \to (\xi^1_1\mathtt{p}_i\xi^1_2\xi^2_1\mathtt{q}_j\xi^2_2)(A, B)$ for $(i,j) \in \mathcal{R}_{\text{feat}}$, we get $L_A = \{(\mathtt{a}^m, \mathtt{c}^m)\}$, $L_B = \{(\mathtt{b}^n, \mathtt{d}^n)\}$ and $L(G) = L_{\text{feat}}$; the grammar is good, of fan-out and rank two. Intersecting with $\mathtt{a}^+\mathtt{p}_0\mathtt{b}^+\mathtt{c}^+\mathtt{q}_0\mathtt{d}^+$ and erasing the markers gives the classical non-context-free $\{\mathtt{a}^m\mathtt{b}^n\mathtt{c}^m\mathtt{d}^n\}$.

Typed substitutability. Let $Q = \mathtt{a}^+\{\mathtt{p}_0, \mathtt{p}_1\}\mathtt{b}^+\mathtt{c}^+\{\mathtt{q}_0, \mathtt{q}_1\}\mathtt{d}^+$ (strictly 2-local), $K_{\text{feat}} := \{w \in \Sigma^+_{\text{feat}} : g(w) \in \mathcal{R}_{\text{feat}}\}$, and let φ map $\mathtt{a}, \mathtt{c}, \mathtt{b}, \mathtt{d}$ to $(1,0), (-1,0), (0,1), (0,-1) \in \mathbb{Z}^2$ and the markers to 0. On Q, $g(w) = (i,j)$ for the markers $\mathtt{p}_i, \mathtt{q}_j$ of w, so $L_{\text{feat}} = Q \cap K_{\text{feat}} \cap \varphi^{-1}(0)$. The three languages are (f,s)-tuple-substitutable by Lemma 6.2, by item (c) after Definition 2.8 (K_{feat} is a union of s-classes), and by Proposition 2.9(ii); Proposition 2.9(i) gives the intersection.

Failure without g. For $x = \mathtt{a}\mathtt{p}_0\mathtt{b}$, $y = \mathtt{a}\mathtt{p}_1\mathtt{b}$, $E = \Box_1\mathtt{c}\mathtt{q}_0\mathtt{d}$ and $F = \Box_1\mathtt{c}\mathtt{q}_1\mathtt{d}$ we have $E[x], E[y], F[x] \in L_{\text{feat}}$ but $F[y] \notin L_{\text{feat}}$, since $(1,1) \notin \mathcal{R}_{\text{feat}}$. Here $s_{1,1}(x) = s_{1,1}(y)$ but $g(x) \neq g(y)$. □

Corollary 6.4 (Uniform finite feature licensing)**.** *With the* same *typing s, for every $R \subseteq \{0,1\}^2$ the language $L_R := \{\mathtt{a}^m\mathtt{p}_i\mathtt{b}^n\mathtt{c}^m\mathtt{q}_j\mathtt{d}^n : m, n \geq 1,\ (i,j) \in R\}$ lies in $\mathcal{C}^{\text{mcf}}_{f,s}$ for all $f \geq 2$, and for $R \neq \emptyset$ it is not context-free. The typing is fixed once for the whole family; the same holds for any fixed inventories $\{0, \ldots, p-1\} \times \{0, \ldots, q-1\}$ with g into $\mathbb{Z}_p \times \mathbb{Z}_q$.*

Proof. Keep the start rules indexed by R, replace $\mathcal{R}_{\text{feat}}$ by R in K_{feat} (still a union of s-classes), and repeat the proof; for $(i,j) \in R$ use $\mathtt{a}^+\mathtt{p}_i\mathtt{b}^+\mathtt{c}^+\mathtt{q}_j\mathtt{d}^+$ for non-context-freeness. □

Proposition 6.5 (Explicit data budget for a common feature typing)**.** *Let $f \geq 2$ and $R \neq \emptyset$. The operator $\mathcal{B}^{\leq 2}_{f,s}$ has a characteristic sample $K_R \subseteq L_R$ with $\|K_R\|_+ = 36|R| + 22$ (for $R = \emptyset$ the empty sample is characteristic). Conversely, if a set-driven operator $\mathcal{D}$ has characteristic samples for* all *$L_{R'}$, $R' \subseteq \{0,1\}^2$, then every characteristic sample K for L_R under $\mathcal{D}$ contains a word with markers $\mathtt{p}_i, \mathtt{q}_j$ for each $(i,j) \in R$; hence $\|K\|_+ \geq 7|R|$.*

Proof. Write $w^{m,n}_{ij} := \mathtt{a}^m\mathtt{p}_i\mathtt{b}^n\mathtt{c}^m\mathtt{q}_j\mathtt{d}^n$, of length $2m + 2n + 2$, fix $(i_0, j_0) \in R$, and let

$$K_R := \{w^{m,n}_{ij} : (i,j) \in R,\ m, n \in \{1,2\}\} \cup \{w^{3,1}_{i_0 j_0}, w^{1,3}_{i_0 j_0}\}.$$

The encoded costs are $7 + 9 + 9 + 11 = 36$ per pair and $11 + 11$ for the two extra words.

Typed refinement. For the grammar G of Proposition 6.3 with start rules indexed by R, the s-type of $(\mathtt{a}^m, \mathtt{c}^m)$ depends only on whether $m = 1$ (short words are recorded exactly), so A has two typed copies A_1, A_+, and likewise B_1, B_+; the type of $w^{m,n}_{ij}$ is determined by (i,j), giving one typed start nonterminal S_{ij} per $(i,j) \in R$. With $\rho_A := (\mathtt{a}\xi^1_1, \mathtt{c}\xi^2_1)$, the typed rules are $A_1 \to (\mathtt{a}, \mathtt{c})$, $A_+ \to \rho_A(A_1)$, $A_+ \to \rho_A(A_+)$, their B-analogues, and four start rules $S_{ij} \to \rho_{ij}(A_\bullet, B_\circ)$ per $(i,j) \in R$.

Witnesses. Take $\iota(A_1) = (\mathtt{a}, \mathtt{c})$, $\iota(A_+) = (\mathtt{a}^2, \mathtt{c}^2)$, $\iota(B_1) = (\mathtt{b}, \mathtt{d})$, $\iota(B_+) = (\mathtt{b}^2, \mathtt{d}^2)$, $\iota(S_{ij}) = w^{1,1}_{ij}$, the contexts $\chi(A_\bullet) = \Box_1\mathtt{p}_{i_0}\mathtt{b}\Box_2\mathtt{q}_{j_0}\mathtt{d}$, $\chi(B_\circ) = \mathtt{a}\mathtt{p}_{i_0}\Box_1\mathtt{c}\mathtt{q}_{j_0}\Box_2$ and $\chi(S_{ij}) = \Box_1$. The required words of Proposition 3.11 are: $w^{1,1}_{i_0 j_0}$ for A_1, B_1 and their constant rules; $w^{2,1}_{i_0 j_0}$ and $w^{1,2}_{i_0 j_0}$ for A_+, B_+ and the rules from A_1, B_1; $w^{3,1}_{i_0 j_0}$ and $w^{1,3}_{i_0 j_0}$ for the rules from A_+, B_+; and $w^{m,n}_{ij}$, $m, n \in \{1,2\}$, for S_{ij} and its four rules. All lie in K_R. Since $K_R \subseteq L_R \in \mathcal{C}^{\text{mcf}}_{f,s}$, Propositions 3.11 and 3.10 make K_R characteristic for $\mathcal{B}^{\leq 2}_{f,s}$.

Lower bound. If K contains no word with markers $\mathtt{p}_i, \mathtt{q}_j$, then $K \subseteq L_{R'}$ for $R' = R \setminus \{(i,j)\}$. For a characteristic sample F of $L_{R'}$, the set $K \cup F$ lies between K and L_R and between F and $L_{R'}$, so $L(\mathcal{D}(K \cup F))$ would equal both $L_R \neq L_{R'}$. Distinct pairs need distinct words, each of length at least 6. □

The example models a finite case-licensing or feature-compatibility constraint; no claim about a specific natural language is made. The relation R varies with the target, while the morphism g and the operator do not.

Group-valued agreement. Let A be a finite group, Δ_1 a finite alphabet whose letter images generate A under a morphism $g_1 : \Delta_1^* \to A$, and $\Delta_2 = \{\bar{\delta} : \delta \in \Delta_1\}$ a disjoint copy. Put $\Sigma_{\mathrm{agr}} := \{\mathtt{a}, \mathtt{b}, \mathtt{c}, \mathtt{d}\} \uplus \Delta_1 \uplus \Delta_2$, and define $g : \Sigma_{\mathrm{agr}}^* \to A \times A$ by $g(\delta) = (g_1(\delta), 1)$, $g(\bar{\delta}) = (1, g_1(\delta))$, and $g(\mathtt{a}) = g(\mathtt{b}) = g(\mathtt{c}) = g(\mathtt{d}) = (1,1)$. For $R \subseteq A \times A$, let

$$L_R^A := \{\mathtt{a}^m u \mathtt{b}^n \mathtt{c}^m v \mathtt{d}^n : m, n \geq 1,\ u \in \Delta_1^+,\ v \in \Delta_2^+,\ g(uv) \in R\},$$

let $L_{\mathrm{agr}}^A := L_{\{(x,x):x\in A\}}^A$, and fix the corestricted positive typing $s_{\mathrm{agr}} := (s_{1,1}, g)|_{\Sigma_{\mathrm{agr}}^+}$. Here the finite group factor of the typing is $A \times A$, not A.

Proposition 6.6 (Agreement of ordered group-valued features)**.** (i) *For every $R \subseteq A \times A$, $L_R^A \in \mathsf{MCFL}(2,2) \cap \mathcal{C}_{f,s_{\mathrm{agr}}}^{\mathrm{mcf}}$ for all $f \geq 2$. If $R \neq \emptyset$, then L_R^A is not context-free.*

(ii) *If the commutator subgroup $[A,A]$ is not contained in the centre $Z(A)$, then $L_{\mathrm{agr}}^A \notin \mathcal{C}_{f,t}^{\mathrm{mcf}}$ for every $f \geq 2$ and every typing $t \preceq (s_{k,\ell}, g')|_{\Sigma_{\mathrm{agr}}^+}$, where $k, \ell \geq 0$ and g' maps into a finite abelian group. In particular $L_{\mathrm{agr}}^A \notin \mathsf{YSub}(f)$.*

For $A = S_3$ the hypotheses of (ii) *hold, while Theorem 4.14 applies uniformly to $\mathcal{C}_{f,s_{\mathrm{agr}}}^{\mathrm{mcf}}$ with a window–$(S_3 \times S_3)$ typing.*

Proof. (i) For each $x \in A$ take a good right-linear grammar U_x for $g_1^{-1}(x) \cap \Delta_1^+$ and let $\bar{U}_x$ generate its copied language over Δ_2. These exist because every group element has a nonempty positive-word representative. With the fan-out-two rules

$$\begin{aligned}
&P \to (\xi_1^1, \xi_2^1)(U_x, \bar{U}_y) && ((x,y) \in R),\\
&C \to (\mathtt{a}, \mathtt{c}) \mid (\mathtt{a}\xi_1^1, \mathtt{c}\xi_1^2)(C),\\
&D \to (\mathtt{b}, \mathtt{d}) \mid (\mathtt{b}\xi_1^1, \mathtt{d}\xi_1^2)(D),\\
&T \to (\xi_1^1\xi_2^1, \xi_1^2\xi_2^2)(C, P),\\
&S \to (\xi_1^1\xi_2^1\xi_1^2\xi_2^2)(T, D),
\end{aligned}$$

we generate L_R^A with fan-out and rank at most two; the adjacent variables in each component of T and S belong to different children. For $R = \emptyset$ the language is empty. Otherwise choose $(x,y) \in R$ and positive $p, q \in \Delta_1^+$ with $g_1(p) = x$, $g_1(q) = y$. Intersection with $\mathtt{a}^+ p \mathtt{b}^+ \mathtt{c}^+ \bar{q} \mathtt{d}^+$, followed by erasing the feature blocks, yields $\{\mathtt{a}^m \mathtt{b}^n \mathtt{c}^m \mathtt{d}^n : m, n \geq 1\}$; thus L_R^A is not context-free.

For typed substitutability write $L_R^A = Q \cap g^{-1}(R) \cap \varphi^{-1}(0)$, where $Q = \mathtt{a}^+ \Delta_1^+ \mathtt{b}^+ \mathtt{c}^+ \Delta_2^+ \mathtt{d}^+$ is strictly 2-local and $\varphi : \Sigma_{\mathrm{agr}}^* \to \mathbb{Z}^2$ sends $\mathtt{a}, \mathtt{c}, \mathtt{b}, \mathtt{d}$ to $(1,0), (-1,0), (0,1), (0,-1)$, respectively, and all feature letters to 0. Lemma 6.2 handles Q; $g^{-1}(R)$ is a union of s_{agr}-classes; and Proposition 2.9 handles the abelian filter and intersection. Hence $L_R^A \in \mathcal{C}_{f,s_{\mathrm{agr}}}^{\mathrm{mcf}}$.

(ii) Since $[A,A] \not\subseteq Z(A)$, some commutator $z := yxy^{-1}x^{-1}$ is noncentral. Choose $h \in A$ with $hz \neq zh$, and nonempty $p, q, w \in \Delta_1^+$ whose g_1-values are x, y, h. Let $N = |A|$, $K \geq \max\{k+\ell, 1\}$, choose $\delta \in \Delta_1$ and put $\pi := \delta^{KN}$. Set

$$u = \pi pqq^{N-1}p^{N-1}\pi, \qquad u' = \pi qpq^{N-1}p^{N-1}\pi.$$

Then $g_1(u) = 1$, $g_1(u') = z$, while u, u' have identical Parikh vectors and agree on their length-k prefixes and length-ℓ suffixes. Thus every window–abelian typing as in the statement gives $t(u) = t(u')$, and likewise $t(\bar{u}) = t(\bar{u}')$. The tuples $X := (u, \bar{u})$ and $Y := (u', \bar{u}')$ therefore have equal t-type. Both are admitted by $E := \mathtt{a}\Box_1\mathtt{bc}\Box_2\mathtt{d}$, since their two group values are respectively $(1,1)$ and (z,z). Choose also $F := \mathtt{a}w\Box_1\mathtt{bc}\Box_2\bar{w}\mathtt{d}$. Then $F[X] \in L^A_{\mathrm{agr}}$ has values (h,h), but $F[Y] \notin L^A_{\mathrm{agr}}$ has values (hz, zh). Thus the common context E does not imply equal distributions, contradicting (f,t)-substitutability at arity two. Trivial typing is a special case. For $A = S_3$, $[S_3, S_3] = A_3$ is not central. □

Remark 6.7. The separating contexts above use a noncentral commutator. No corresponding claim is made here for groups whose commutator subgroup is central, such as Q_8.

6.2 Separating the Typing Classes

The feature example is covered by a window and an abelian group. Two further fan-out-two targets separate the regimes of Section 4: L_{A} is admitted by a local-group typing but by no window–group-determined typing, and L_{B} by a window–S_3 typing but by no window–abelian one. Both have the form $\sigma w\$c$: a flag $\sigma = \mathtt{r}$ leaves w unconstrained, $\sigma = \mathtt{l}$ imposes a typing-dependent condition on w, and $c \in C_\times := \{\mathtt{1}^m\mathtt{2}^n\mathtt{3}^m\mathtt{4}^n : m, n \geq 1\}$ makes the target non-context-free. Let $\varphi_\times$ map $\mathtt{1}, \mathtt{3}, \mathtt{2}, \mathtt{4}$ to $(1,0), (-1,0), (0,1), (0,-1) \in \mathbb{Z}^2$ and all other letters to 0, and let $\#_{\mathtt{ab}}(w)$ be the number of occurrences of the factor $\mathtt{ab}$ in w.

Lemma 6.8 (Flag separation)**.** *Let t be a typing, $u, v \in \Sigma^+$ with $t(u) = t(v)$, and $z \in \Sigma^*$. If L contains $\mathtt{r}uz$, $\mathtt{r}vz$ and $\mathtt{l}uz$ but not $\mathtt{l}vz$, then $L \notin \mathcal{C}^{\mathrm{mcf}}_{f,t}$ for every $f \geq 1$.*

Proof. $\mathtt{r}\Box_1 z \in \mathcal{E}_1$ accepts u and v, while $\mathtt{l}\Box_1 z$ accepts u but not v; so $\mathcal{D}^{(1)}_L(u) \neq \mathcal{D}^{(1)}_L(v)$, and every (f,t) condition includes arity one. □

The targets. Let $\Sigma_{\mathrm{A}} = \{\mathtt{l}, \mathtt{r}, \mathtt{a}, \mathtt{b}, \mathtt{c}, \$, \mathtt{1}, \mathtt{2}, \mathtt{3}, \mathtt{4}\}$,

$$L_{\mathrm{A}} := \{\sigma w\$c : \sigma \in \{\mathtt{l}, \mathtt{r}\},\ w \in \{\mathtt{a}, \mathtt{b}, \mathtt{c}\}^+,\ c \in C_\times,\ \sigma = \mathtt{r} \text{ or } \#_{\mathtt{ab}}(w) \text{ even}\},$$

and $s_{\mathrm{A}} := \lambda_{1,\nu}|_{\Sigma_{\mathrm{A}}^+}$ with $\nu : \Sigma_{\mathrm{A}}^2 \to \mathbb{Z}_2$, $\nu(\mathtt{ab}) = 1$ and $\nu = 0$ otherwise (Definition 4.18); it records the first and last letters and, for $|w| \geq 2$, the parity of $\#_{\mathtt{ab}}(w)$. Let $\Sigma_{\mathrm{B}} = \Sigma_{\mathrm{A}} \setminus \{\mathtt{c}\}$, $\alpha = (12)$, $\beta = (23) \in S_3$, $g_{\mathrm{B}} : \Sigma_{\mathrm{B}}^* \to S_3$ with $\mathtt{a} \mapsto \alpha$, $\mathtt{b} \mapsto \beta$ and all other letters $\mapsto 1$,

$$L_{\mathrm{B}} := \{\sigma w\$c : \sigma \in \{\mathtt{l}, \mathtt{r}\},\ w \in \{\mathtt{a}, \mathtt{b}\}^+,\ c \in C_\times,\ \sigma = \mathtt{r} \text{ or } g_{\mathrm{B}}(w) = 1\},$$

and $s_{\mathrm{B}} := (s_{1,1}, g_{\mathrm{B}})|_{\Sigma_{\mathrm{B}}^+}$, corestricted. **Ab** is the pseudovariety of finite abelian groups, and $\mathbf{LI} \vee \mathbf{Ab}$ consists of the divisors of $T \times A$ with $T \in \mathbf{LI}$ and A finite abelian.

Proposition 6.9 (A target separating local-group and window–group typings)**.** (i) $L_{\mathrm{A}} \in \mathsf{MCFL}(2,2) \setminus \mathrm{CFL}$.

(ii) $L_{\mathrm{A}} \in \mathcal{C}^{\mathrm{mcf}}_{f,s_{\mathrm{A}}}$ *for every $f \geq 2$, and* $s_{\mathrm{A}}(\Sigma_{\mathrm{A}}^+) \in \mathbf{LG}$.

(iii) *If* $t \preceq (s_{k,\ell}, g'|_{\Sigma_{\mathrm{A}}^+})$ *for some* k, ℓ *and a morphism* g' *into a finite group (in particular if* $t(\Sigma_{\mathrm{A}}^+) \in \mathbf{LI} \vee \mathbf{Grp}$*), then* $L_{\mathrm{A}} \notin \mathcal{C}_{f,t}^{\mathrm{mcf}}$ *for all* $f \geq 1$.

Hence $s_{\mathrm{A}}(\Sigma_{\mathrm{A}}^+) \in \mathbf{LG} \setminus (\mathbf{LI} \vee \mathbf{Grp})$*, and* L_{A} *is covered by Theorem 4.19 but, for no typing, by Theorems 4.13, 4.14, 4.15 or Corollary 4.17.*

Proposition 6.10 (A target separating nonabelian and window–abelian typings)**.** (i) $L_{\mathrm{B}} \in \mathsf{MCFL}(2,2) \setminus \mathrm{CFL}$.

(ii) $L_{\mathrm{B}} \in \mathcal{C}_{f,s_{\mathrm{B}}}^{\mathrm{mcf}}$ *for every* $f \geq 2$*, and* s_{B} *is window–group-determined with group* S_3.

(iii) *If* $t \preceq (s_{k,\ell}, g'|_{\Sigma_{\mathrm{B}}^+})$ *for some* k, ℓ *and a morphism* g' *into a finite* abelian *group (in particular if* $t(\Sigma_{\mathrm{B}}^+) \in \mathbf{LI} \vee \mathbf{Ab}$*), then* $L_{\mathrm{B}} \notin \mathcal{C}_{f,t}^{\mathrm{mcf}}$ *for all* $f \geq 1$.

Hence L_{B} *is covered by Theorem 4.14 with the nonabelian constant* $c_{S_3,f} = M_{S_3,f} - 1$*, but for no typing by Theorem 4.13 or 4.15.*

Proof of Propositions 6.9 and 6.10. Write $X \in \{\mathrm{A}, \mathrm{B}\}$, let Σ_X° be $\{\mathtt{a}, \mathtt{b}, \mathtt{c}\}$ or $\{\mathtt{a}, \mathtt{b}\}$, and let $P_X(w)$ be "$\#_{\mathtt{ab}}(w)$ even" or "$g_{\mathrm{B}}(w) = 1$".

(i) Let W be a right-linear (fan-out-one, rank-one, good) grammar for the regular language $\{\sigma w : \sigma \in \{\mathtt{l}, \mathtt{r}\}, w \in (\Sigma_X^\circ)^+, \sigma = \mathtt{r} \text{ or } P_X(w)\}$; it tracks the last letter and the parity, respectively the value in S_3. Add

$$C \to (\xi_1^1 \$ \mathtt{1}, \mathtt{3})(W) \mid (\xi_1^1 \mathtt{1}, \xi_1^2 \mathtt{3})(C), \quad D \to (\mathtt{2}, \mathtt{4}) \mid (\xi_1^1 \mathtt{2}, \xi_1^2 \mathtt{4})(D), \quad S \to (\xi_1^1 \xi_2^1 \xi_1^2 \xi_2^2)(C, D).$$

Then $L_C = \{(\sigma w \$ \mathtt{1}^m, \mathtt{3}^m)\}$, $L_D = \{(\mathtt{2}^n, \mathtt{4}^n)\}$, and S generates L_X; the grammar is good (adjacent variables in the start template belong to different children). Intersecting with $\mathtt{ra}\$\mathtt{1}^+\mathtt{2}^+\mathtt{3}^+\mathtt{4}^+$ and erasing $\mathtt{ra}\$$ gives the non-context-free $C_\times$.

(ii) $L_X = Q_X \cap \Pi_X \cap \varphi_\times^{-1}(0)$ with $Q_X = \{\mathtt{l}, \mathtt{r}\}(\Sigma_X^\circ)^+\$\mathtt{1}^+\mathtt{2}^+\mathtt{3}^+\mathtt{4}^+$ and Π_X the set of nonempty words that begin with $\mathtt{r}$, or begin with $\mathtt{l}$ and satisfy P_X for the whole word. On Q_X the condition P_X of the whole word is that of w: no factor $\mathtt{ab}$ meets σ, \$ or the digits, and g_{B} is trivial on all other letters. Q_X is strictly 2-local (its pairs are $\vdash \sigma$, σx, xy, $x\$$, $\$\mathtt{1}$, $\mathtt{11}, \mathtt{12}, \mathtt{22}, \mathtt{23}, \mathtt{33}, \mathtt{34}, \mathtt{44}$, $\mathtt{4} \dashv$ for $\sigma \in \{\mathtt{l}, \mathtt{r}\}$, $x, y \in \Sigma_X^\circ$) and $s_{1,1} \preceq s_X$, so Lemma 6.2 applies. Π_X is a union of s_X-classes: s_X records the first letter, and for words of length at least two the parity of $\#_{\mathtt{ab}}$, respectively the value of g_{B} (the one-letter words $\mathtt{l}, \mathtt{r}$ lie in Π_X); so item (c) after Definition 2.8 applies. Proposition 2.9 handles $\varphi_\times^{-1}(0)$ and the intersection; with (i), $L_X \in \mathcal{C}_{f,s_X}^{\mathrm{mcf}}$ for $f \geq 2$. s_{A} is local-group-determined, so its positive image lies in $\mathbf{LG}$ by Theorem B.4.

(iii) for A. Let $\hat{k} = \max\{k, \ell, 1\}$, $q \geq 2$ an even multiple of the exponent of the group of g', $u = \mathtt{a}^{\hat{k}}(\mathtt{abc})^q\mathtt{a}^{\hat{k}}$ and $v = \mathtt{a}^{\hat{k}}(\mathtt{bac})^q\mathtt{a}^{\hat{k}}$. Both are long for $s_{k,\ell}$ with prefix $\mathtt{a}^k$ and suffix $\mathtt{a}^\ell$, and $g'(u) = g'(\mathtt{a})^{2\hat{k}} = g'(v)$; so $t(u) = t(v)$. But $\#_{\mathtt{ab}}(u) = q$ is even and $\#_{\mathtt{ab}}(v) = 1$ (only at $\mathtt{a}^{\hat{k}}\mathtt{b}$). With $z = \$\mathtt{1234}$, Lemma 6.8 applies. A typing with positive image in $\mathbf{LI} \vee \mathbf{Grp}$ satisfies the hypothesis by Lemma 4.16.

(iii) for B. Let $\hat{k} \geq \max\{k, \ell, 1\}$ be even, $u = \mathtt{a}^{\hat{k}}(\mathtt{ab})^3\mathtt{a}^{\hat{k}}$ and $v = \mathtt{a}^{\hat{k}}(\mathtt{ab})^2\mathtt{ba}\,\mathtt{a}^{\hat{k}}$. They have the same prefix $\mathtt{a}^k$, suffix $\mathtt{a}^\ell$ and Parikh vector, so $t(u) = t(v)$ for abelian g'. Since $\alpha^2 = \beta^2 = 1$, $\alpha\beta$ has order 3 and $\beta\alpha = (\alpha\beta)^{-1}$, we get $g_{\mathrm{B}}(u) = (\alpha\beta)^3 = 1$ and $g_{\mathrm{B}}(v) = (\alpha\beta)^2(\alpha\beta)^{-1} = \alpha\beta \neq 1$. With $z = \$\mathtt{1234}$, Lemma 6.8 applies. For $\mathbf{LI} \vee \mathbf{Ab}$, the proof of Lemma 4.16 with an abelian group factor gives an abelian g'.

Consequences. If $s_{\mathrm{A}}(\Sigma_{\mathrm{A}}^+)$ were in $\mathbf{LI} \vee \mathbf{Grp}$, (iii) with $t = s_{\mathrm{A}}$ would contradict (ii). The window, window–group and window–abelian theorems and Corollary 4.17 assume a typing of the excluded kind. □

Remark 6.11 (Scope of the typing separations). The first two separations already occur at arity one (Lemma 6.8); the group-agreement separation uses arity-two contexts. Refining the typing by a finite morphism recognizing the strictly 2-local structural language writes each target as $\widehat{s'}^{-1}(\mathsf{Acc}') \cap \varphi^{-1}(0)$ for an abelian φ, so Theorem 5.4 gives polynomial data for the *different* operator $\mathcal{B}^{\leq 2}_{f,s'}$, with degree $3|\Sigma| + 2$ depending on the alphabet. The window–group and local-group bounds instead hold uniformly for the original fixed-typing class with degree independent of $|\Sigma|$.

6.3 Why the fan-out-one lower bound does not lift

Lemma 6.12 (Power transfer across two tuple components)**.** *Let $f \geq 2$, $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$, $u \in \Sigma^+$, and let $i_0, \pi \geq 1$ be an eventual index and period of $s(u)$, i.e. $s(u)^{i+\pi} = s(u)^i$ for $i \geq i_0$. Suppose $a, c \geq i_0$, $d \in \pi\mathbb{Z}_{>0}$, and some word of L contains the factor $u^{a+d+c+2}$. Then for all $X, Z \in \Sigma^*$ and $Y \in \Sigma^+$,*

$$Xu^aYu^{d+c}Z \in L \quad \Longleftrightarrow \quad Xu^{a+d}Yu^cZ \in L.$$

Proof. Write the given word as $Pu^{a+d+c+2}Q$. The tuples $\vec{x} = (u^a, u^{d+c})$ and $\vec{y} = (u^{a+d}, u^c)$ have the same componentwise s-types by eventual periodicity. They also have the common admissible accepting context $E = Pu\Box_1 u\Box_2 Q$, since $E[\vec{x}] = E[\vec{y}] = Pu^{a+d+c+2}Q \in L$. The (f, s)-condition thus equates their two-component distributions. Test them in the admissible context $X\Box_1 Y\Box_2 Z$; the condition $Y \neq \varepsilon$ is essential. □

Theorem 6.13 (The bracketed doubling family does not lift)**.** *Let $s : \Gamma^+_{\mathrm{br}} \twoheadrightarrow \mathsf{S}$ be any fixed finite-semigroup typing and let L_n be the family of Proposition 4.22. If $n \geq 2|\mathsf{S}| + 3$, then $L_n \notin \mathcal{C}^{\mathrm{mcf}}_{f,s}$ for every $f \geq 2$.*

Proof. Choose an eventual index $i_0 \geq 1$ and period $\pi \geq 1$ for $s(\lfloor)$, with $i_0 + \pi \leq |\mathsf{S}| + 1$. Thus $n \geq 2i_0 + \pi + 2$. Put $a = i_0$, $d = \pi$, $c = n - a - d - 2 \geq i_0$. The unique ceiling-free word $W_n = \lfloor W_{n-1}W_{n-1} \rfloor \in L_n$ starts with exactly n occurrences of $\lfloor$, so contains the factor $\lfloor^{a+d+c+2}$. In this same W_n, let $\lfloor^a$ be the a letters immediately after the first $\lfloor$, and let $\lfloor^{d+c}$ be the last $d + c$ letters of the initial run of the second copy of W_{n-1}. This is possible since $d + c = n - a - 2 \leq n - 1$. We can write

$$W_n = X\lfloor^a Y \lfloor^{d+c} Z,$$

where $X = \lfloor$ and $Y \neq \varepsilon$ contains the first $\mathtt{a}$. By Lemma 6.12, membership in L_n would force $X\lfloor^{a+d}Y\lfloor^c Z \in L_n$. The resulting word is ceiling-free, but it begins with $\lfloor^{n+d}\mathtt{a}$ rather than $\lfloor^n\mathtt{a}$. This contradicts uniqueness of the ceiling-free member W_n (Proposition 4.22(iii)). Higher f include the arity-two substitutability condition. □

Remark 6.14. Power transfer preserves the total length and Parikh vector of the word. It does not prove that all rare-type representatives are short; rather it shows why the particular exponential lower-bound family of Proposition 4.22 is excluded by every fixed finite typing at fan-out two.

6.4 An Intrinsically Rank-Two Boundary-Typed Target

Let $\Sigma_{\mathrm{intr}} := \{\mathtt{a}, \mathtt{b}, \mathtt{l}, \mathtt{r}, \mathtt{p}, \mathtt{q}, \mathtt{c}, \mathtt{d}, \#, \$, \mathtt{u}, \mathtt{v}, \mathtt{w}\}$, $P_{\mathrm{intr}} := \{\mathtt{lpc}, \mathtt{lpd}, \mathtt{lqc}, \mathtt{rpc}, \mathtt{rpd}\}$,

$$\mathrm{MIX}^+_2 := \{x \in \{\mathtt{a}, \mathtt{b}\}^+ : |x|_{\mathtt{a}} = |x|_{\mathtt{b}}\}, \quad T^+_3 := \{\mathtt{u}^n\mathtt{v}^n\mathtt{w}^n : n \geq 1\}, \quad L_{\mathrm{intr}} := P_{\mathrm{intr}} \# \mathrm{MIX}^+_2 \,\$\, T^+_3.$$

The factor MIX^+_2 forces rule rank two, T^+_3 forces fan-out two, and the wrapper P_{intr} separates the target from $\mathsf{YSub}(f)$.

An EDT0L system uses deterministic parallel-rewriting tables and a regular set of permitted table sequences. Its *index is at most k* if every generated terminal word has a derivation with at most k nonterminal occurrences in each intermediate sentential form.

Lemma 6.15 (Rank-one MCFLs are finite-index EDT0L)**.** *Every language in* $\mathsf{MCFL}(f', 1)$ *has an EDT0L system of index at most* f'*.*

Proof. Let H be reduced, of fan-out $\le f'$ and rank ≤ 1. Use a symbol $[B, j]$ for each nonterminal B and component j, axiom $[S, 1]$, and for each rule $A \to \rho(B)$ (or $A \to \rho$) a table mapping $[A, j]$ to the jth component of ρ with ξ_1^k replaced by $[B, k]$, and fixing every other symbol. Since each rule has at most one child and is linear, every reachable sentential form contains symbols of a single nonterminal, each at most once, so at most f' nonterminal occurrences. Table sequences that produce terminal words correspond to derivations of H (components of the child not used by a deleting rule are completed by any derivation, as H is reduced), with the same terminal word. □

Theorem 6.16 (A non-context-free intrinsic-rank-two typed witness)**.** $L_{\mathrm{intr}} \in \mathsf{MCFL}(2, 2) \setminus \mathrm{CFL}$*, and* $L_{\mathrm{intr}} \in \mathcal{C}^{\mathrm{mcf}}_{f, s_{1,1}} \setminus \mathsf{YSub}(f)$ *for every* $f \ge 2$*, but* $L_{\mathrm{intr}} \notin \mathsf{MCFL}(f', 1)$ *for every finite* f'*. Thus its minimum fan-out is two and its minimum rule rank over all finite-fan-out presentations is two: Theorem 4.13 applies to it, while Theorem 5.1 does not at any finite fan-out.*

Proof. Grammar. The fan-out-one rules $B \to \mathtt{a}B\mathtt{b}B \mid \mathtt{b}B\mathtt{a}B \mid \mathtt{a}B\mathtt{b} \mid \mathtt{ab}B \mid \mathtt{ab} \mid \mathtt{b}B\mathtt{a} \mid \mathtt{ba}B \mid \mathtt{ba}$ give $L_B = \mathrm{MIX}_2^+$: every word they generate is balanced, and conversely a balanced word decomposes at the first return of $|\cdot|_{\mathtt{a}} - |\cdot|_{\mathtt{b}}$ to zero as $\mathtt{a}y\mathtt{b}z$ or $\mathtt{b}y\mathtt{a}z$ with y, z balanced and possibly empty. With $A \to (\mathtt{u}, \mathtt{w}) \mid (\mathtt{u}\xi_1^1\mathtt{v}, \mathtt{w}\xi_1^2)(A)$ we get $L_A = \{(\mathtt{u}^n\mathtt{v}^{n-1}, \mathtt{w}^n)\}$, and the rules $S \to (\alpha\#\xi_1^1\$\xi_2^1\mathtt{v}\xi_2^2)(B, A)$, $\alpha \in P_{\mathrm{intr}}$, generate L_{intr}. The grammar is reduced and good (the components of A are separated by $\mathtt{v}$), of fan-out and rank two.

Not context-free. Intersecting with $\mathtt{lpc}\#\mathtt{ab}\$\mathtt{u}^+\mathtt{v}^+\mathtt{w}^+$ and erasing the prefix gives T_3^+.

Typed membership. $L_{\mathrm{intr}} = Q_{\mathrm{intr}} \cap \Phi^{-1}(0)$, where $Q_{\mathrm{intr}} := P_{\mathrm{intr}}\#\{\mathtt{a}, \mathtt{b}\}^+\$\,\mathtt{u}^+\mathtt{v}^+\mathtt{w}^+$ is strictly 2-local, with pairs $\vdash \mathtt{l}$, $\vdash \mathtt{r}$, $\mathtt{lp}, \mathtt{lq}, \mathtt{rp}, \mathtt{pc}, \mathtt{pd}, \mathtt{qc}, \mathtt{c}\#, \mathtt{d}\#$, $\#x$, xy, $x\$$ ($x, y \in \{\mathtt{a}, \mathtt{b}\}$), $\$\mathtt{u}, \mathtt{uu}, \mathtt{uv}, \mathtt{vv}, \mathtt{vw}, \mathtt{ww}, \mathtt{w} \dashv$ (the first six letter pairs admit exactly the words of P_{intr}), and $\Phi : \Sigma_{\mathrm{intr}}^* \to \mathbb{Z}^3$ maps $\mathtt{a}, \mathtt{b}, \mathtt{u}, \mathtt{v}, \mathtt{w}$ to $(1,0,0), (-1,0,0), (0,1,0), (0,-1,1), (0,0,-1)$ and other letters to 0. Lemma 6.2 and Proposition 2.9 give $L_{\mathrm{intr}} \in \mathcal{C}^{\mathrm{mcf}}_{f, s_{1,1}}$ for $f \ge 2$.

Outside $\mathsf{YSub}(f)$*.* For $x = \mathtt{p}$, $y = \mathtt{q}$, $E = \mathtt{l}\square_1\mathtt{c}\#\mathtt{ab}\$\mathtt{uvw}$, $F = \mathtt{r}\square_1\mathtt{d}\#\mathtt{ab}\$\mathtt{uvw}$: $E[x], E[y], F[x] \in L_{\mathrm{intr}}$ but $F[y] \notin L_{\mathrm{intr}}$; the boundary typing separates x and y.

Rank one is impossible. $\mathsf{MCFL}(f', 1)$ is closed under letter-to-word homomorphisms (apply them to the templates). If $L_{\mathrm{intr}} \in \mathsf{MCFL}(f', 1)$, erasing all letters except $\mathtt{a}, \mathtt{b}$ would put MIX_2^+ in $\mathsf{MCFL}(f', 1)$, hence make it EDT0L by Lemma 6.15, and then also $\mathrm{MIX}_2 = \mathrm{MIX}_2^+ \cup \{\varepsilon\}$ (EDT0L is closed under union). But MIX_2 is the word problem of $\mathbb{Z}$, which is not EDT0L [22]; Lemma 6.15 agrees with Gallot's characterization of non-branching MCFGs [21, Definition 38 and Theorems 28–29]. □

The two lower bounds come from separate factors, and the non-branching lower bound is prior work. What is specific here is that a strictly local wrapper and the $(1, 1)$ boundary typing place the target inside the typed class while the substitution implication keeps it outside Yoshinaka's baseline.

7 Conclusion

This paper turns the qualitative positive-data learning principle for typed MCFGs into upper bounds on the *total encoded length* of characteristic samples, measured against a given presentation. For a

reduced good grammar G and fixed alphabet, fan-out, rank and typing, Theorem 4.4 reduces the principal uniform encoded-length bounds to the computable $t_{\mathrm{typ}}(G, s)$ and $N_{\mathrm{typ}} = O(\|G\|)$:

- window and window–finite-group typings give $O(\|G\|^2 \bar{\theta}_G)$, for noncommutative groups and interleaved tuple components alike (Theorems 4.13 and 4.14);
- local-group typings give $O(\|G\|^3 \bar{\theta}_G)$, by combining classical sliding-factor coordinates with owner-wise block shortening (Theorem 4.19);
- rank one gives $O(\|G\|^3)$ under every typing (Theorem 5.1), and arbitrary standard input grammars are covered through component-pattern thickness (Corollary 4.26).

The refinement-thickness dichotomy (Corollary 4.20) places the exact boundary for polynomial thickness of the full yield-type refinement at **LG**, at every fan-out. The rare-type lower bound (Proposition 4.6) makes this an exact boundary for characteristic-data length at fan-out one, at the level of pseudovarieties (Remark 4.23). The exponential lower-bound family was introduced in the companion CFG manuscript [25, Proposition 6.4]; the properties used here are proved in Proposition 4.22 and Appendix G. At fan-out two the same family fails tuple-substitutability under every fixed finite typing (Theorem 6.13). The feature-licensing family of Section 6.1 gives separate examples of fixed finite biases for non-context-free cross-serial targets.

Open Problem 7.1 (Characteristic-data length at higher fan-out). Fix $f \geq 2$ and $r_{\max} \geq 2$. Is there a typing s with $\mathsf{S} \notin \mathbf{LG}$ and a polynomial q such that, for every $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$ and every reduced good fan-out-$\leq f$, rank-$\leq r_{\max}$ presentation G of $L \cap \Sigma^+$, the *fixed* operator $\mathcal{B}^{\leq r_{\max}}_{f,s}$ has a characteristic sample of total length at most $q(\|G\|, \bar{\theta}_G)$?

For the opposite direction, take a fixed surjective typing s with positive image U_1. Do there exist fixed $f \geq 2$, $r_{\max} \geq 2$ and targets $L_n \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$ with reduced good presentations G_n of fan-out $\leq f$ and rank $\leq r_{\max}$, such that $\|G_n\| \bar{\theta}_{G_n} = n^{O(1)}$, but *every* characteristic sample for $\mathcal{B}^{\leq r_{\max}}_{f,s}$ on L_n has total length $n^{\omega(1)}$? This would be a lower bound for the specified operator, not automatically for every set-driven learner. The refinement-thickness dichotomy (Corollary 4.20) does not answer either characteristic-data question. By Proposition 4.6, producing a family with $\mu^\star(L_n, s) = n^{\omega(1)}$ would suffice, and would give the same lower bound for any set-driven operator that has characteristic samples throughout the fixed-typing class. Theorem 6.13 excludes the bracketed doubling family of Proposition 4.22 as such a construction.

A more focused, unproved question is whether, for every fixed typing s and rank bound $r_{\max}$, a polynomial $p_{s,r_{\max},\Sigma}$ always bounds $\mu^\star(L, s)$ by $p(\|G\|, \bar{\theta}_G)$ for all $L \in \mathcal{C}^{\mathrm{mcf}}_{2,s}$ with reduced good fan-out-≤ 2, rank-$\leq r_{\max}$ presentations G. This holds for local-group typings by (4.1) and Theorem 4.19, and for the abelian-filtered targets of Theorem 5.4; at fan-out one the analogous statement fails. Neither power transfer nor these special cases settles the question.

Tightness of the exponents. The polynomial exponents are not claimed optimal, and the source of each factor is explicit. In $O(\|G\|^2 \bar{\theta}_G)$, one factor $\|G\|$ counts the witness words ($N_{\mathrm{typ}} \leq |\mathsf{S}^\sharp|^{f r_{\max}} \|G\|$, Theorem 4.4), and the other bounds the length of a root-to-leaf path in a λ-minimal derivation by the number $|V||N|^f$ of node labels (Lemma 4.9). The further factor in the local-group bound $O(\|G\|^3 \bar{\theta}_G)$ arises because heavy subtrees are first exchanged for window representatives of length $O(\|G\| \bar{\theta}_G)$ (Corollary 4.12 and Lemma B.6); it may be an artefact of this two-stage argument. On the other side, $\bar{\theta}_G$ cannot be removed: the size-$O(n)$ grammar for $\{\mathsf{c}^{2^n}\}$ of Section 5.1 has $\bar{\theta}_G = 2^n$, and every characteristic sample contains c^{2^n}. For typings with $\mathsf{S} \in \mathbf{LG}$ the examples of this paper force no more than $\Omega(\bar{\theta}_G)$, so the remaining gap is a factor $\|G\|^2$ for window–group

typings and $\|G\|^3$ for local-group typings. Is there, for some fixed typing (possibly trivial), a family of targets with reduced good presentations G_n whose characteristic samples need total length $\Omega(\|G_n\|\,\bar{\theta}_{G_n})$? The rare-type argument cannot give this: for a fixed typing it yields at most $\sum_{m\in M_L}(\mu_m(L)+1) \le |\mathsf{S}|(t_{\mathrm{typ}}+1)$ by (4.1). A factor $\|G\|$ requires $\Theta(\|G\|)$ words that each separate two targets, and over a fixed alphabet and typing such words must be encoded; the operator's generalization across equal-type factors then tends to reconstruct the whole family from a few of them.

Thus the outstanding issue concerns characteristic-data complexity for a fixed reconstruction operator across a typed target class, not whether an individual grammar has a thick full yield-type refinement.

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Data availability

No empirical dataset is associated with this theoretical paper.

A Simultaneous Block Replacement in Finite Groups

This appendix proves Proposition 4.10, used in Lemmas 4.11 and B.6. The abelian case is a zero-sum argument. The general case never commutes factors; it has three parts: an order classification for homogeneous point configurations (Lemma A.4), an interval selection by boundary states (Lemma A.5), and a Ramsey-type extraction of homogeneous indices (Lemma A.7).

Definition A.1 (Block-replacement family)**.** Let A be a finite group and $f \ge 1$. A *block-replacement family* of width f over A with index set $[m]$ consists of $d \le f$ *lines*; on line c, a left-to-right sequence of blocks $\beta_{c,1},\dots,\beta_{c,q_c}$ and constants $a_{c,0},\dots,a_{c,q_c} \in A$; for every block β an *owner* $o(\beta) \in [m]$ and two values $x_\beta, x'_\beta \in A$; every index owns at most f blocks. For $J \subseteq [m]$ put

$$\mathrm{val}_c(J) := a_{c,0}\, y_{c,1}\, a_{c,1} \cdots y_{c,q_c}\, a_{c,q_c}, \qquad y_{c,j} := \begin{cases} x'_{\beta_{c,j}}, & o(\beta_{c,j}) \in J, \\ x_{\beta_{c,j}}, & o(\beta_{c,j}) \notin J. \end{cases}$$

Lemma A.2 (Block replacement)**.** *In a block-replacement family of width f over A there is a nonempty $J \subseteq [m]$ with $\mathrm{val}_c(J) = \mathrm{val}_c(\emptyset)$ for every line c, provided that either* (a) *A is abelian and $m > \mathsf{d}(A^f)$, or* (b) *A is arbitrary and $m \ge M_{A,f}$, with d and $M_{A,f}$ as in Section 4.3.*

Proof of (a). Write A additively. By commutativity $\mathrm{val}_c(J) - \mathrm{val}_c(\emptyset) = \sum_{i\in J} \delta_{i,c}$, where $\delta_{i,c}$ is the sum of $x'_\beta - x_\beta$ over the blocks β of line c owned by i. The sequence $(\delta_i)_{i\in[m]}$ with $\delta_i := (\delta_{i,c})_c \in A^d$ has length $m > \mathsf{d}(A^f) \geq \mathsf{d}(A^d)$ (A^d embeds in A^f), so it has a nonempty zero-sum subsequence $(\delta_i)_{i\in J}$. □

Remark A.3 (Size of the zero-sum constant). The prefix-sum pigeonhole gives $\mathsf{d}(A^f) \leq |A|^f - 1$, but much more is true. Write $A \cong C_{n_1} \oplus \cdots \oplus C_{n_s}$ and $n = \exp(A) \geq 2$. Then

$$f\sum_{i=1}^{s}(n_i - 1) \;\leq\; \mathsf{d}(A^f) \;\leq\; (n-1) + n\log\frac{|A|^f}{n},$$

where the lower bound is attained by a basis sequence and the upper bound is due to van Emde Boas and Kruyswijk [33, Proposition 5.1.8 and Theorem 5.5.5]; for a p-group the lower bound is an equality [33, Theorem 5.5.9]. Hence $\mathsf{d}(A^f) = O(\exp(A)\, f \log|A|)$; for $A = C_2$ the antichain bound drops from $2^f - 1$ to f. Since zero-sum-free sequences of length $\mathsf{d}(A^d)$ exist, the argument cannot do better; we do not claim that a grammar realizes them.

A.1 Arbitrary Finite Groups: Order Homogenization

Lemma A.4 (A homogeneous block-order lemma)**.** *Let $e \geq 1$ and $n \geq 3$, and let (t,i), $t \in [e]$, $i \in [n]$, be distinct points on a line. Suppose that* (i) *the order of (t,i) and (u,i) does not depend on i, and* (ii) *for each ordered pair (t,u), including $t = u$, the order of (t,i) and (u,j) is the same for all $i < j$. Then $[e]$ is partitioned into classes such that: the points of each class occupy a consecutive region containing no other points; each class has a direction (increasing or decreasing in i); and its region consists of n successive* chunks*, one per index in that direction, each containing exactly one point of every type of the class, in an order independent of the index.*

Proof. By (ii) with $t = u$, the points of each type are monotone in i. For distinct t, u let X mean $(t,i) < (u,j)$ and Y mean $(u,i) < (t,j)$ for $i < j$; both are constant by (ii).

Both increasing. $X \wedge Y$ puts all points of index i of both types before those of index $j > i$ (*interleaved by rank*). Under $X \wedge \neg Y$ all comparisons of different indices put t left of u, and so do equal indices, since $(u,i) < (t,i)$ would give $(t,1) < (t,2) < (u,1) < (t,1)$; so t and u are *separated*. $\neg X \wedge Y$ is symmetric, and $\neg X \wedge \neg Y$ gives $(t,1) < (t,2) < (u,1) < (u,2) < (t,1)$.

t increasing, u decreasing. $X \wedge Y$ gives $(t,2) < (u,3) < (u,1) < (t,2)$ and $\neg X \wedge \neg Y$ gives $(t,3) < (u,2) < (t,1) < (t,3)$. Under $X \wedge \neg Y$ the comparisons of different indices put t left of u, and $(u,i) < (t,i)$ would give $(t,1) < (u,2) < (u,1) < (t,1)$; so they are separated, and $\neg X \wedge Y$ is symmetric. Reversing indices covers two decreasing types.

Thus two types are separated, or have the same direction and are interleaved. Interleaving is transitive: if t, u and u, v are interleaved (all increasing, say) but t, v are separated, then $(v,1) < (u,2) < (t,3) < (v,1)$ if t is left of v, and $(t,1) < (u,2) < (v,3) < (t,1)$ otherwise. So interleaving is an equivalence relation. A type outside a class is separated from every type of the class and lies on the same side of all of them, since otherwise two interleaved types would be separated; hence class regions contain no other points. Inside a class, interleaving gives the chunks, and (i) gives the within-chunk order. □

Lemma A.5 (Simultaneous finite-group interval shortening)**.** *Let A be a finite group and consider a block-replacement family of width f over A with $n \geq \max\{3, |A|^f\}$ indices, in which all indices have the same number of blocks, on the same lines, in the same within-index order, and in which, for every two block positions t, u on a common line, the truth value of "block t of index i lies left*

of block u of index j" is the same for all $i < j$. Then a nonempty interval J of indices satisfies $\mathrm{val}_c(J) = \mathrm{val}_c(\emptyset)$ *for every line c.*

Proof. Apply Lemma A.4 on each line; there are $q \le f$ class regions in total. Fix a region $\mathcal{Q}$ and number its chunks $1, \dots, n$ from left to right. Assign constants between chunks to the chunk on their left, and absorb fixed constants at the ends of the region into its first or last chunk. Each chunk has original and replacement products with the *same* fixed constants in the same order. Let P_p be the original value of chunks $1, \dots, p$ and Q_p their value when all of them are replaced ($P_0 = Q_0 = 1_A$), and put $\kappa_p^{\mathcal{Q}} := P_p Q_p^{-1}$. Replacing exactly chunks $p+1, \dots, p'$ gives the ordered value $P_p\,(Q_p^{-1}Q_{p'})\,(P_{p'}^{-1}P_n)$, which equals P_n iff $\kappa_p^{\mathcal{Q}} = \kappa_{p'}^{\mathcal{Q}}$; no factors are commuted. If $\mathcal{Q}$ is increasing, the owners in $(a, b]$ occupy exactly chunks $(a, b]$; if it is decreasing, chunks $(n-b, n-a]$. For $j = 0, \dots, n$ let the state be the vector in A^q whose $\mathcal{Q}$-entry is $\kappa_j^{\mathcal{Q}}$ for increasing and $\kappa_{n-j}^{\mathcal{Q}}$ for decreasing regions. Since $n + 1 > |A|^q$, two states coincide, at $a < b$. Replacing the owners $a+1, \dots, b$ then preserves every region value, and nothing outside the regions changes. □

Example A.6 (Opposite orientations and noncommuting values). In S_3 let $\alpha = (12)$, $\beta = (23)$. Three owners have blocks U_i, V_i on row 1 and W_i on row 2:

$$\text{row 1: } (U_1\beta V_1)\,\alpha\,(U_2\beta V_2)\,\alpha\,(U_3\beta V_3), \qquad \text{row 2: } W_3\,\beta\,W_2\,\beta\,W_1,$$

so row 1 is one increasing class and row 2 one decreasing class. Originally $U_i = \beta$, $V_i = W_i = 1$, and both rows have value 1. Replace exactly owners $J = \{2, 3\}$ by $U_2' = 1$, $W_2' = \beta\alpha\beta$, $U_3' = \alpha\beta\alpha\beta$, $W_3' = \alpha$, $V_i' = 1$. Then row 1 becomes $\alpha \cdot \beta\alpha \cdot \alpha\beta\alpha = 1$ and row 2 becomes $\alpha\beta \cdot \beta\alpha\beta \cdot \beta = 1$, computed in order. In the notation of the proof, $(a, b] = (1, 3]$ and $\kappa_1 = \kappa_3 = 1$ on row 1, while $\kappa_{3-b} = \kappa_0 = 1 = \kappa_2 = \kappa_{3-a}$ on row 2: one owner interval matches two different boundary-index orientations, although no owner preserves its own contribution.

Lemma A.7 (Polynomial order homogenization)**.** *Let $N \ge 3$, $m_0 := N$ and $m_{s+1} := (m_s - 1)^2 + 1$, so $m_k \le N^{2^k}$. Let T be a finite set of types, each assigned to a line, and for $i \in [n]$ and $t \in T$ let $p_t(i)$ be a point on the line of t, the points on each line being distinct. Let Π be the set of ordered pairs (t, u), $t = u$ allowed, of types on a common line, and $k := |\Pi|$. If $n \ge m_k$, there is $I \subseteq [n]$ with $|I| = N$ such that for every $(t, u) \in \Pi$ the truth value of $p_t(i) < p_u(j)$ is the same for all $i < j$ in I.*

Proof. Treat the pairs one at a time, the pairs (t, t) first; each step passes from an index set of size at least $(m-1)^2 + 1$ to a subset of size at least m, and earlier homogeneity persists in subsets. For (t, t), Erdős–Szekeres gives a monotone subsequence of length m. For $t \ne u$ (both now monotone), relabel the current indices $1, \dots, m'$ and put $X(i, j) := [p_t(i) < p_u(j)]$ for $i < j$; rows and columns of X are monotone, with directions fixed by those of p_u and p_t. Replacing X by $1 - X$ if necessary, rows are nondecreasing in j.

If columns are nonincreasing in i, let $J(i) := \min\{j > i : X(i, j) = 1\}$ (∞ if none); J is nondecreasing. The chain $i_1 = 1$, $i_{s+1} = J(i_s)$ satisfies $X(i_a, i_b) = 1$ for $a < b$; if it has m terms we are done. Otherwise its at most $m - 1$ terms cut $[m']$ into at most $m - 1$ intervals $[i_s, i_{s+1})$, $[i_L, m']$, one of which has at least m indices, and inside it $j < J(i_s) \le J(i)$ gives $X(i, j) = 0$.

If columns are nondecreasing, let κ be the least index with $X(\kappa, \kappa + 1) = 1$ ($\kappa = m'$ if none). Then $X(i, j) = 1$ for $\kappa \le i < j$ and $X(i, j) \le X(j - 1, j) = 0$ for $i < j \le \kappa$, and one of $[1, \kappa]$, $[\kappa, m']$ has at least $(m' + 1)/2 \ge m$ indices because $(m-1)^2 + 1 \ge 2m - 1$. □

Proof of Lemma A.2(b). An index owning no block can be chosen alone, so assume every index owns a block. The *pattern* of an index records the number of its blocks, the line of each, and their order on shared lines; there are at most $f \cdot f^f \cdot f! = U_f$ patterns, so at least $N_{A,f}^{2^{f^2}}$ indices share one.

Absorb the blocks of all other indices into the constants. The blocks of the common pattern give at most f types and at most f^2 ordered pairs on common lines, so Lemma A.7 with $N = N_{A,f}$ yields $N_{A,f}$ indices satisfying the hypotheses of Lemma A.5, which gives the interval. □

For fixed f, $M_{A,f} = U_f \max\{3, |A|^f\}^{2^{f^2}}$ is polynomial in $|A|$; its degree is doubly exponential in f.

B Local-Group Coordinates and MCFG Block Shortening

LG is always tested on the positive image $S = \widehat{s}(\Sigma^+)$. The classical identity $\mathbf{LG} = \mathbf{Grp} * \mathbf{D}$ places it within the sliding-window and delay theory [37, 38], and the exclusion of $U_1 = \{1, \zeta\}$ ($\zeta^2 = \zeta$) is [36, Proposition 2.3]. The coordinates below are a self-contained implementation of this algebra, not a new characterization; what is specific to MCFG learning is the factorization of group values into fixed separators and replaceable blocks (Lemma B.2). Throughout, $k \geq 1$, A is a finite group, $\lambda_{k,\nu}$ is as in Definition 4.18, a word is *short* if $|w| < 2k$ (recorded exactly by $\widehat{s}_{k,k}$) and *long* otherwise, and for $s, p \in \Sigma^k$ we put $J_\nu(s,p) := \prod_{i=1}^{k} \nu((sp)[i..i+k])$, the ordered product over the k factors of length $k+1$ of sp that start in s.

Lemma B.1 (Local-group typings are morphisms)**.** *$\lambda_{k,\nu}(uv)$ is a function of $\lambda_{k,\nu}(u)$ and $\lambda_{k,\nu}(v)$; hence $\lambda_{k,\nu}$ is a morphism onto a finite monoid $M_{k,\nu}$ with at most $|\Sigma|^{<2k} + |\Sigma|^{2k}|A|$ elements. For long u, v,*

$$\lambda_{k,\nu}(uv) = \big(\langle \mathrm{pref}_k\, u, \mathrm{suf}_k\, v\rangle,\ \gamma_{k,\nu}(u)\, J_\nu(\mathrm{suf}_k\, u, \mathrm{pref}_k\, v)\, \gamma_{k,\nu}(v)\big).$$

Proof. The window component is a morphism. For long u, v the factors of length $k+1$ of uv are, in order, those of u, the k factors starting in the last k positions of u (inside $\mathrm{suf}_k(u)\,\mathrm{pref}_k(v)$), and those of v. If u is short and v long, the factors starting in u lie inside $u\,\mathrm{pref}_k(v)$ and the others are those of v; the case of long u and short v is symmetric; and if both are short, uv is determined exactly. □

Lemma B.2 (Block decomposition)**.** *Let $w = x_0\beta_1 x_1 \cdots \beta_q x_q$ and $w' = x_0\beta'_1 x_1 \cdots \beta'_q x_q$ with $\widehat{s}_{k,k}(\beta'_j) = \widehat{s}_{k,k}(\beta_j)$ for all j. Then $\widehat{s}_{k,k}(w') = \widehat{s}_{k,k}(w)$, and if w is long there are $c_0, \ldots, c_q \in A$, depending only on the x_j and the window types of the β_j, with*

$$\gamma_{k,\nu}(w) = c_0\, \gamma_{k,\nu}(\beta_1)\, c_1 \cdots \gamma_{k,\nu}(\beta_q)\, c_q, \qquad \gamma_{k,\nu}(w') = c_0\, \gamma_{k,\nu}(\beta'_1)\, c_1 \cdots \gamma_{k,\nu}(\beta'_q)\, c_q.$$

Proof. The window claim holds because $\widehat{s}_{k,k}$ is a morphism. A short block equals its replacement and is absorbed into the fixed words, so let all blocks be long. A factor of length $k+1$ either lies inside some block or meets every block it touches in a prefix or suffix of length at most k; factors of the second kind are spelled by the fixed words and the first and last k letters of the blocks, so they are the same in w and w' and occur in the same order. Ordered by starting position, the factors inside β_j form one run with product $\gamma_{k,\nu}(\beta_j)$; let c_j be the ordered product of the factors between consecutive runs. □

Lemma B.3 (Sliding-factor coordinates)**.** *Let S be a finite semigroup whose idempotents all lie in its minimal ideal $\mathsf{K} = \mathcal{M}(H; I, \Lambda; P)$ (a Rees matrix semigroup with P over H), let $n := |S|$, and let $h : \Sigma^+ \to S$ be a morphism. There is $\nu : \Sigma^{n+1} \to H$ such that every w with $|w| \geq 2n$ satisfies*

$$h(w) = \big(\mathrm{row}(h(\mathrm{pref}_n\, w)),\ \mathrm{grp}(h(\mathrm{pref}_n\, w))\, \gamma_{n,\nu}(w),\ \mathrm{col}(h(\mathrm{suf}_n\, w))\big), \qquad (\star)$$

where row, grp, col *are the coordinates of $(i, a, \lambda) \in \mathsf{K}$. In particular $h(w)$ is determined by its first and last n letters and an ordered product in H.*

Proof. (1) Products of n letters lie in K. The Rees quotient S/K has no idempotent but 0, so each element has a power equal to 0. If a product of $m = |S/\mathsf{K}| \le n$ elements were nonzero, two of its m nonzero prefix products would coincide, $p_i = p_j$ with $i < j$; then $p_i = p_i x^q$ for $x = s_{i+1} \cdots s_j$ and all q, impossible for $x^q = 0$.

(2) Coordinates. With $(i, a, \lambda)(j, b, \mu) = (i, a p_{\lambda j} b, \mu)$, fix $i_0 \in I$ and $e_\lambda := (i_0, p^{-1}_{\lambda i_0}, \lambda)$, a right identity for elements with Λ-coordinate λ. For $s \in S$ write $e_\lambda s = (j, c, \mu)$ and $\rho(\lambda, s) := p_{\lambda j} c$; then $(i, a, \lambda)s = (i, a\rho(\lambda, s), \mu)$, so right multiplication fixes the I-coordinate. Dually, left multiplication fixes the Λ-coordinate of an element of K (write $s(i, a, \lambda) = (sf)(i, a, \lambda)$ for an idempotent $f \in \mathsf{K}$ fixing (i, a, λ) on the left).

(3) Group coordinate. Let $w = a_1 \cdots a_m$, $m \ge 2n$, and $h(a_1 \cdots a_n) = (i, a, \lambda_n)$. For $j \ge n$ the Λ-coordinate λ_j of $h(a_1 \cdots a_j)$ equals that of $h(a_{j-n+1} \cdots a_j)$ by (1) and (2). Put $\nu(F) := \rho(\mathrm{col}(h(F[1..n])), h(F[n+1]))$. Applying the right-action identity for $j = n, \ldots, m-1$ in order gives $h(w) = (i,\, a \prod_{j=n}^{m-1} \nu(a_{j-n+1} \cdots a_{j+1}),\, \lambda_m)$, and the product is $\gamma_{n,\nu}(w)$. □

Theorem B.4 (Local-group criteria in sliding-factor coordinates)**.** *For a typing s with positive image S the following are equivalent:* (i) $S \in \mathbf{LG}$*;* (ii) *S has no subsemigroup isomorphic to U_1;* (iii) *every idempotent of S lies in the minimal ideal of S;* (iv) *s is local-group-determined, and one may take $k = |S|$.*

Proof. By [36, Proposition 2.3], (i) and (ii) are equivalent to $S^m = \mathsf{K}$ for some m. If $S^m = \mathsf{K}$ then $e = e^m \in \mathsf{K}$ for each idempotent e; conversely, if $E(S) \subseteq \mathsf{K}$ then $S^n = SE(S)S \subseteq \mathsf{K}$ for $n = |S|$ [36, Lemma 2.2], and $S^n \supseteq \mathsf{K}$ since S^n is an ideal. So (i)–(iii) are equivalent.

(iii)⇒(iv): Lemma B.3 with $h = s$ and $n = |S|$ determines $s(w)$ for $|w| \ge 2n$ from $\lambda_{n,\nu}(w)$, and $\widehat{s}_{n,n}$ records shorter words and ε exactly; so $\widehat{s} \preceq \lambda_{n,\nu}$.

(iv)⇒(i): S is a homomorphic image of $T := \lambda_{k,\nu}(\Sigma^+)$ and $\mathbf{LG}$ is closed under homomorphic images, so it suffices that $T \in \mathbf{LG}$. No short element is idempotent (squaring changes the exactly recorded word or makes it long). Let $e = (\langle p, s\rangle, \gamma_e)$ be idempotent. By Lemma B.1, every element of eTe has window $\langle p, s\rangle$, and long elements with this window multiply by $(\langle p, s\rangle, \alpha)(\langle p, s\rangle, \beta) = (\langle p, s\rangle, \alpha J \beta)$, $J := J_\nu(s, p)$. This operation makes $\{\langle p, s\rangle\} \times A$ a group isomorphic to A via $\alpha \mapsto \alpha J$, and eTe is a finite nonempty subsemigroup of it, hence a subgroup. □

Proposition B.5 (Window–group typings are local-group typings, strictly)**.** (i) *If $\widehat{s} \preceq (\widehat{s}_{k,\ell}, g)$ for a morphism g into a finite group, then $\widehat{s} \preceq \lambda_{\hat{k},\nu}$ with $\hat{k} = \max\{k, \ell, 1\}$ and $\nu(F) := g($last letter of $F)$. Hence $\mathbf{LI} \vee \mathbf{Grp} \subseteq \mathbf{LG}$.*

(ii) *Already for $\Sigma = \{\mathtt{a}, \mathtt{b}\}$ and $h_{\mathtt{ab}} := \lambda_{1,\nu}$ with $\nu(\mathtt{ab}) = 1 \in \mathbb{Z}_2$ and $\nu = 0$ otherwise, $h_{\mathtt{ab}}(\Sigma^+) \in \mathbf{LG} \setminus (\mathbf{LI} \vee \mathbf{Grp})$.*

Proof. (i) For $|w| \ge 2\hat{k}$, $g(w) = g(\mathrm{pref}_{\hat{k}}\, w)\, \gamma_{\hat{k},\nu}(w)$, and $\mathrm{pref}_{\hat{k}}, \mathrm{suf}_{\hat{k}}$ determine $\widehat{s}_{k,\ell}(w)$; shorter words are recorded exactly. For the inclusion, a semigroup $S \in \mathbf{LI} \vee \mathbf{Grp}$ is the positive image of the evaluation typing over the alphabet S, which is window–group-determined by Lemma 4.16, hence local-group-determined, so $S \in \mathbf{LG}$ by Theorem B.4. (ii) Membership in $\mathbf{LG}$ follows from Theorem B.4. Let $k, \ell \ge 0$ and let $g' : \Sigma^* \to H$ map into an arbitrary finite group. Choose $K \ge k + \ell + 1$ and $N \ge 1$ divisible by the order of $g'(\mathtt{a})$, and put $u = \mathtt{b}^K \mathtt{a}^N \mathtt{b}^K$ and $v = \mathtt{b}^{2K}$. They have equal length-k prefix and length-ℓ suffix, are long for $s_{k,\ell}$, and satisfy $g'(u) = g'(\mathtt{b})^{2K} = g'(v)$, but $\#_{\mathtt{ab}}(u) = 1$ and $\#_{\mathtt{ab}}(v) = 0$. Thus $h_{\mathtt{ab}}$ is not window–group-determined, and Lemma 4.16 excludes $\mathbf{LI} \vee \mathbf{Grp}$. □

Lemma B.6 (Short tuples with prescribed local-group type)**.** *Let G be reduced and good of fan-out $\le f$ and rank $\le r_{\max}$, and $\lambda = \lambda_{k,\nu}$. Let $b_{\mathrm{w}} := b_{k,k} = (r_{\max} + 1)\bar{\theta}_G\, f(2k+1)\, |V||M_{k,k}|^f$*

(Corollary 4.12) and $n_\lambda := |V||M_{k,\nu}|^f$. *For every* $B \in V$ *and every realizable* $\mathbf{t} \in M_{k,\nu}^{\mathrm{fan}(B)}$ *there is* $\vec{u} \in L_B(G)$ *with* $\lambda^{(\mathrm{fan}(B))}(\vec{u}) = \mathbf{t}$ *and*

$$\|\vec{u}\|_1 \le \beta_\lambda := (r_{\max}+1)\, b_{\mathrm{w}} \max\{1, c_{A,f}\, n_\lambda\}.$$

Proof. Let T be λ-minimal for $(B, \mathbf{t})$ with heavy threshold $b = b_{\mathrm{w}} \ge \bar{\theta}_G$. For a heavy node v with nonterminal C and yield $\vec{x}$, Corollary 4.12 gives $\vec{x}' \in L_C(G)$ with the same $\widehat{s}_{k,k}$-vector and $\|\vec{x}'\|_1 \le b_{\mathrm{w}}$. For an antichain $v_1, \dots, v_m$ of heavy nodes and $J \subseteq [m]$, let T_J replace the subtrees at v_i, $i \in J$; their components are disjoint contiguous blocks of the root components. By Lemma B.2, every T_J has the root window vector of T, short root components are unchanged, and on each long root component $\gamma_{k,\nu}$ is an ordered product of block values between constants independent of J: a block-replacement family over A with owners $v_1, \dots, v_m$. If $m > c_{A,f}$, Lemma A.2 gives a nonempty J with λ-vector $\mathbf{t}$, and T_J is strictly shorter, contradicting minimality. Lemma 4.9 with $c = c_{A,f}$ and $N = M_{k,\nu}$ gives the bound. □

C Pattern-Thickness Transfer for General MCFG Presentations

Proof of $t_G \le t_{\mathrm{pat}}(H_{\mathrm{red}})$ *in Proposition 4.25.* For a nonterminal A of H_{red} and a realized pattern $I \in \mathsf{Pat}_{H_{\mathrm{red}}}(A)$ let $F_{A,I} := \{\vec{u} \in L_A(H_{\mathrm{red}}) : \mathrm{supp}(\vec{u}) = I\}$; it contains a tuple of length at most $t_{\mathrm{pat}}(H_{\mathrm{red}})$, and $L_A(H_{\mathrm{red}}) = \bigcup_I F_{A,I}$. Call a map *admissible* if it is a composition of projections π_J, permutations σ and merges μ; it does not increase total length, and the support of $\phi(\vec{u})$ depends only on that of $\vec{u}$. Follow the stages of Lemma F.1 applied to H_{red}. After (a) and (b), every nonterminal X has $L_X = \phi_X(L_A)$ for an original A and an admissible ϕ_X. Stage (c) creates $X^{I'}$, for realized I', with $L_{X^{I'}} = \pi_{I'}\{\vec{u} \in L_X : \mathrm{supp}(\vec{u}) = I'\}$; as supports of images depend only on supports, this contains $\pi_{I'}\phi_X(F_{A,I})$ for some realized I. Stage (d) replaces languages by their images under merges, and final trimming does not change surviving languages. Hence every nonterminal Y of G satisfies $\psi(F_{A,I}) \subseteq L_Y$ for some A, realized I and admissible ψ, so $\min_{\vec{u} \in L_Y} \|\vec{u}\|_1 \le t_{\mathrm{pat}}(H_{\mathrm{red}})$. □

Proposition C.1 (Thickness under good-form conversion of λ-free inputs)**.** *Let* G_{in} *be a reduced* λ*-free fan-out-*$\le f$*, rank-*$\le r_{\max}$ *MCFG, possibly with deleting rules, with* $L(G_{\mathrm{in}}) \cap \Sigma^+ \ne \emptyset$. *The good normalization yields a reduced good* G *with the same bounds and*

$$\|G\| = O_{f,r_{\max}}(\|G_{\mathrm{in}}\|), \qquad t_G \le t_{G_{\mathrm{in}}}, \qquad \bar{\theta}_G = O_{f,r_{\max}}(\|G_{\mathrm{in}}\| \max\{1, t_{G_{\mathrm{in}}}\}).$$

Proof. For reduced λ-free G_{in} only full patterns occur, so $t_{\mathrm{pat}}(G_{\mathrm{in}}) = t_{G_{\mathrm{in}}}$; apply Proposition 4.25. □

Thus deleting rules alone do not obstruct thickness transfer; genuinely empty components can (Proposition F.2).

D Auxiliary Verification of Qualitative Reconstruction

Proof of Lemma 3.2. Put $\vec{z}_j := \rho(\vec{u}_1, \dots, \vec{u}_j, \vec{x}_{j+1}, \dots, \vec{x}_r)$. For $E \in \mathcal{E}_e$, Lemma 3.1 gives an admissible E_j with $E[\vec{z}_{j-1}] = E_j[\vec{x}_j]$ and $E[\vec{z}_j] = E_j[\vec{u}_j]$; since $\vec{u}_j \equiv_{L,s} \vec{x}_j$, E accepts $\vec{z}_{j-1}$ iff it accepts $\vec{z}_j$. So $\mathcal{D}_L(\vec{z}_0) = \mathcal{D}_L(\vec{z}_r)$. The types agree because $\widehat{s}^{(e)}(\rho(\cdots)) = \mathrm{out}_\rho$ of the children's types. □

Proof of Proposition 3.10. Induction on the height of a derivation from $[\vec{x}]$. Const derives $\vec{x}$. Link $[\vec{x}] \to [\vec{y}]$ has same-type $\vec{x}, \vec{y}$ with a common context in $K \subseteq L$, so $\vec{x} \equiv_{L,s} \vec{y}$ by substitutability; combine with the hypothesis for $[\vec{y}]$. For Comp $[\vec{z}] \to \rho([\vec{u}_1], \dots, [\vec{u}_r])$ with $\vec{z} = \rho(\vec{u}_1, \dots, \vec{u}_r)$, the children derive tuples equivalent to the $\vec{u}_j$, and Lemma 3.2 transfers this to $\vec{z}$. For $\widehat{S} \to [(w)]$ with

$w \in K \subseteq L$, a derived (v) satisfies $(v) \equiv_{L,s} (w)$, and $\square_1$ accepts (w), so $v \in L$; $\widehat{S} \to \varepsilon$ occurs only if $\varepsilon \in K \subseteq L$. The uncapped grammar is the case of cap $\|K\|_+$. □

Proof of Proposition 3.11. This is the argument of [15, Lemma 3] for the concrete grammar. Each $\iota(X)$ and each $\vec{z}_R := \rho(\iota(Y_1), \ldots, \iota(Y_q))$ for a typed rule $R : X \to \rho(Y_1, \ldots, Y_q)$ is observed, via $\chi(X)$. They have the same type (out_ρ of the children's types is the type of X) and the common context $\chi(X)$, so Link gives $[\iota(X)] \to [\vec{z}_R]$. If $q = 0$, Const gives $[\vec{z}_R] \to \vec{z}_R$; otherwise $(\vec{z}_R, \rho, (\iota(Y_j))_j)$ is a composition witness of rank $q \le r_G$ (and $q \le \|K\|_+$ by (3.1)), so Comp gives $[\vec{z}_R] \to \rho([\iota(Y_1)], \ldots, [\iota(Y_q)])$. By induction, $X \Rightarrow^*_{\widetilde{G}_0} \vec{u}$ implies $[\iota(X)] \Rightarrow^* \vec{u}$. For $w \in L(G) = L(\widetilde{G}_0)$ (Proposition 3.6(iii)) a successful derivation starts with a start link $\widetilde{S} \to X$; then $\chi(X) = \square_1$, so $\iota(X) = (w')$ with $w' \in K$, and Start gives $\widehat{S} \to [\iota(X)] \Rightarrow^* (w)$. □

Proof of Theorem 3.13. If $L \cap \Sigma^+ = \emptyset$, then $K = L \subseteq \{\varepsilon\}$ and Definition 3.4 gives the claim. Otherwise $\mathcal{F}_L \subseteq L$ is finite, Proposition 3.11 with $\iota = \omega$ (canonical tuples have the right type by Proposition 3.6(i)) gives $L \cap \Sigma^+ \subseteq L(\mathcal{B}(K))$, Proposition 3.10 gives $L(\mathcal{B}(K)) \subseteq L$, and $\varepsilon \in L(\mathcal{B}(K))$ iff $\varepsilon \in K$ iff $\varepsilon \in L$. Rank-preserving normalization gives the last sentence. □

Proof of Theorem 3.14. Put $n_K = \max\{1, \|K\|_+\}$. An occurrence is a sample word with at most $2f$ cut positions, so there are $n_K^{O(f)}$ of them, each typed in linear time; grouping occurrences by context and type lists the Link rules. A composition witness of rank $\le r_{\max}$ is a parent occurrence with at most $2fr_{\max}$ further cut positions and a labelling whose number depends only on $f, r_{\max}$; goodness is checked directly. This gives $n_K^{O(fr_{\max})}$ Comp rules, and all other rules are fewer. □

Proof of Corollary 3.16. Let $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$ (for (ii), also $L \in \mathsf{MCFL}(f, r_{\max})$), let $\mathcal{F}_L$ be the characteristic sample of Theorem 3.13 for the operator used, and consider a text for L; membership in MCFLs is decidable [29], so the learner is computable. Let n_0 be a stage by which all of $\mathcal{F}_L$ has appeared. If the learner changes its hypothesis at some stage $n \ge n_0$, the new hypothesis $\mathcal{B}(K_n)$ generates exactly L, so it is never changed again. Otherwise the hypothesis H held at n_0 is final. If $H = H_0$, every datum was generated by $\emptyset$, so $L = \emptyset$. Else $H = \mathcal{B}(K_m)$ for some $m \le n_0$, so $L(H) \subseteq L$ by Proposition 3.10; every word of L appears in the text, before m (then it lies in $K_m \subseteq L(H)$ by sample consistency) or after it (then it is generated by H, which is never changed). So $L(H) = L$. For (ii), each update is a construction (Theorem 3.14) plus a membership test, both polynomial in $\|K\|_+$ for fixed f and $r_{\max}$ [14, Proposition 4]. □

These bounds concern the work per datum. They do not bound the stage of convergence, since a text may postpone any example arbitrarily long; short characteristic data alone gives no presentation-independent convergence stage.

E Auxiliary Proofs for Typed Witness Construction

Proof of Proposition 3.6. At every rule application the morphism property gives

$$\widehat{s}^{(\mathrm{fan}(A))}(\rho(\vec{u}_1, \ldots, \vec{u}_r)) = \mathrm{out}_\rho(\widehat{s}^{(d_1)}(\vec{u}_1), \ldots, \widehat{s}^{(d_r)}(\vec{u}_r)).$$

Induction on height gives (i) and the existence and uniqueness of the lift in (ii), since each typed label is forced by its yield. Erasing labels gives $L(G^{\widehat{s}}) \subseteq L(G)$, (ii) gives the converse, and trimming keeps every rule of every successful derivation. □

Definitions 2.8, 3.4 and 3.5 apply verbatim with any finite-monoid morphism $h : \Sigma^* \to N$ in place of $\widehat{s}$ (tuple types being $h^{(d)}$); write $\mathcal{C}^{\mathrm{mcf}}_{f,h}$ for the class. The class and the operator use types of nonempty components only, so they depend only on $h|_{\Sigma^+}$.

Lemma E.1 (Monoid implementations)**.** *Let $h_0 : \Sigma^* \to M_0$ be a finite-monoid morphism, $S_0 := h_0(\Sigma^+)$ and $s_0 := h_0|_{\Sigma^+}$.*

(i) *On $(\Sigma^+)^d$, equality of $h_0^{(d)}$-, $s_0^{(d)}$- and $\widehat{s_0}^{(d)}$-values coincide. Hence $\mathcal{C}^{\mathrm{mcf}}_{f,h_0} = \mathcal{C}^{\mathrm{mcf}}_{f,s_0}$, and the operators built with h_0 and with $\widehat{s_0}$ return the same grammar on every sample.*

(ii) *For every good MCFG G, the trimmed refinements of G under h_0 and under $\widehat{s_0}$ coincide (the untrimmed ones need not); hence the same canonical tuples, contexts and witness words may be chosen.*

Proof. (i) The three maps agree on Σ^+. (ii) By λ-freeness and Proposition 3.6(i), every typed nonterminal of a successful derivation under either morphism is $A_{\mathbf{t}}$ with $\mathbf{t} \in S_0^{\mathrm{fan}(A)}$, and every start link used is $\widetilde{S} \to S_{(\eta)}$ with $\eta \in S_0$. For children indexed in S_0, each nonempty template component is evaluated in S_0 identically by both morphisms, so both refinements have the same rules among S_0-indexed symbols, which carry all successful derivations. □

Proof of Lemma 3.7. Take a successful typed derivation containing X and follow the path from the root to an occurrence of X, starting with $\Box_1$; at each rule apply Lemma 3.1 with the (nonempty) sibling yields fixed. This gives $\chi(X) \in \mathcal{E}_d$, and replacing the X-subtree by any X-derivation gives a successful derivation of $\chi(X)[\vec{u}]$. If $\widetilde{S} \to X$ is a rule, take the path of length zero. □

Proof of Lemma 3.8. For $\vec{u}, \vec{v} \in L_X$, Proposition 3.6(i) gives equal types, $\chi(X)$ accepts both, and the arity is at most f; substitutability gives $\mathcal{D}_L(\vec{u}) = \mathcal{D}_L(\vec{v})$. □

F Details of the Good Normalization

The four conversions below give Proposition 3.12; each satisfies a tuple-language identity from which the size bound and the thickness transfer of Appendix C are read off. Correctness of the nondeleting and nonmerging conversions is standard [14, 29, 31]; the size and thickness bounds are not stated there and are derived here. For $\vec{w} = (w_1, \ldots, w_d)$ and nonempty $I \subseteq [d]$, $\pi_I(\vec{w})$ keeps the components indexed by I in order, $\sigma\vec{w}$ permutes components and $\mu(\vec{w})$ concatenates consecutive blocks of components; so $\|\pi_I(\vec{w})\|_1 \le \|\vec{w}\|_1 = \|\sigma\vec{w}\|_1 = \|\mu(\vec{w})\|_1$.

Lemma F.1 (Stagewise tuple languages)**.** *Let G_0 be a reduced f-MCFG of rank at most $r_{\max}$. Each conversion below preserves linearity, does not increase fan-out, rank or any template, and multiplies the numbers of nonterminals and rules by a factor depending only on f and $r_{\max}$.*

(a) Nondeletion *(construction (f3) in the proof of [29, Lemma 2.2]). Introduce A^I for nonempty $I \subseteq [\mathrm{fan}(A)]$. For $R : A \to \rho(B_1, \ldots, B_q)$ let J_j be the components of B_j used by the I-indexed outputs of ρ; keep these outputs, use $B_j^{J_j}$ if $J_j \neq \emptyset$ and omit B_j otherwise. The rules are nondeleting and $L_{A^I} = \pi_I(L_A(G_0))$.*

(b) Nonpermutation. *Decorate a fan-out-d nonterminal by $\sigma \in \mathfrak{S}_d$ (the symmetric group on $\{1, \ldots, d\}$), the child's permutation being determined by the order in which the parent template uses its variables. The decorated nonterminal generates σL_A.*

(c) Empty components. *Let* $\mathsf{Pat}(A)$ *be the set of support patterns of* L_A*, computed by saturation. For realizable child patterns, replace variables of empty components by* ε*, delete empty parent components, omit all-empty children, and introduce* A^I *for each nonempty* $I \in \mathsf{Pat}(A)$*. Then* $L_{A^I} = \pi_I\{\vec{u} \in L_A : \mathrm{supp}(\vec{u}) = I\}$*, the result is* λ*-free, and* $S^{\{1\}}$ *generates the nonempty part of the language.*

(d) Nonmerging *(construction in the proof of [14, Section 2.3, Lemma 1]). Decorate a nonterminal by a composition* μ *of its fan-out, the child's composition being given by the maximal runs of its consecutive variables in the parent template. The decorated nonterminal generates the image* $\mu(L_A)$*.*

Stages (b)–(d) preserve the properties established by earlier stages.

Proof. (a) For any child tuples, $\pi_I(\rho(\vec{v}_1, \ldots, \vec{v}_q)) = \rho^I((\pi_{J_j}\vec{v}_j)_{J_j \neq \emptyset})$; induction gives $\subseteq$, and for $\supseteq$ the omitted children are completed by terminal derivations, which exist because G_0 is reduced. There are at most 2^f copies of each nonterminal and rule. (b) Linearity and nondeletion make the child permutations well defined; induction gives σL_A. There are at most $f!$ copies. (c) Retained variables occur once and in order; induction gives the identity, and omitted all-empty children are completed by derivations of their empty pattern. There are at most 2^f patterns per nonterminal and $2^{f r_{\max}}$ choices per rule. (d) Nonpermutation makes the runs compatible with each child's order and nondeletion makes them cover its components; then $\rho^\mu(\mu_1(\vec{v}_1), \ldots, \mu_q(\vec{v}_q)) = \mu(\rho(\vec{v}_1, \ldots, \vec{v}_q))$, and induction gives the image identity (merging need not be injective, so no inverse-image claim is made). The size is linear for fixed $f, r_{\max}$ [14, Section 2.3, after Example 2]. □

Proof of Proposition 3.12. Apply (a)–(d) to the trimmed input and trim again. Each stage multiplies the size by a factor depending only on $f, r_{\max}$, so $\|G_{\mathrm{good}}\| = O_{f,r_{\max}}(\|G_{\mathrm{in}}\|)$, and $L(G_{\mathrm{good}}) = L(G_{\mathrm{in}}) \setminus \{\varepsilon\}$ by (c). If this language is nonempty, the final trimming leaves a reduced grammar, and trimming preserves goodness. □

Proposition F.2 (Empty-component elimination can blow up thickness)**.** *For every* $n \geq 1$ *there is a reduced fan-out-two, rank-two MCFG* E_n *of size* $O(n)$ *with* $t_{E_n} \leq 2$ *and* $\theta_{E_n} = O(n)$ *whose stage-(c) conversion contains a nonterminal and a rule of thickness at least* 2^n*; in particular* $t_{\mathrm{pat}}(E_n) \geq 2^n$*.*

Proof. Let E_n have rules $A_0 \to (\mathtt{a}, \varepsilon)$, $A_i \to (\varepsilon, \mathtt{c}) \mid (\xi^1_1\xi^1_2, \xi^2_1\xi^2_2)(A_{i-1}, A_{i-1})$ for $1 \leq i \leq n$, and $S \to (\xi^1_1 \mathtt{d}\, \xi^2_1)(A_n)$. It is reduced, nondeleting, nonpermuting and nonmerging. Completing every A_i by $(\varepsilon, \mathtt{c})$ (or A_0 by $(\mathtt{a}, \varepsilon)$) gives $t_{E_n} \leq 2$ and, along a path of length at most n, $\theta_{E_n} = O(n)$. An A_i-tuple of support $\{1\}$ cannot use $(\varepsilon, \mathtt{c})$, so it is $(\mathtt{a}^{2^i}, \varepsilon)$. Hence $A_n^{\{1\}}$ and the rule $S \to (\xi^1_1 \mathtt{d})(A_n^{\{1\}})$ have thickness at least 2^n. □

This differs from Proposition 4.8: here a support pattern of the source forces an exponentially long tuple, there the typed refinement does. Neither is a characteristic-data lower bound.

G Substitutability of the Bracketed Doubling Family

This appendix proves Proposition 4.22(ii), adapting the deterministic-parser argument of [25]. Fix $n \geq 1$, put $L^{(0)} := \{\mathtt{a}\}$ and $L^{(i)} := \lceil L^{(i-1)} \rceil \cup \lfloor L^{(i-1)} L^{(i-1)} \rfloor$ for $i \geq 1$, so that $L_{E_i}(G_n) = L^{(i)}$ and $L_n = L^{(n)}$. For $x, y \in \Gamma^*_{\mathrm{br}}$ write $x \equiv y$ if $\{(u, v) : uxv \in L_n\} = \{(u, v) : uyv \in L_n\}$; this is a monoid congruence.

The parser. A *stack* is a word over $E_0, \dots, E_n, \rceil, \rfloor$, with its top on the left; ε is the empty stack. For every stack ψ and $1 \le i \le n$ the moves are

$$E_0\psi \xrightarrow{\mathtt{a}} \psi, \quad E_i\psi \xrightarrow{\lceil} E_{i-1}\rceil\psi, \quad E_i\psi \xrightarrow{\lfloor} E_{i-1}E_{i-1}\rfloor\psi,$$
$$\rceil\psi \xrightarrow{\rceil} \psi, \qquad \rfloor\psi \xrightarrow{\rfloor} \psi,$$

and there are no others; in particular no move leaves ε. Write $\sigma \xrightarrow{w} \theta$ if reading w from σ ends in θ. Three properties are used.

(P1) *Determinism and injectivity.* For each letter c, the c-moves form a partial injective map on stacks: an opening bracket reveals the expanded E_i and the untouched suffix, and $\mathtt{a}$, $\rceil$, $\rfloor$ identify the popped top symbol. Hence every word acts as a partial injection on stacks.

(P2) *Weight.* Give E_i weight 4^i and each closing marker weight 1, and let $\mathrm{wt}(\sigma)$ be the total weight. Every move lowers wt, since $4^i > 2 \cdot 4^{i-1} + 1$; hence $\sigma \xrightarrow{w} \sigma$ only for $w = \varepsilon$.

(P3) *Locality.* Moves read and rewrite only the top symbol, and closing markers are only popped. Hence $\sigma \xrightarrow{w} \theta$ implies $\sigma\psi \xrightarrow{w} \theta\psi$. Conversely, if $\sigma_1 \neq \varepsilon$ and $\sigma_1\psi \xrightarrow{w} \theta$, then either (a) the run reaches ψ: $w = w_1w_2$ with $\sigma_1 \xrightarrow{w_1} \varepsilon$ and $\psi \xrightarrow{w_2} \theta$, so $\mathrm{wt}(\theta) \le \mathrm{wt}(\psi)$; or (b) it does not: $\theta = \theta_1\psi$ with $\sigma_1 \xrightarrow{w} \theta_1 \neq \varepsilon$, so $\mathrm{wt}(\theta) > \mathrm{wt}(\psi)$, and θ_1 ends with the last symbol of σ_1 whenever that symbol is a closing marker. For fixed ψ, comparing $\mathrm{wt}(\theta)$ with $\mathrm{wt}(\psi)$ decides the case.

Lemma G.1. *For $0 \le i \le n$ and every stack ψ, $E_i\psi \xrightarrow{w} \psi$ if and only if $w \in L^{(i)}$. Moreover, if $\sigma \xrightarrow{w} \theta$ and $w \in L^{(i)}$, then $\sigma = E_i\theta$.*

Proof. A run $E_i\psi \xrightarrow{w} \psi$ is in case (a) of (P3), since case (b) ends in $\theta_1\psi \neq \psi$; so $w = w_1w_2$ with $E_i \xrightarrow{w_1} \varepsilon$ and $\psi \xrightarrow{w_2} \psi$, and $w_2 = \varepsilon$ by (P2). With the lifting in (P3), it suffices to show $E_i \xrightarrow{w} \varepsilon$ iff $w \in L^{(i)}$, by induction on i. From E_0 only the $\mathtt{a}$-move applies, and ε admits none. For $i \ge 1$, the first letter is $\lceil$ or $\lfloor$. After $\lceil$, the stack $E_{i-1}\rceil$ must be emptied; case (a) of (P3) with $\sigma_1 = E_{i-1}$, $\psi = \rceil$ splits the rest as zz' with $E_{i-1} \xrightarrow{z} \varepsilon$ and $z' = \rceil$. After $\lfloor$, two such splittings of $E_{i-1}E_{i-1}\rfloor$ give $z_1z_2\rfloor$ with $E_{i-1} \xrightarrow{z_j} \varepsilon$. Induction gives $E_i \xrightarrow{w} \varepsilon$ iff $w \in L^{(i)}$. For the last claim, $E_i\theta \xrightarrow{w} \theta$ by the first part, and (P1) gives $\sigma = E_i\theta$. □

Lemma G.2. *If $0 \le i \le n$ and $w, w' \in L^{(i)}$, then $w \equiv w'$.*

Proof. If $pwq \in L_n$, Lemma G.1 gives a run $E_n \xrightarrow{p} \sigma \xrightarrow{w} \theta \xrightarrow{q} \varepsilon$ with $\sigma = E_i\theta$, and also $E_i\theta \xrightarrow{w'} \theta$; so $pw'q \in L_n$. Exchange w and w' for the converse. □

Lemma G.3 (Branch synchronization)**.** *Let $\sigma \xrightarrow{x} \theta$ and $\sigma \xrightarrow{y} \theta$, where x, y are nonempty and begin with different letters. Then $\sigma = E_i\psi$ for some $1 \le i \le n$ and stack ψ, and $x = wx'$, $y = w'y'$ with $w, w' \in L^{(i)}$, $\psi \xrightarrow{x'} \theta$ and $\psi \xrightarrow{y'} \theta$.*

Proof. Only a top E_i with $i \ge 1$ admits two different letters, namely $\lceil$ and $\lfloor$. Say $x = \lceil x_0$ and $y = \lfloor y_0$, so $E_{i-1}\rceil\psi \xrightarrow{x_0} \theta$ and $E_{i-1}E_{i-1}\rfloor\psi \xrightarrow{y_0} \theta$. By (P3) both runs are in the same case. Case (b) is impossible: it would give $\theta = \theta_1\psi = \theta_2\psi$ with θ_1 ending in $\rceil$ and θ_2 in $\rfloor$. In case (a), $x_0 = x_1x'$ with $E_{i-1}\rceil \xrightarrow{x_1} \varepsilon$ and $\psi \xrightarrow{x'} \theta$, so $w := \lceil x_1$ satisfies $E_i \xrightarrow{w} \varepsilon$ and $w \in L^{(i)}$ by Lemma G.1. The same argument gives $y = w'y'$. □

Proposition G.4. *If $\sigma \xrightarrow{x} \theta$ and $\sigma \xrightarrow{y} \theta$, then $x \equiv y$. Consequently L_n is Clark–Eyraud substitutable.*

Proof. Induct on $|x| + |y|$. If x or y is empty, then $\sigma = \theta$, and the other is empty by (P2). If $x = cx'$ and $y = cy'$ for a letter c, determinism gives $\sigma \xrightarrow{c} \sigma'$ with $\sigma' \xrightarrow{x'} \theta$ and $\sigma' \xrightarrow{y'} \theta$, and induction applies. Otherwise Lemma G.3 gives $x = wx'$, $y = w'y'$; induction gives $x' \equiv y'$, Lemma G.2 gives $w \equiv w'$, and $\equiv$ is a congruence.

For the consequence, let $x, y \in \Gamma^+_{\mathrm{br}}$ and $uxv, uyv \in L_n$. Lemma G.1 gives runs $E_n \xrightarrow{u} \sigma \xrightarrow{x} \theta \xrightarrow{v} \varepsilon$ and $E_n \xrightarrow{u} \sigma' \xrightarrow{y} \theta' \xrightarrow{v} \varepsilon$. By (P1), $\sigma = \sigma'$ and $\theta = \theta'$, so $x \equiv y$. □

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During preparation of this work, the author used ChatGPT (OpenAI) and Claude (Anthropic) to assist with organization of the manuscript, expression and translation of text, cross-checks of references and notation, and editing of LaTeX source. After using these tools, the author reviewed and edited the material as needed and takes full responsibility for the content of the publication.

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